



The ENVIRONMENTAL DEFENCE SOCIETY is a charitable organisation which has been operating since 1971. It is comprised of resource management professionals who are committed to improving environmental outcomes throughout New Zealand. EDS has long sought to protect New Zealand's outstanding coastal and marine areas, through providing thought leadership and best practice guidance, and taking matters of national importance to the courts. This book is the latest in a series of publications on key environmental issues facing the country.

Further information about EDS and its work can be found at www.eds.org.nz.

Front cover: Marine studies student photographing underwater diversity at the Poor Knights Islands *(Daniel Sharp)*

Back cover: Diver enjoying school of demoiselle at Tuhua (Mayor Island), Bay of Plenty *(Daniel Sharp)*

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Preface

Blue is the new green.

As our attention shifts to the sea as the new resource frontier for energy, minerals, aquaculture, tourism and other components of a burgeoning 'blue economy', awareness deepens of the need to defend the marine environment from overexploitation. Hence the straight-to-the-point title of this timely book, *Sustainable Seas: Managing the Marine Environment*, which addresses the challenge of sustainable marine management head-on.

Conservation of the sea has lagged far behind that of land, in New Zealand as everywhere else in the world. While a third of Aotearoa's land is in conservation areas, less than one per cent of the coastal marine area around the main islands is protected by marine reserves.

Yet scientists say that some of the most serious impacts of climate change will occur in the marine realm. Rising sea levels, ocean acidification, higher sea temperatures — this menacing trio is already having a damaging impact on marine ecosystems, coastal land and human shoreline infrastructure, and the effects will worsen as the century proceeds. Some scientists are predicting mass extinctions in the oceans as a result.

If the goal is sustainability, the path must include understanding the environmental limits of the sea. One of the great difficulties in identifying those limits has been the invisibility of what goes on in the more than two-thirds of the planet that is ocean. Fortunately, this is changing, and with advanced underwater imaging and remote-operated vehicles even the deepest ocean trenches (including some in New Zealand waters) are revealing their secrets.

And what they show is startling: ecosystems of extraordinary diversity and often extreme fragility. Biologists tell us that four-fifths of New Zealand's total living diversity is marine, and most of it is found nowhere else in the world. This is a heritage that's as important to preserve as kakapo or kauri forests.

Sustainable seas demand responsible governance, which means integrated governance — not easy to achieve in New Zealand's case, where jurisdiction of the marine environment is apportioned among multiple agencies with multiple policies and pieces of legislation. When it comes to understanding and negotiating this jurisdictional thicket, *Sustainable Seas* could be called 'a guide for the perplexed'. It explores and explains relevant legislation and policy, as well as providing case studies and guides to best practice.

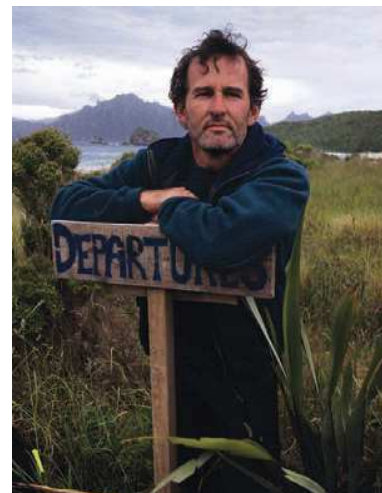
A particular focus is the new tool known as 'marine spatial planning', which is already showing its worth in the New Zealand setting. It is a way of resolving conflicts of use in the sea — between conservation and fisheries, fisheries and mining, aquaculture and amenity preservation, and so on — by taking an ecosystem-wide approach to marine activities. The aim is to apportion zones for various uses, and zones for protection, based on identifying the 'ecological backbone' of a marine area.

Despite tools such as marine spatial planning, there are genuine concerns that marine governance may not be adequately precautionary in the face of the significant demands of commercial development. The pressure to extract from the oceans will only increase, and strong voices are needed to speak for the blue wilderness.

My grandfather was a big game fishing captain in the Bay of Islands in the 1930s. Those were the days of catching a dozen marlin before breakfast. He used to speak of schools of kahawai and trevally stretching from one side of the Bay to the other. I will never see such schools, nor the monster black marlin that fed on them. And perhaps my grandchildren won't see the marine abundance I have had the privilege of seeing.

In my grandfather's day, it was inconceivable that ocean ecosystems could be endangered. The sea was an icon of permanence, a realm immune from lasting human damage, an inexhaustible larder.

We know differently now. What is required — and what this book speaks to — is an ethic of kaitiakitanga, that vital expression of reciprocity between human and non-human elements of the environment, between us and the encircling sea.



Kennedy Warne

Marine writer and editor-at-large of New Zealand Geographic magazine

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Conservation Act 1987
Continental Shelf Act 1964
Crown Minerals Act 1991
Environment Act 1986
Environmental Protection Authority Act 2011
Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012
Exclusive Economic Zone and Continental Shelf (Environmental Effects) Amendment Act 2013
Exclusive Economic Zone and Continental Shelf (Environmental Effects – Non-Notified Activities) Regulations 2014
Exclusive Economic Zone and Continental Shelf (Environmental Effects – Permitted Activities) Regulations 2013
Fiordland (Te Moana o Atawhenua) Marine Management Act 2005
Fisheries Act 1996
Fisheries (Benthic Protection Areas) Regulations 2007
Fisheries (Kaimoana Customary Fishing) Regulations 1998
Fisheries (Quota Operations Validation) Act 1997
Fisheries (South Island Customary Fishing) Regulations 1999
Harbours Act 1950
Hauraki Gulf Marine Park Act 2000
Health and Safety in Employment (Petroleum Exploration and Extraction) Regulations 2013
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Local Government Act 2002
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Māori Commercial Aquaculture Claims Settlement Act 2004
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Marine Mammals Protection Act 1978
Marine Mammals Protection Regulations 1992
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National Parks Act 1980
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Reserves Act 1977
Resource Management Act 1991
Resource Management (Marine Pollution) Regulations 1998
Subantarctic Islands Marine Reserves Act 2014
Submarine Cables and Pipelines Protection Act 1996
Sugar Loaf Islands Marine Protected Area Act 1991
Te Ture Whenua Act 1993
Town and Country Planning Act 1977
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Topography

New Zealand's seabed has a very diverse topography. This is because the country sits astride two colliding tectonic plates. The movement of the plates has resulted in arcs of volcanic activity and the uplift and depression of large areas of coastal land and seabed.

Many coastal features have been formed from the remnants of volcanic activity including cones, sunken craters and lahars. New Zealand's seabed features chains of underwater volcanoes and deep ocean trenches and ridges. The mountains which are entirely under the sea are called 'seamounts'.

Closer to land, the New Zealand coastline is punctuated by many estuaries and harbours. These shallow sheltered areas were formed after the last ice age, when the rising seas flooded river valleys and other low-lying areas. They have continued to infill with sediment washed off the land.

Rocky reefs have formed around the more exposed parts of the coast, with soft rock being eroded away by waves to form bays, and harder rock remaining as headlands and reefs. Sand spits and barrier islands are also common around the coast, forming across the mouths of harbours and bays, where rivers have dropped their sediment load.

Movement of water

In terms of the sea itself, New Zealand's marine area is influenced by two major bodies of oceanic water: subtropical surface water which travels from the central Pacific Ocean via the eastern coast of Australia; and subantarctic surface water which originates from the Southern Ocean.

Both of these water bodies move in an easterly direction and they meet along a subtropical front which bisects the southern part of New Zealand's marine area. This oceanic meeting point is located along the same latitude as Fiordland on the west coast and along the Chatham Rise to the east (see Figure 1.2). It is a very productive area for marine life, particularly over the Chatham Rise. More localised currents move the water around the New Zealand coast.

Marine habitats

The marine environment is subjected to intense geological processes, which have led to the formation of some unique marine habitats, including chains of underwater volcanoes and deep oceanic trenches. It is a highly interconnected system, with ocean currents transporting species over large distances, and lifecycles extending over wide geographic areas. As a result, marine habitats vary considerably, with geology, ocean currents and climate affecting distribution.³

New Zealand has a high diversity of marine habitats, some of which are particularly important to the ecological health and productivity of the marine area. A broad-scale gap analysis of marine protected areas in the territorial sea (the marine area extending out to 12 nautical miles from land), prepared by the Department of Conservation and the then Ministry of Fisheries in 2011, provides some useful information on the location of different marine habitats, including maps.⁴

Estuaries

An estuary is a partially enclosed body of water, that is either permanently or periodically open to the sea, and that is affected by both runoff from the land and inflow from the sea.⁵ Estuaries are located in the brackish water at river mouths and are one of the most ecologically productive



Mangrove-flanked estuaries, such as the one shown here at Pataua, Northland, provide important habitat for juvenile fish and seabirds (Raewyn Peart)

Soft sediments

Supporting a wide variety of organisms, soft sediment habitats cover 70 per cent of the world's seafloor, and are found throughout New Zealand harbours, estuaries and open coastal environments. They play a key role in marine ecosystem functioning.

Bottom-dwelling animals burrow within the sediment column (up to 2 metres below the sediment surface in the case of some crabs and shrimps). Because sediment is an accumulation of particles that have settled to the seabed, it is generally rich in organic matter. Soft sediments are inherently complex: bacteria, microalgae and invertebrates all influence oxygen and nutrient concentrations simultaneously, via direct and indirect pathways.⁸

The sediment–seawater interface is permeable, so water fills the small gaps between particles and carries oxygen into the sediment. The concentration of oxygen in pore water decreases rapidly with depth in the sediment, so animals living beneath the surface must maintain access to an oxygen-rich supply through maintaining open pores. Also, concentrations of nutrients (released by the breakdown of organic matter by bacteria) are generally much greater in pore water than they are in overlying seawater. The nutrients in pore water act as a fertiliser for algal growth, and the algae can only flourish with a combination of both sunlight and essential nutrients.⁹



Many animals burrow in soft sediments including this tube worm living within a scallop bed at Ahuahu (Great Mercury Island), Hauraki Gulf (Daniel Sharp)

Biogenic reefs

A biogenic reef is composed of the hard parts of living and dead organisms which create structure above the seafloor, and is a highly specialised habitat. Such reefs often occur in areas of strong water movement where there is a good food source and little sediment deposition.¹⁰ Biogenic reefs are formed by colonial tube worms, bivalves such as oysters and mussels, sponges, corals, bryozoans and coralline algae, and can form extensive areas of three-dimensional structures up to 2 metres tall. Formations such as horse-mussel beds and mangrove habitats are also classed as biogenic reefs.¹¹ These living reefs are important as they provide refuges for juvenile fish and a stable home for other marine life in an otherwise featureless seabed.

The distribution of known biogenic reefs around New Zealand is patchy, but they are primarily found in the far south of New Zealand on the Catlins coast and around Stewart Island; in the middle of New Zealand from Kaikōura across to the northwest tip of the South Island, in Cook Strait, and along the Kapiti and Wairarapa coasts, East Cape and the east coast of Northland. Stewart Island and the Chatham Islands support the greatest proportions of biogenic reef habitat in New Zealand.¹²

One of the most biodiverse biogenic reef habitats in New Zealand is rare, and occurs where gravel lying on the seabed is intermixed with the hard calcium remains of shellfish and bryozoans. These areas are called 'calcareous gravels' and are home to rich thickets of bryozoans and sponges. A very ecologically important area of these gravels occurs around the northern tip of the North Island near Spirits Bay, where more than 330 species of bryozoans and 220 species of sponges have been found. They also occur near the southern coast of the South Island, off the Otago coast, in the Foveaux Strait and in Tasman Bay.

Hydrothermal vents support complex ecosystems of unusual organisms that have developed unique biochemical adaptations to high temperatures and the highly toxic (to land-based creatures) environment. The fluids emanating from the vents contain chemicals that feed microbes at the base of a unique food web that survives without any interaction with the sun.¹⁷ These microbes use chemicals such as hydrogen sulphide to provide the energy source that drives their metabolic processes. They ultimately support a wide range of other organisms such as tubeworms, shrimp and mussels.¹⁸

Scientists have recently discovered new forms of life deep under the sea within these vents. Despite the often very high temperatures, large numbers of micro-organisms grow around the vents, typically as bacterial mats. Many animals are unique to particular vent sites and are not seen even a few hundred metres away.¹⁹ Deepwater hydrothermal vents in the Kermadec Islands support extensive beds of giant vent mussels, which are unique to the area, and which in turn provide habitat for deepwater crabs and an endemic eel-like fish.

Marine species

As much as 80 per cent of New Zealand's total biodiversity lives in the sea.²⁰ It is estimated that there might be up to 65,000 different marine species, of which only 25 per cent have been described.²¹ Marine research is significantly behind its terrestrial counterpart, making assessment of ecological health and measurement of biodiversity difficult because information is so scarce.²² However, we do know that the high diversity of marine habitats within New Zealand's waters is critical to the ecological health and productivity of the marine area, and is relied on by an extraordinarily high number of species.

New Zealand is considered internationally to be an important contributor to global marine diversity. For example, endemic species include:

- About 95 per cent of the 733 known sponge species
- 84 per cent of bivalves and gastropods (types of marine molluscs)
- 75 per cent of ascidians (commonly known as sea squirts)²³

The greatest biodiversity of fish and invertebrates is found along the northeast coast of the North Island, where known hotspots of species endemism and/or species diversity occur at the Three Kings Islands, in the soft sediment assemblages of Spirits Bay (Piwhane) and Tom Bowling Bay (over 700 invertebrate species).²⁴

In a recent paper by marine biologist Dennis Gordon, it was recorded that the animal kingdom Animalia²⁵ (including fish and invertebrates) is by far the richest in species numbers in New Zealand with 13,415 marine species, followed by Chromista (including algae) with 2,644 species, Plantae (mostly red and green seaweeds) with 702 species, Fungi with 89 and Protozoa²⁶ with 43 species.²⁷ Gordon noted that seaweeds comprise approximately 900 species, including at least 109 that are new to science and not yet described.²⁸

Plankton

Plankton comprises the largest group of organisms in the ocean. The term refers to organisms that rely on the movement of the water to carry them around, because the vast majority of plankton are too weak to swim against the current. They are divided into two basic groups: phytoplankton (or plant plankton) and zooplankton (or animal plankton). Phytoplankton are the primary producers for the oceans. Around half of worldwide primary productivity is achieved by these microscopically small organisms which grow and divide in the ocean.²⁹ The animal components of plankton are the zooplankton.

Plankton are believed to be the most important life form on Earth. Through photosynthesis, phytoplankton convert water and carbon dioxide into organic material and oxygen. They are at the base of the oceanic food chain, and without them few other life forms in the ocean could exist.³⁰ Plankton are also responsible for manufacturing a significant amount of the Earth's oxygen.

Specific species of plankton occupy particular marine habitats, with each species being uniquely adapted to factors such as light, temperature, turbulence and salinity. Plankton are sensitive to their environment and a change in concentration can indicate a subtle environmental change. They are often highly responsive to nutrient levels, temperatures, pollution, food that is not nutritious, levels of light and increased predation.³¹

brightly coloured sea slugs (nudibranchs), shellfish-type animals which have evolved to live without shells. Some sea slugs protect themselves with stinging cells recycled from their prey.

There are numerous types of shellfish found in New Zealand's marine environment. Of these, many bivalves, such as mussels, cockles, oysters, pipi and scallops, feed by straining plankton from the water. Some species of bivalves can reach phenomenal densities at over 20,000 per square metre. Most bivalves are sedentary or slow-moving animals. Some, such as pipi and cockles, spend their life buried in seafloor sediment, while others like oysters and mussels remain anchored to one spot by attaching themselves to rocks and solid structures.³⁵ Scallops, on the other hand, can move quickly through the water if they sense danger.

Seamounts can support very large deepwater sponges and corals that may live for hundreds of years. In these habitats, sponges can reach several metres in height and corals have been found growing up to 10 metres tall. Deepwater coral banks, large sea fans, sea pens and sponges are particularly vulnerable as they are fragile and slow-growing. Once destroyed, coral formations appear to need 200 to 400 years to recover to some degree but will never be quite the same if their endemic species have been lost.³⁶

Fish

Fish are found in all kinds of marine habitats. Rocky reefs support some 250 species of reef fish. Some of these, including parore, silver drummer and butterfish, graze on seaweed. Others including maomao, trevally and demoiselle sift plankton from the seawater. Other fish commonly targeted by fishers, including snapper, blue cod, kahawai, John Dory and moki, predate on smaller fish and invertebrates living in the rocky reef environment or on adjacent areas of sediments. Some unusual species also make New Zealand's rocky reefs their home including sea horses and sea dragons. The sea horse species found in New Zealand is one of the largest in the world. The sea horses anchor themselves to seaweeds or other structures on the reef with their prehensile tails, and ambush their food, mainly crustaceans. Male sea horses incubate fertilised eggs in a specialised pouch.

Bottom-feeding fish such as flounder, terakihi, snapper, red cod and warehou predate on common residents of coastal soft sediments including a variety of worms, crabs, shellfish, starfish, sea urchins and sea cucumber. Some fish, such as the stargazer, completely bury themselves in the sand and pounce on unsuspecting passers-by.³⁷ Seamounts offer a refuge for deepwater fish species including orange roughy, black oreo and black cardinalfish.

Many of the fish found in New Zealand waters are present in other parts of the world. However, there are a number of fish species that are very rare, and found only in certain habitats, such as the morse-code leatherjacket which is generally only spotted in the Kermadec Islands.³⁸ There is a much higher level of endemism in New Zealand rock pool fish which are less mobile. It is thought that 62 per cent of New Zealand's 94 species of rock pool fish are only found in this country.³⁹

Fish are an important food source for marine mammals and birds, as well as for the larger predatory migratory fish species. New Zealand waters are visited by numerous migratory fish and they provide a critical habitat for many of them. Species such as the southern bluefin tuna and striped marlin migrate seasonally across the vast ocean basins.



Snapper, shown here at the Poor Knights Islands, Northland, are carnivorous and prey on small fish and invertebrates whereas the smaller two spot demoiselles surrounding them feed on plankton (Daniel Sharp)

Marine productivity

New Zealand's marine environment is influenced by its geological history, its isolation, and the action of major ocean currents. Longer-term weather cycles affect New Zealand's oceans. Together, the atmosphere and the rotation of the earth drive major ocean circulation patterns and climate systems across the world, affecting deepwater and surface currents, sea temperature, sea levels, seawater chemistry, and the productivity of our oceans.⁴⁰

Productivity is a measure of the amount of life that can be supported in a given area. It is driven by the availability of nutrients and light for phytoplankton growth and reproduction.⁴¹ There are a number of key processes that are responsible for marine productivity levels. Upwelling occurs where the deep, cold ocean water rises to the surface, both in the open ocean and along coastlines. This water is replacing ocean surface water pushed away from the land by offshore winds. The water coming from the deeper ocean is naturally rich in nutrients, primarily nitrate and phosphate. This helps to fertilise the surface water and promotes phytoplankton growth, ultimately resulting in high biological productivity in these areas.⁴² There is a reverse of the process called downwelling, where the wind pushes surface water towards the coast where it eventually sinks down.

Natural inflows of nutrients and sediments into coastal ecosystems are an important part of biological, physical and chemical cycles in New Zealand's inshore area. These are derived both from deepwater upwellings and inputs from the land. Such cycles are the basis for productivity in these areas, although excess nutrients coming from the land can cause eutrophication, which in turn results in oxygen depletion and acidification of seawater. The productivity of the offshore ocean area is primarily influenced by major oceanic currents.⁴³

As discussed previously, habitats contribute to marine productivity in different ways and to differing degrees. For example, estuaries are highly productive environments. They have a continuous source of nutrients from their freshwater inputs and are important sites of nutrient recycling in the coastal environment. Nutrients within the estuary are used by plants such as algae, seagrasses, seaweeds and mangroves and they enter other parts of the food web when these plants are grazed on by invertebrates. When the plants, or parts of them die, nutrients are recycled within the estuary by the actions of invertebrates and bacteria.⁴⁴

Different species contribute to marine productivity in various ways. For example, seagrass and kelp store nutrients in their biomass for a long time, including nitrogen and phosphorous compounds transported by rivers from agricultural areas to the sea. This means that they function as a kind of biological purification system in coastal ecosystems.⁴⁵

Food webs and lifecycles

Food webs are complex networks through which energy and materials move within and among trophic levels in an environment, from microbes all the way through to top predators.⁴⁶ Marine food webs or chains refer to a group of organisms linked through feeding relationships. All of the plants and animals living in the sea, from algae to whales, form part of the marine food web. In addition, marine life is intricately linked to the physical processes within the ocean, including factors such as temperature and sunlight.

Some species will spend all their lives in one habitat whereas others will use a range of different habitats during their lifecycle. For example, adult snapper generally spawn their eggs in highly productive areas close to estuaries and harbours, such as the Hauraki Gulf. The juvenile snapper move into sheltered coastal areas to mature, and then once they have grown to adult size, they often move further out to sea. Studies have shown that the larvae of longfin eels actually hatch far away from New Zealand, possibly near Tonga, and one tagged female longfin eel took 161 days to swim from Canterbury's Lake Ellesmere to a point 160 kilometres north-east of New Caledonia.⁴⁷

The wide variety of lifecycles for different species, from spawning and pupping, to migrations over vast distances, means that taking an ecosystem approach to understanding food webs is critical. This looks at the relationships between all species using the oceans, rather than considering species as independent entities.⁴⁸ An ecosystem approach recognises that all species are interlinked, so that a change to one species will have an impact on other species in the food web to some degree. One example of this is fish stock productivity, which is dependent on recruitment of small fish into the fishery, and this in turn is dependent on the availability of food and suitable habitats to enable juvenile fish to avoid excessive predation.⁴⁹

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and minerals, and energy produced from water, currents and winds. Other countries retain the freedom of navigation and overflight and can lay submarine cables and pipelines within New Zealand's EEZ.

Where the outer edge of the continental shelf (which includes the continental slope and rise) extends beyond the EEZ, New Zealand has more limited rights. These only cover the exploitation of minerals, gas and other non-living resources within the seabed and subsoil as well as sedentary species on the seabed, but excluding mobile fish. When taking non-living resources such as minerals and petroleum from this part of the continental shelf, payments must be made to the International Seabed Authority. This authority was established, under UNCLOS, to control activities within areas of the seabed outside national jurisdiction.²

The United Nations Commission on the Limits of the Continental Shelf confirmed the boundary of New Zealand's continental shelf in 2008 (see Figure 2.2). It includes approximately 1.7 million square kilometres of seabed outside the existing EEZ (about six times the area of our land mass).³ Because some parts of New Zealand's EEZ and continental shelf overlap with areas claimed by other countries, the boundaries need to be agreed by treaty. New Zealand has signed a treaty with Australia to settle joint maritime boundaries but has yet to conclude maritime treaties with Fiji, Tonga and France (regarding New Caledonia).

The water column outside the territorial sea and EEZ of New Zealand and other countries is the high seas. All states have 'freedom of the high seas' which includes freedom of navigation, overflight, laying cables and pipelines, constructing artificial installations, fishing and scientific research.

The seabed and subsoil beyond the continental shelves of New Zealand and other countries is called 'The Area'. This is vested in humankind as a whole and administered by the International Seabed Authority. No state can claim or exercise sovereignty or sovereign rights over the Area.



Under international treaty, New Zealand has jurisdiction over an extensive marine area, but other nations retain rights of passage through the area, including that where this ship, seen here leaving Otago Harbour, is registered (*Raewyn Peart*)

Marine management agencies

Marine management in New Zealand is split amongst multiple management agencies with overlapping responsibilities (see Figure 2.3).

Agency/organisation	Areas of management focus	Key legislation
Minister and Department of Conservation	Conservation of natural and historic resources Protected species Marine protection Coastal management	Conservation Act 1987 Marine Mammals Protection Act 1978 Marine Reserves Act 1971 Resource Management Act 1991 Wildlife Act 1952 Marine and Coastal Area (Takutai Moana) Act 2011 Reserves Act 1977 Fisheries Act 1996 Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012
Minister and Ministry for Primary Industries	Fisheries Aquaculture Biosecurity	Fisheries Act 1996 Treaty of Waitangi (Fisheries Claims) Settlement Act 1992 Fisheries (Quota Operations Validation) Act 1997 Māori Fisheries Act 2004 Māori Commercial Aquaculture Claims Settlement Act 2004 Biosecurity Act 1993
Minister of Energy and Resources and Ministry of Business, Innovation and Employment	Allocation of minerals and petroleum resources	Crown Minerals Act 1991 Continental Shelf Act 1964
Minister and Ministry for the Environment	Environmental policy development, including regulations under EEZ legislation	Resource Management Act 1991 Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012 Environmental Reporting Act – when enacted
Environmental Protection Authority	Specified activities in the exclusive economic zone and extended continental shelf including minerals and petroleum activities	Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012
Maritime New Zealand	Shipping and marine pollution Oil spill preparedness and response	Maritime Transport Act 1994
Regional Councils	Catchment and marine management (excluding fisheries) to 12 nautical miles Regional marine pest management	Resource Management Act 1991 Local Government Act 2002 Reserves Act 1977 Biosecurity Act 1993

the coastal marine area which extends from mean high water springs⁵ to the outer edge of the territorial sea, and land near the coast that is influenced by the sea (although the extent of the landward boundary is not defined in the RMA).

The Minister approves regional coastal plans to the extent that they apply to the coastal marine area. The Department of Conservation is the unitary authority for the Subantarctic and Kermadec islands, so is responsible for writing and implementing the regional coastal plans for these marine areas. The Department may also become involved in making submissions on plans and resource consents under the RMA as part of its advocacy role and to support the Minister's wider coastal management responsibilities.

Minister and Ministry for Primary Industries

The Ministry for Primary Industries was established in 2012 as a result of a merger between the former Ministry of Fisheries and Ministry of Agriculture and Forestry. The Minister and new Ministry manage fisheries resources under the Fisheries Act 1996, which include fish, shellfish, seaweed and other aquatic life. The Minister determines a wide range of matters including what quantity of fish can be caught and what controls are placed on where, when and how fishing is to be undertaken. The Ministry supports the Minister's decision-making role by commissioning research, managing stakeholder inputs and providing advice. In addition the Ministry also enforces fisheries regulations.

The Ministry is also the Government's principal advisor on aquaculture. The Minister plays a role in the allocation of space for marine farms. This is through determining requests for stays on new applications and for aquaculture applications to be processed and heard together. The Minister has the power to amend aquaculture provisions in regional coastal plans through regulation.⁶ The Minister also assesses whether a proposed marine farm will have an undue adverse effect on fishing. The Ministry maintains a register of fish farms, which is managed by FishServe on its behalf.

The Minister for Primary Industries (along with the Minister of Transport) has a concurrence role in deciding whether or not proposed marine reserves should be established under the Marine Reserves Act. The Minister applies the decision-making criteria in section 5 of the Act. For fisheries, this includes determining whether declaring the area a marine reserve would interfere unduly with commercial fishing or recreational usage in the area. If concurrence is not provided, the proposed marine reserve cannot be created.

The Ministry for Primary Industries is also charged with leadership of the New Zealand biosecurity system under the Biosecurity Act 1993, taking over the role of the former Biosecurity New Zealand. This includes preventing new organisms from entering the country and eradicating or controlling them once they have arrived.



The Ministry for Primary Industries manages the harvest and sale of New Zealand's fisheries resources, including the fish sales shown here on the Akaroa wharf, Banks Peninsula (Raewyn Peart)

Minister of Energy and Resources and Ministry of Business, Innovation and Employment

The Minister of Energy and Resources manages the allocation of minerals resources within New Zealand's marine area. The Minister is supported by the Ministry of Business, Innovation and Employment which, through New Zealand Petroleum and Minerals, processes prospecting, exploration and mining permits, collects royalties and enforces mining regulations.

Minister and Ministry for the Environment

The Ministry for the Environment was established under the Environment Act 1986. It is the principal advisor to the Government on environmental matters within New Zealand and on international matters that affect the country's environment. The Ministry is responsible for developing environmental policy and it assists with legislative amendments.

issues, regional coastal plans, which apply to the coastal marine area and sometimes set out objectives and policies relating to coastal land, and regional plans which can address catchment issues such as runoff into the marine area.

Regional councils are consent authorities for coastal permits, which apply to activities within the coastal marine area. They also determine in the first instance water and discharge permits within catchments, and these can address runoff from land into the marine area amongst other things.

Regional councils play a role in biosecurity management under the Biosecurity Act 1993. Historically this role has been primarily on land with less involvement in the marine area. More recently, however, councils have taken on a stronger marine biosecurity role.

Some regional councils manage regional parks which border the sea under the Reserves Act 1977 and other legislation. Regional council activities and expenditure are guided by long-term council community plans prepared under the Local Government Act 2002.

Territorial authorities

Territorial authorities, which include both city and district councils, are also democratically elected bodies constituted under the Local Government Act. There are 61 territorial authorities operating throughout the country in addition to the six unitary authorities.⁹

These bodies have a more limited role in marine management under the RMA, being primarily focused on land subdivision, use and development. This role is significant, however, as land development can have major impacts on the marine environment, particularly through generating sediment and contaminated runoff. Territorial authorities prepare district plans under the RMA and make first-instance decisions on subdivision and land use consents.

Many territorial authorities manage coastal reserve land and esplanade reserves under the Reserves Act. Similar to regional councils, the activities and expenditure of territorial authorities is guided by long-term council community plans prepared under the Local Government Act. These plans can have a significant impact on the marine environment as they determine the levels of expenditure on infrastructure such as stormwater and wastewater treatment facilities. The Local Government Act also empowers territorial authorities to implement bylaws to regulate the management of stormwater and wastewater systems.

Heritage New Zealand Pouhere Taonga

Heritage New Zealand Pouhere Taonga (Heritage NZ), previously named the Historic Places Trust, is an independent body established under the Heritage New Zealand Pouhere Taonga Act 2014.

Heritage NZ is charged with identifying, recording and protecting historic heritage areas throughout New Zealand including sites within the marine area. It maintains a register of historic and wāhi tapu places and has a direct consenting role in relation to archaeological sites. Heritage NZ also directly manages some historic places and buildings.

The Act also continues the Māori Heritage Council, which advises Heritage NZ, and advocates for Māori heritage issues.



Heritage NZ is responsible for identifying, recording and protecting marine historic heritage such as the hulk of the *Rewa* at Moturekareka Island, Hauraki Gulf (Kevin L Jones)

Iwi and hapū

Iwi and hapū are increasingly applying contemporary forms of kaitiakitanga in the coastal environment through greater involvement in management decision-making. The meaning and implementation of kaitiakitanga is described more fully in Chapter 4: Kaitiakitanga. As well as engaging in kaitiakitanga through statutory processes and co-governance agreements, iwi and hapū are becoming major marine resource managers in their own right through fisheries, aquaculture and individual iwi settlements.

Management area	Legislation	Spatial area	Managing authorities
Coastal reserves	Reserves Act 1977	Land	Department of Conservation Regional councils Territorial authorities
Marine and coastal area	Marine and Coastal Area (Takutai Moana) Act 2011	Marine and coastal area	Ministry of Justice responsible for administering protected customary right and customary marine title applications Minister of Conservation has various administrative functions not conferred on a local authority or other person

Figure 2.4 Summary of marine management legislation

Resource Management Act

Most activities that have an environmental impact on the territorial sea, with the major exception of fishing, are managed under the RMA. These include land-based activities which impact on the marine area such as farming, forestry and urban development; activities within the marine area such as boating and aquaculture; and major marine works such as reclamation, dredging and the construction of wharves, jetties, sea walls and outfalls.

Part 2 of the RMA provides a purpose and set of principles to govern all management under the Act. The purpose of the RMA is to promote 'sustainable management' of natural and physical resources. This is defined in the legislation to mean (section 5(2)):

... managing the use, development and protection of natural and physical resources in a way, or at a rate, which enables people and communities to provide for their social, economic, and cultural wellbeing and for their health and safety while –

- (a) Sustaining the potential of natural and physical resources (excluding minerals) to meet the reasonable foreseeable needs of future generations;*
- (b) Safeguarding the life-supporting capacity of air, water, soil, and ecosystems; and*
- (c) Avoiding, remedying or mitigating any adverse effects of activities on the environment.*

The exclusion of minerals in subsection (a) means that finite mineral resources can be exploited without concern about retaining them for future use. However, the consent authority is still required to assess if the mining activity would promote the sustainable management of natural and physical resources in other respects.

Matters of national importance identified in section 6 of the RMA, which decision-makers are required to recognise and provide for, and which are relevant to marine management include:

- a) the preservation of the natural character of the coastal environment (including the coastal marine area), wetlands, and lakes and rivers and their margins, and the protection of them from inappropriate subdivision, use and development;*
- b) the protection of outstanding natural features and landscapes from inappropriate subdivision, use and development;*
- c) the protection of areas of significant indigenous vegetation and significant habitats of indigenous fauna;*
- d) the maintenance and enhancement of public access to and along the coastal marine area, lakes, and rivers;*
- e) the relationship of Māori and their culture and traditions with their ancestral lands, water, sites, wāhi tapu, and other taonga;*
- f) the protection of historic heritage from inappropriate subdivision, use, and development;*
- g) the protection of protected customary rights.*

New Zealand Coastal Policy Statement

The NZCPS sets out a number of objectives and policies to be applied to the coastal environment. These relate to a wide range of matters including the precautionary approach, integrated management, aquaculture, ports, reclamation, biodiversity, natural character, landscapes, surf breaks, heritage, public access, water quality and coastal hazards. These policies are discussed in detail in Section Two of this Guide where they relate to specific activities.

The following objectives and policies have particular importance for managing the marine environment.

Biodiversity

Objective 1 of the NZCPS seeks to protect representative or significant natural ecosystems and sites of biological diversity in the coastal environment. This objective also seeks to maintain the diversity of indigenous coastal flora and fauna. Other elements of Objective 1 are important for coastal biodiversity including 'maintaining or enhancing natural biological and physical processes', and 'maintaining coastal water quality'.

NZCPS, Objective 1

To safeguard the integrity, form, functioning and resilience of the coastal environment and sustain its ecosystems, including marine and intertidal areas, estuaries, dunes and land, by:

- maintaining or enhancing natural biological and physical processes in the coastal environment and recognising their dynamic, complex and interdependent nature;
- protecting representative or significant natural ecosystems and sites of biological importance and maintaining the diversity of New Zealand's indigenous coastal flora and fauna; and
- maintaining coastal water quality, and enhancing it where it has deteriorated from what would otherwise be its natural condition, with significant adverse effects on ecology and habitat, because of discharges associated with human activity.

Policy 11 requires district and regional plans to protect indigenous biological diversity in the coastal environment. There is an internal hierarchy built into the Policy. It requires the avoidance of all adverse effects of activities on the matters referred to in part (a) which includes species that are listed in New Zealand and internationally as threatened or at risk. Significant adverse effects are to be avoided and other adverse effects are to be avoided, remedied or mitigated on the matters listed in part (b). The focus of these matters is on the protection of important habitats. Policy 11 protects threatened species as well as ecosystems. Marine managers now need to familiarise themselves with the New Zealand Threat Classification System lists and the International Union for Conservation of Nature and Natural Resources lists.¹¹



Under the NZCPS, councils are required to avoid adverse effects of activities on marine reserves, including through the generation of sediment. Fine sediment from rivers flowing into Mercury Bay has been identified as a potential threat to the Te Whanganui-A-Hei (Cathedral Cove) Marine Reserve on the Coromandel Peninsula (Raewyn Peart)

Biosecurity

In terms of protecting marine biodiversity, Policy 12 recognises the importance of marine biosecurity. It requires regional councils to control, as far as practicable, activities which could cause harmful aquatic organisms to be released or spread. It identifies a number of activities which might cause this including the introduction of marine structures, vessels, dredging and aquaculture.

NZCSP, Policy 12: Harmful aquatic organisms

1. Provide in regional policy statements and in plans, as far as practicable, for the control of activities in or near the coastal marine area that could have adverse effects on the coastal environment by causing harmful aquatic organisms⁷ to be released or otherwise spread, and include conditions in resource consents, where relevant, to assist with managing the risk of such effects occurring.
2. Recognise that activities relevant to (1) include:
 - a. the introduction of structures likely to be contaminated with harmful aquatic organisms;
 - b. the discharge or disposal of organic material from dredging, or from vessels and structures, whether during maintenance, cleaning or otherwise; and whether in the coastal marine area or on land;
 - c. the provision and ongoing maintenance of moorings, marina berths, jetties and wharves; and
 - d. the establishment and relocation of equipment and stock required for or associated with aquaculture.

7 Harmful aquatic organisms – as defined in the Glossary.

Water quality

Many of the issues associated with degradation of marine water quality are the result of land use and discharges within catchments. This means that coastal water quality is primarily managed through provisions in regional plans applied above mean high water springs and through the resource consent process. Policies 21, 22 and 23 are discussed further in Chapter 17: Catchment-based activities.

The Freshwater NPS also guides the management of freshwater prior to it entering the marine environment. This is also discussed in more detail in Chapter 17.



The biosecurity risk created by marinas, such as the one shown here at Gulf Harbour, Auckland, is specifically addressed by the NZCPS (Raewyn Peart)

NZCPS, Policy 13: Preservation of natural character

1. To preserve the natural character of the coastal environment and to protect it from inappropriate subdivision, use, and development:
 - a. avoid adverse effects of activities on natural character in areas of the coastal environment with outstanding natural character; and
 - b. avoid significant adverse effects and avoid, remedy or mitigate other adverse effects of activities on natural character in all other areas of the coastal environment; including by:
 - c. assessing the natural character of the coastal environment of the region or district, by mapping or otherwise identifying at least areas of high natural character; and
 - d. ensuring that regional policy statements, and plans, identify areas where preserving natural character requires objectives, policies and rules, and include those provisions.
2. Recognise that natural character is not the same as natural features and landscapes or amenity values and may include matters such as:
 - a. natural elements, processes and patterns;
 - b. biophysical, ecological, geological and geomorphological aspects;
 - c. natural landforms such as headlands, peninsulas, cliffs, dunes, wetlands, reefs, freshwater springs and surf breaks;
 - d. the natural movement of water and sediment;
 - e. the natural darkness of the night sky;
 - f. places or areas that are wild or scenic;
 - g. a range of natural character from pristine to modified; and
 - h. experiential attributes, including the sounds and smell of the sea; and their context or setting.



The natural character of the coastal environment is protected under the NZCPS. This includes the natural movement of water and sediment and absence of artificial structures, as shown here at Mitimiti, Northland (*Raewyn Peart*)

Coastal landscapes

Policy 15 provides guidance on how landscapes in the coastal environment are to be managed:

- It directs councils to 'avoid adverse effects' of activities on outstanding natural features and outstanding natural landscapes
- Councils are to avoid 'significant' adverse effects on all other natural features and natural landscapes
- Outside of outstanding natural features and landscapes, and where adverse effects are not significant, councils must 'avoid, remedy or mitigate' adverse effects

This means that landscapes in the coastal environment have higher protection under the law than those found elsewhere. It also means that the case law applying to coastal landscapes prior to the NZCPS coming into effect in 2010 may not be fully applicable with a higher standard of protection now required.

NZCPS, Policy 15: Natural features and natural landscapes

To protect the natural features and natural landscapes (including seascapes) of the coastal environment from inappropriate subdivision, use, and development:

- a. avoid adverse effects of activities on outstanding natural features and outstanding natural landscapes in the coastal environment; and
- b. avoid significant adverse effects and avoid, remedy, or mitigate other adverse effects of activities on other natural features and natural landscapes in the coastal environment; including by:
- c. identifying and assessing the natural features and natural landscapes of the coastal environment of the region or district, at minimum by land typing, soil characterisation and landscape characterisation and having regard to:
 - i. natural science factors, including geological, topographical, ecological and dynamic components;
 - ii. the presence of water including in seas, lakes, rivers and streams;
 - iii. legibility or expressiveness – how obviously the feature or landscape demonstrates its formative processes;
 - iv. aesthetic values including memorability and naturalness;
 - v. vegetation (native and exotic);
 - vi. transient values, including presence of wildlife or other values at certain times of the day or year;
 - vii. whether the values are shared and recognised;
 - viii. cultural and spiritual values for tangata whenua, identified by working, as far as practicable, in accordance with tikanga M ori; including their expression as cultural landscapes and features;
 - ix. historical and heritage associations; and
 - x. wild or scenic values;
- d. ensuring that regional policy statements, and plans, map or otherwise identify areas where the protection of natural features and natural landscapes requires objectives, policies and rules; and
- e. including the objectives, policies and rules required by (d) in plans.



Outstanding natural landscapes, such as on the Purerua Peninsula in the Bay of Islands, have few structures and are highly protected under the NZCPS (Raewyn Peart)

Hauraki Gulf Marine Park Act

As well as being managed under the RMA and other legislation, the Hauraki Gulf has its own dedicated piece of legislation – the Hauraki Gulf Marine Park Act 2000. This legislation seeks to improve the environmental management of the Gulf through achieving better integration of the numerous statutory authorities which impact on the area and providing better recognition of the relationships of tangata whenua with the Gulf. Boundaries of the Hauraki Gulf Marine Park are shown in Figure 2.5.

The Act establishes a Hauraki Gulf Marine Park and a Hauraki Gulf Forum. Members of the Forum include representatives of the Ministers of Conservation, Primary Industries and Māori Affairs, seven representatives of Auckland Council, five representatives of other local authorities, and six representatives from the tangata whenua of the Hauraki Gulf and its islands appointed by the Minister of Conservation.

Functions of the Forum include integrating the management of the Gulf, facilitating communication, co-operation and co-ordination of members' statutory functions exercised within the Gulf, preparing a list of strategic issues, determining priorities for action on each issue and publishing a report on the state of the environment every three years.

When undertaking its functions, the Forum must have particular regard to the historic, traditional, cultural and spiritual relationship of tangata whenua with the Gulf. Day-to-day management of the Gulf continues to be carried out by individual statutory agencies.

The Act sets out matters of national significance and management objectives in sections 7 and 8 which are to be incorporated into decision-making affecting the Gulf under other legislation. Under the RMA, these must be treated as though they were a national policy statement and given 'effect to' by all regional policy statements, regional plans and district plans which apply within the Gulf area. In addition, consent authorities must have regard to the objectives when considering an application for a resource consent within that area.

Sections 7 and 8 seek to promote the protection and enhancement of:

- The life-supporting capacity of the Gulf's environment
- The historic, traditional, cultural and spiritual relationship of tangata whenua with the Gulf
- The Gulf's natural, historic and physical resources
- People's relationship to, use and enjoyment of, these resources



Figure 2.5 Hauraki Gulf Marine Park (Auckland Council)

- Avoiding, remedying, or mitigating any adverse effects of fishing on the aquatic environment

‘Utilisation’ is defined as meaning ‘conserving, using, enhancing, and developing fisheries resources to enable people to provide for their social, economic and cultural wellbeing’. This means that the purpose of the Act includes both utilisation and conservation of fisheries resources.

Three environmental principles are to be taken into account when decisions are made under the legislation:²⁰

- Associated or dependent species should be maintained above a level that ensures their long-term viability
- Biological diversity of the aquatic environment should be maintained
- Habitat of particular significance for fisheries management should be protected

In addition, four principles are to be taken into account when making decisions under the Act:²¹

- Decisions should be based on the best available information
- Decision-makers should consider any uncertainty in the information available in any case
- Decision-makers should be cautious when information is uncertain, unreliable, or inadequate
- The absence of, or any uncertainty in, any information should not be used as a reason for postponing or failing to take any measure to achieve the purpose of the Fisheries Act

MAIN SPECIES	MINIMUM LENGTH	DAILY LIMIT PER FISHER
SNAPPER	27cm	9
KINGFISH	75cm	Combined limit of 5 with no more than 3 Kingfish
GROPER/HAPUKU/BASS	None	
* BLUE COD	30cm	Combined limit of 20*
* FLOUNDER	25cm	
* RED GURNARD	25cm	
* TERAKIHI	25cm	
* TREVALLY	25cm	
SPINY ROCK LOBSTER (CRAYFISH)	Male: 54mm tail width Female: 60mm tail width	6 total
COCKLE/TUATUA/PIPI	None	150 of each
ORDINARY PAUA	125mm	10
SCALLOPS (1 Sept to 31 March Only)	100mm	20
KINA	None	50
* MUSSEL	None	50 *Except where 25 applies

*A daily limit of 25 mussel applies to Mount Maunganui Maitaitai Reserve area including Moturiki (leisure) Island, and Motuotau (rabbit) Island.

*SUBJECT TO COMBINED TOTAL OF 20 WITH CERTAIN OTHER FINFISH

MAIN SPECIES GUIDE ONLY. FURTHER RESTRICTIONS APPLY!

ADDITIONAL INFORMATION:

• www.fish.govt.nz (information on fishing)

• 800 4 POACHER (0800 4 76224) (Report illegal fishing)

newzealand.govt.nz

The harvest of fisheries resources in New Zealand is managed under the Fisheries Act. Recreational fishing is often managed through restrictions on the size and number of fish taken, as shown here (Daniel Sharp)

The High Court has held that it is sufficient for the Minister to be accurately informed as to what information is available, and at what cost and in what timeframe. The Minister need not actually obtain that information. However, if the Minister decides not to base his or her decision on the best available information there would have to be a reasonable basis for not doing so.²²

There are three main categories of fishing activity which are managed under the legislation: commercial (discussed in Chapter 9: Management of fisheries), customary non-commercial (discussed in Chapter 4: Kaitiakitanga) and recreational (discussed in Chapter 16: Marine recreation and tourism).

Crown Minerals Act

The Continental Shelf Act 1964 previously governed the allocation of rights to prospect and mine for minerals in the seabed and subsoil of the continental shelf. However, from 24 May 2013 no further licences were granted under that Act²³ and the Crown Minerals Act 1991 now applies to all new applications.²⁴ The purpose of this legislation is to ‘promote prospecting for, exploration for, and mining of Crown owned minerals for the benefit of New Zealand.’

This Act is discussed further in Chapter 11: Sand mining, Chapter 12: Deep sea minerals exploration and mining, and Chapter 13: Oil and gas exploration and mining.

Marine Mammals Protection Act

Marine mammals are managed by the Department of Conservation under the Marine Mammals Protection Act. A permit is required to 'take' (the definition includes to harm, harass, injure and attract) a marine mammal in captivity or to remove one from its natural habitat. Any fishing operation using a purse seine net is required to incorporate an escape panel or aperture in the net which allows any dolphin or porpoise to readily escape.²⁵ When a marine mammal is accidentally injured or killed, the legislation provides a defence against prosecution, so long as the incident is reported to the relevant authority.

The legislation makes provision for the preparation of population management plans for marine mammal species. Where a species is threatened, such plans can determine the maximum allowable level of fishing-related mortality for the species that would 'allow the species to achieve non-threatened status as soon as reasonably practicable, and in any event within a period not exceeding 20 years'. Such plans require the approval of the Minister of Conservation and the concurrence of the Minister for Primary Industries before they take effect. The Minister for Primary Industries considers the impact of the plan on commercial fishing. No population management plans have been approved since these provisions were inserted into the legislation in 1996. In their absence, provisions in the Fisheries Act have been used to manage the effect of fishing on protected species (discussed in Chapter 9: Management of fisheries).

Marine mammal sanctuaries can be established by the Minister of Conservation under the Act for the purpose of protecting, conserving and managing marine mammals and these are managed by the Department of Conservation. There are currently six marine mammal sanctuaries: Auckland Islands, Banks Peninsula, Catlins Coast, Clifford and Cloudy Bay, Te Waewae Bay and West Coast North Island.²⁶ The *New Zealand Gazette* notice for a marine mammal sanctuary specifies the restrictions that apply to activities within the sanctuary. For example, the West Coast North Island marine mammal sanctuary puts in place restrictions on seabed mining activities and acoustic seismic surveying.

Wildlife Act

The Wildlife Act identifies protected species and it is an offence to hunt or kill such species within the territorial sea or EEZ.²⁷ However, protected species may be captured, injured or killed inadvertently or as part of any fishing operation and the legislation provides a defence against prosecution when this happens, so long as the incident is reported to the relevant authority.

Wildlife Act protected species

Species protected under the Wildlife Act include:²⁸

- black coral (all species in the order of Antipatharia)
- gorgonian coral
- stony coral
- hydrocorals
- oceanic whitetip shark
- basking shark
- deepwater nurse shark
- white pointer shark
- whale shark
- manta ray
- spinetail devil ray
- giant grouper
- spotted black grouper
- marine reptiles such as turtles and sea snakes and marine birds



The Wildlife Act protects most seabirds, including this pied shag drying its wings at Coppermine Bay, Kawau Island, Hauraki Gulf (Raewyn Peart)

All seabirds are protected under the Act except for the black-backed gull. Sooty shearwaters (mutton bird), grey-faced petrels (northern mutton bird) and the black, little and pied shag may be killed subject to conditions set by the Minister of Conservation and their chicks can be legally harvested by iwi and mana whenua at identified sites.

Fiordland (Te Moana o Atawhenua) Marine Management Act

The Fiordland (Te Moana o Atawhenua) Marine Management Act 2005 implements the Fiordland Marine Conservation Strategy completed in 2003 by the Guardians of Fiordland's Fisheries and Marine Environment. This group consisted of commercial fishers, recreational fishers, charter boat operators, iwi, environmentalists and scientists. The legislation created eight new marine reserves

Reserves Act

The Reserves Act has a number of purposes, including:³⁰

- Providing for the preserving and managing areas of public enjoyment
- Ensuring, as far as possible, the survival of indigenous species
- Preserving representative samples of natural ecosystems and landscapes
- Promoting the protection of the coast's natural character

Reserves can be established on the coast (and inland) under the Reserves Act and these can help protect catchment areas draining into the sea. Reserve land is categorised on the basis of its primary purpose as either recreation, historic, scenic, nature, scientific, government or local. Each reserve is required to have a reserve management plan and to be managed in accordance with the purpose for which it is classified.

The Reserves Act also provides mechanisms for the protection of privately-owned land including the declaration of protected private land, the establishment of conservation covenants and the establishment of Ngā Whenua Rāhui kawenata on Māori land.



Land-based reserves can assist with marine management, especially where land and marine reserves are contiguous. This is the situation at Tawharanui, Auckland, where a predator-free regional park is contiguous with the Tawharanui marine reserve (Craig Potton)

Marine and Coastal Area (Takutai Moana) Act

The Marine and Coastal Area (Takutai Moana) Act 2011 established a new regime for the recognition of customary rights and title over the common marine and coastal area. The legislation defines a 'common marine and coastal area' which includes the marine and coastal area, excluding existing freehold title and areas owned by the Crown as conservation areas, national parks or reserves. It states that the common marine and coastal area has a 'special status' and that neither the Crown nor any other person owns, or is capable of owning it.³¹ Every person has the right to enter, pass over, and engage in recreational activities in the common marine and coastal area.³²

This legislation is discussed in detailed in Chapter 4: Kaitiakitanga.

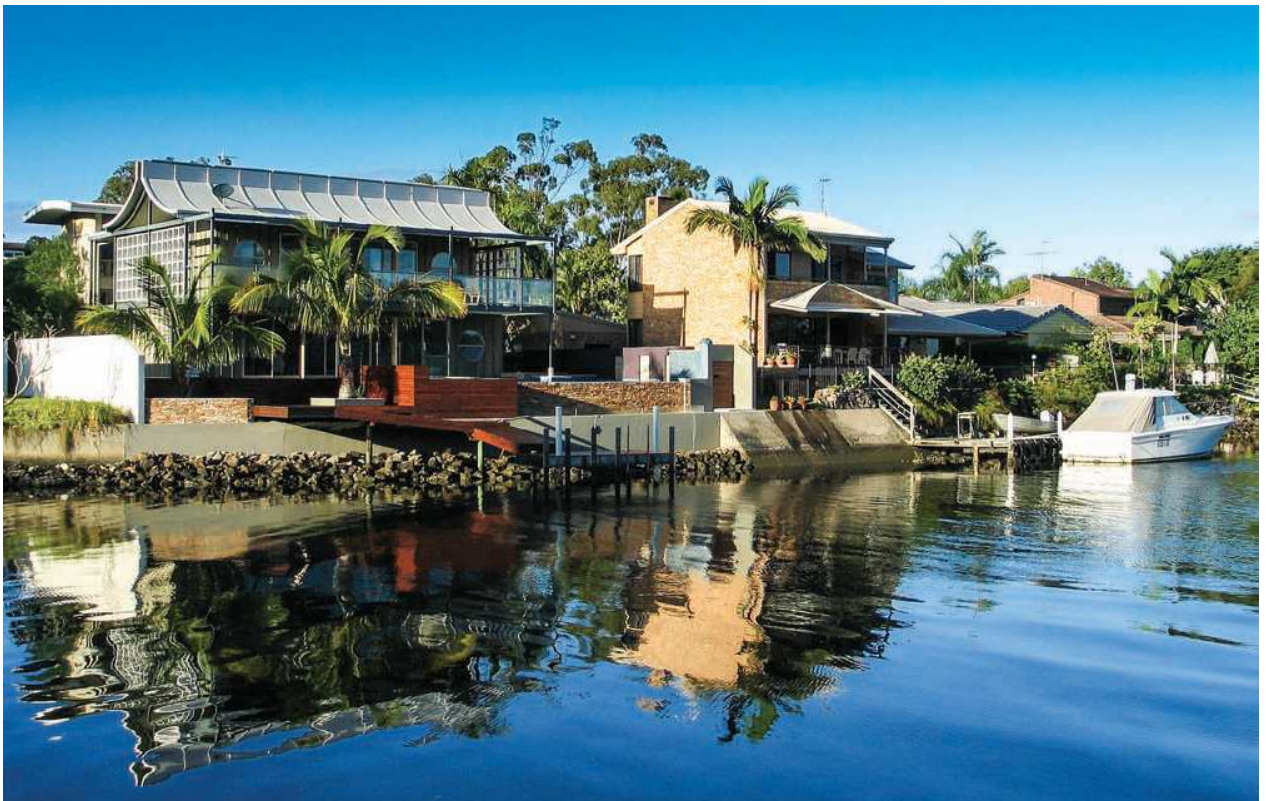
Queensland Marine Fish Habitats Offset Policy

The Queensland Government Environmental Offset Policy provides the principles for biodiversity offsetting in Queensland. This is supported by four specific policies, one of which is the Marine Fish Habitats Offset Policy.⁴⁰ This particular policy sets out the requirements of Fisheries Queensland for environmental offset conditions to counterbalance permanent or temporary impacts or loss on fisheries resources or fish habitat. It applies to fisheries development approval decisions under the Fisheries Act 1994 and Sustainable Planning Act 2009.⁴¹

The principles specified for biodiversity offsetting include:⁴²

- Offsets will not replace or undermine existing environmental standards or regulatory requirements, or be used to allow development in areas otherwise prohibited through legislation or policy
- Environmental impacts must first be avoided, then minimised, before considering the use of offsets for any remaining impact
- Offsets must achieve an equivalent or better environmental outcome
- Offsets must provide environmental values as similar as possible to those being lost
- Offset provision should minimise the time-lag between the impact and delivery of the offset
- Offsets must provide additional protection to environmental values at risk, or additional management actions to improve environmental values
- Offsets must be legally secured for the duration of the offset requirement

Fisheries Queensland assesses all impacts of proposed marine fish habitat losses and gains through a mitigation and offset hierarchy of 'avoid, minimise, mitigate and offset'. All approved, temporary or permanent fish habitat losses require the use of offsets to balance residual impacts. This helps to ensure that there is no net loss.



The Queensland Government applies a mitigation and offset hierarchy to development that impacts on fish habitat, which the Noosa canal development shown here has the potential to do *(Raewyn Peart)*

Endnotes

- 1 WWF New Zealand, 2009, 11
- 2 See <http://www.isa.org.jm/en/about>
- 3 <http://www.mfe.govt.nz/issues/oceans/jurisdictional.html>
- 4 <http://www.lin.govt.nz/hydro/projects-programmes/continental-shelf/undersea-image>
- 5 The term 'mean high water springs' is not defined in the RMA. *The New Zealand Nautical Almanac* defines it as 'the average of the levels of each pair of successive high waters during that period of about 24 hours in each semi-lunation (approximately every 14 days), when the range of the tide is greatest (Spring Range).'
- 6 RMA 1991, section 28B
- 7 http://www.localcouncils.govt.nz/lqip.nsf/wpg_URL/Profiles-Councils-by-Type-Index?OpenDocument
- 8 RMA 1991, section 30(2)
- 9 http://www.localcouncils.govt.nz/lqip.nsf/wpg_URL/Profiles-Councils-by-Type-Index?OpenDocument
- 10 *Environmental Defence Society Incorporated v The New Zealand King Salmon Company Limited* [2014] NZSC 38 at [77]. See also *Clevedon Cares Inc v Manukau City Council* [2010] NZEnvC 211
- 11 <http://www.iucnredlist.org/>
- 12 Department of Conservation, 2013
- 13 <http://www.biodiversity.govt.nz/pdfs/picture/nzbs-whole.pdf>
- 14 *Environmental Defence Society Incorporated v The New Zealand King Salmon Company Limited* [2014] NZSC at [38]
- 15 *Environmental Defence Society Incorporated v The New Zealand King Salmon Company Limited and Ors* [2014] NZSC at [38]
- 16 Ministry for the Environment, 2012
- 17 Fisheries Act 1996, section 11(2)(c)
- 18 Hauraki Gulf Marine Park Act 2000, section 13
- 19 Fisheries Act 1996, section 8
- 20 Fisheries Act 1996, section 9
- 21 Fisheries Act 1996, section 10
- 22 *New Zealand Federation of Commercial Fishermen Inc v Minister of Fisheries* [2010] NZHC 54, at [38] to [40]
- 23 The licences granted under the Continental Shelf Act 1964 will be grandfathered into the Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2013 as these licences contain environmental conditions
- 24 Crown Minerals Act 1991, section 5(8)
- 25 Marine Mammals Protection Act 1978, section 17
- 26 <http://www.doc.govt.nz/conservation/marine-and-coastal/marine-protected-areas/marine-mammal-sanctuaries/>
- 27 Wildlife Act 1952, section 63A
- 28 <http://www.legislation.govt.nz/act/public/1953/0031/latest/DLM278598.html>
- 29 <http://www.teara.govt.nz/en/shipwrecks/page-7>
- 30 <http://www.doc.govt.nz/about-doc/role/legislation/guides-and-bylaws/a-guide-for-reserve-administering-bodies/chapter-2-powers-under-the-reserves-act/>
- 31 Marine and Coastal Area (Takutai Moana) Act 2011, section 11
- 32 Marine and Coastal Area (Takutai Moana) Act 2011, section 26
- 33 Intergovernmental Panel on Climate Change, 2014
- 34 Intergovernmental Panel on Climate Change, 2014, 25
- 35 National Policy Statement on Renewable Electricity Generation 2011, Policy E1
- 36 MacDiarmid A B et al., 2013
- 37 <http://www.doc.govt.nz/publications/conservation/biodiversity-offsets-programme/biodiversity-offsets-programme/what-is-biodiversity-offsetting/>
- 38 <http://www.waikatoregion.govt.nz/PageFiles/21512/March%2015/Mar%2015%20Item%2014%20DOC%20Tabled%20%20Biodiversity%20Offsets%20programme.pdf>
- 39 <http://www.doc.govt.nz/documents/our-work/biodiversity-offsets-the-guidance.pdf>
- 40 http://www.daff.qld.gov.au/__data/assets/pdf_file/0003/68601/Marine-Fish-Habitat-Offset-Policy-12.pdf
- 41 http://www.daff.qld.gov.au/__data/assets/pdf_file/0003/68601/Marine-Fish-Habitat-Offset-Policy-12.pdf
- 42 http://www.daff.qld.gov.au/__data/assets/pdf_file/0003/68601/Marine-Fish-Habitat-Offset-Policy-12.pdf



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Under the EEZ Act, 'the environment' is defined to mean the natural environment, including ecosystems and their constituent parts and all natural resources of New Zealand, the EEZ, the continental shelf, and the waters beyond. This definition is much narrower than the RMA definition of 'environment' which also includes physical resources (e.g. buildings and infrastructure); people and communities; amenity values; and social, economic, aesthetic and cultural conditions. The focus of the EEZ Act on the natural environment and its resources reflects the fact that people and communities have fewer links to the EEZ and continental shelf (because of the distance from shore) and there are very few man-made physical resources there.

Interpretation of the purpose of the EEZ Act

The EPA-appointed Decision-Making Committee, hearing a marine consent application for sand-mining off the North Island's west coast by Trans-Tasman Resources, noted that 'Parliament could have chosen to carry across the RMA definition of sustainable management but has not done so.' As a result, the Committee was careful to consider the evidence and submissions in light of the EEZ Act's purpose, not that of the RMA. It stated:

... it needs to be understood that the many differences between the EEZ Act and the RMA – of which the purpose is but one – mean that the large body of case law and understanding that has built up in relation to the interpretation of the RMA's provisions cannot be applied uncritically to the different environment and statutory contest of the exclusive economic zone and continental shelf.⁴

The term 'economic wellbeing' in the EEZ Act purpose is broader than simply the 'monetary values associated with the activity for which consent is sought'. It includes 'the direct and indirect values of those resources as used by others or for their intrinsic and ecosystem services values'.⁵

The Supreme Court has recently recognised that sections 6, 7 and 8 of the RMA give 'further elaboration' to the purpose of the RMA.⁶ The matters listed in sections 33 and 59 of the EEZ Act (see box below), which guide decision-making, are the closest equivalent to those sections of the RMA, but lack a similar hierarchy.

Information principles

Sections 34 and 61 of the EEZ Act set out 'information principles' which the Minister for the Environment and the EPA are required to apply when making decisions. Section 34 applies to the development of regulations and section 61 to the consideration of marine consent applications.

EEZ Act, section 34 Information principles applying to regulation making

1. When developing regulations under section 27, the Minister must—
 - (a) make full use of the information and other resources available to him or her; and
 - (b) base decisions on the best available information; and
 - (c) take into account any uncertainty or inadequacy in the information available.
2. If, in relation to the making of a decision under this Act, the information available is uncertain or inadequate, the Minister must favour caution and environmental protection.
3. If favouring caution and environmental protection means that an activity is likely to be prohibited, the Minister must first consider whether providing for an adaptive management approach would allow the activity to be classified as discretionary.
4. In this section, best available information means the best information that, in the particular circumstances, is available without unreasonable cost, effort, or time.

Application of cautionary approach

The Trans-Tasman Resources Decision-Making Committee determined that there was considerable uncertainty in the information provided to it as to both the nature of the environment and the way the mining operation might affect it. It also found information on how Māori existing interests would be addressed to be inadequate.¹⁴ As a result, the Committee was required to favour caution and environmental protection. It stated that this requirement is 'an absolute one' and 'this direction is not to be traded off against the attainment of economic wellbeing'. The Committee noted that section 10(3) makes it clear that applying the information principles is one of the ways the purpose of the EEZ Act is to be achieved.¹⁵

Adaptive management

If favouring caution and environmental protection means that an activity is likely to be refused, the EPA must then turn its mind to adaptive management, and consider whether such an approach would allow the activity to be undertaken. The Environment Court has described adaptive management as an:

*... experimental approach to management, or 'structured learning by doing'. It is based on developing dynamic models that attempt to make predictions or hypotheses about the impacts of alternative management policies. Management learning then proceeds by systematic testing of these models, rather than by random trial and error. Adaptive management is most useful when large complex ecological systems are being managed and management decisions cannot wait for final research results.*¹⁶

The Environment Court has identified the following features of adaptive management (in relation to a marine energy project):¹⁷

- Stages of development are set out
- The existing environment is established by robust baseline monitoring
- There are clear and strong monitoring, reporting and checking mechanisms so that steps can be taken before significant adverse effects eventuate
- These mechanisms must be supported by enforceable resource consent conditions which require certain criteria to be met before the next stage can proceed
- There is a real ability to remove all or some of the development that has occurred at that time if the monitoring results warrant it

The EEZ Act requires the decision-maker to 'first consider' whether an adaptive management approach would allow the activity to be undertaken.¹⁸ The necessary implication is that there are situations in which an adaptive management approach would not allow the activity to be undertaken.

In the RMA context, the Supreme Court has set out the following factors to consider when determining whether an adaptive management approach would or would not allow the activity to be undertaken:¹⁹

- a) The extent of the environmental risk (including the gravity of the consequences if the risk is realised)
- b) The importance of the activity (which could in some circumstances be an activity it is hoped will protect the environment)
- c) The degree of uncertainty
- d) The extent to which an adaptive management approach will sufficiently diminish the risk and the uncertainty

The Supreme Court referred to (d) as '[t]he vital part of the test' and accepted that the following matters must be satisfied for an adaptive management approach to sufficiently diminish the risk and the uncertainty:²⁰

- Good baseline monitoring about the receiving environment
- The conditions provide for effective monitoring of adverse effects using appropriate indicators
- Thresholds are set to trigger remedial action before the effects become overly damaging
- Effects that might arise can be remedied before they become irreversible

Interpretation of the international obligations principle

When the Permitted Activity Regulations were released in 2013,³¹ Greenpeace New Zealand and Forest and Bird lodged a complaint with the Regulations Review Committee that the regulations were not in accordance with the general objects and intentions of the EEZ Act and specifically section 11. The Regulations Review Committee rejected this argument and determined that section 11 does not require that regulations made under the EEZ Act must not breach international obligations.

The Trans-Tasman Resources Decision-Making Committee reached a similar conclusion in the context of a marine consent application ... *in our view, the relevant decision-making criteria and information principles to which we must direct ourselves reflect the nature of New Zealand's international obligations... Indeed, that is what section 11 indicates. We do not agree that New Zealand's international obligations require additional consideration to be applied in addition to the decision-making criteria and information principles contained in the EEZ Act.*

Treaty of Waitangi provisions

Unlike section 8 of the RMA, the EEZ Act does not contain a general Treaty of Waitangi principles provision. Instead section 12 identifies aspects of the EEZ Act which 'recognise and respect the Crown's responsibility to give effect to the principles of the Treaty of Waitangi', including provision for a Māori Advisory Committee, iwi feedback on regulations, consideration of existing interests, and notification to iwi about consent applications that may affect them.

EEZ Act, section 12 Treaty of Waitangi

In order to recognise and respect the Crown's responsibility to give effect to the principles of the Treaty of Waitangi for the purposes of this Act,—

- (a) section 18 (which relates to the function of the Māori Advisory Committee) provides for the Māori Advisory Committee to advise the Environmental Protection Authority so that decisions made under this Act may be informed by a Māori perspective; and
- (b) section 32 requires the Minister to establish and use a process that gives iwi adequate time and opportunity to comment on the subject matter of proposed regulations; and
- (c) sections 33 and 59, respectively, require the Minister and the EPA to take into account the effects of activities on existing interests; and
- (d) section 45 requires the Environmental Protection Authority to notify iwi authorities, customary marine title groups, and protected customary rights groups directly of consent applications that may affect them.

It is clear from section 12 that the Crown's responsibility to give effect to the principles of the Treaty of Waitangi has both procedural (the way decisions are made) and substantive (the interests of Māori in the outcome of the decisions made) dimensions.³²

Interpretation of Treaty of Waitangi principle

The Trans-Tasman Resources Decision-Making Committee distinguished between notification and consultation, and noted that the EEZ Act does not expressly require marine consent applicants to consult with iwi however:

... in understanding and addressing existing interests, some level of consultation appears to us to be not just good practice but an important element in compiling a robust proposal. Further, a failure to consult adequately with tangata whenua/tangata moana may be seen as culturally offensive and disrespectful.³³



The Treaty of Waitangi provisions in the EEZ Act recognise the close association that iwi have with the marine environment, including through sailing offshore waka, as shown here at Napier (Raewyn Peart)

Regulations

The EEZ Act allows regulations to be developed to:

- Classify activities as permitted, discretionary (notified or non-notified), or prohibited³⁹
- Identify and provide for areas which are important or vulnerable or require particular management due to competing uses or cumulative adverse environmental effects (thereby allowing for marine spatial planning to be given legal effect under the EEZ Act)⁴⁰
- Prescribe technical standards, methods or requirements⁴¹

The Governor-General may make regulations on the recommendation of the Minister for the Environment. The EEZ Act puts in place a process for the development of regulations which includes notifying the public, iwi authorities, regional councils and persons whose existing interests are likely to be affected by the proposed subject matter of the regulations. These parties must be given adequate time and opportunity to comment,⁴² but no hearing is held. In addition, the Minister must have regard to any comments made and take into account the criteria identified in Figure 3.2.⁴³

Criteria	Additional detail
Any effects on the environment or existing interests of allowing an activity with or without a marine consent	This is to include cumulative effects and effects that may occur in New Zealand or in the waters above or beyond the continental shelf beyond the outer limits of the EEZ
The effects on the environment or existing interests of other activities undertaken in the EEZ or in or on the continental shelf	This is to include the effects of activities that are not regulated under the EEZ Act and effects that may occur in New Zealand or in the waters above or beyond the continental shelf beyond the outer limits of the EEZ
The effects on human health that may arise from effects on the environment	
The importance of protecting the biological diversity and integrity of marine species, ecosystems and processes	
The importance of protecting rare and vulnerable ecosystems and the habitats of threatened species	
New Zealand's international obligations	
The economic benefit to New Zealand of an activity	
The efficient use and development of natural resources	
The nature and effect of other marine management regimes	This would include the RMA, the Fisheries Act and the Maritime Transport Act
Best practice in relation to an industry or activity	
The desirability of allowing the public to be heard in relation to the activity or type of activity	To be considered in relation to whether an activity is classified as permitted, discretionary, non-notified discretionary
Any other relevant matter	

Figure 3.2 Criteria the Minister must take into account when recommending regulations

The Minister must also:⁴⁴

- Make full use of the information and other resources available to him or her
- Base decisions on the best available information
- Take into account any uncertainty or inadequacy in the information available

Activity status

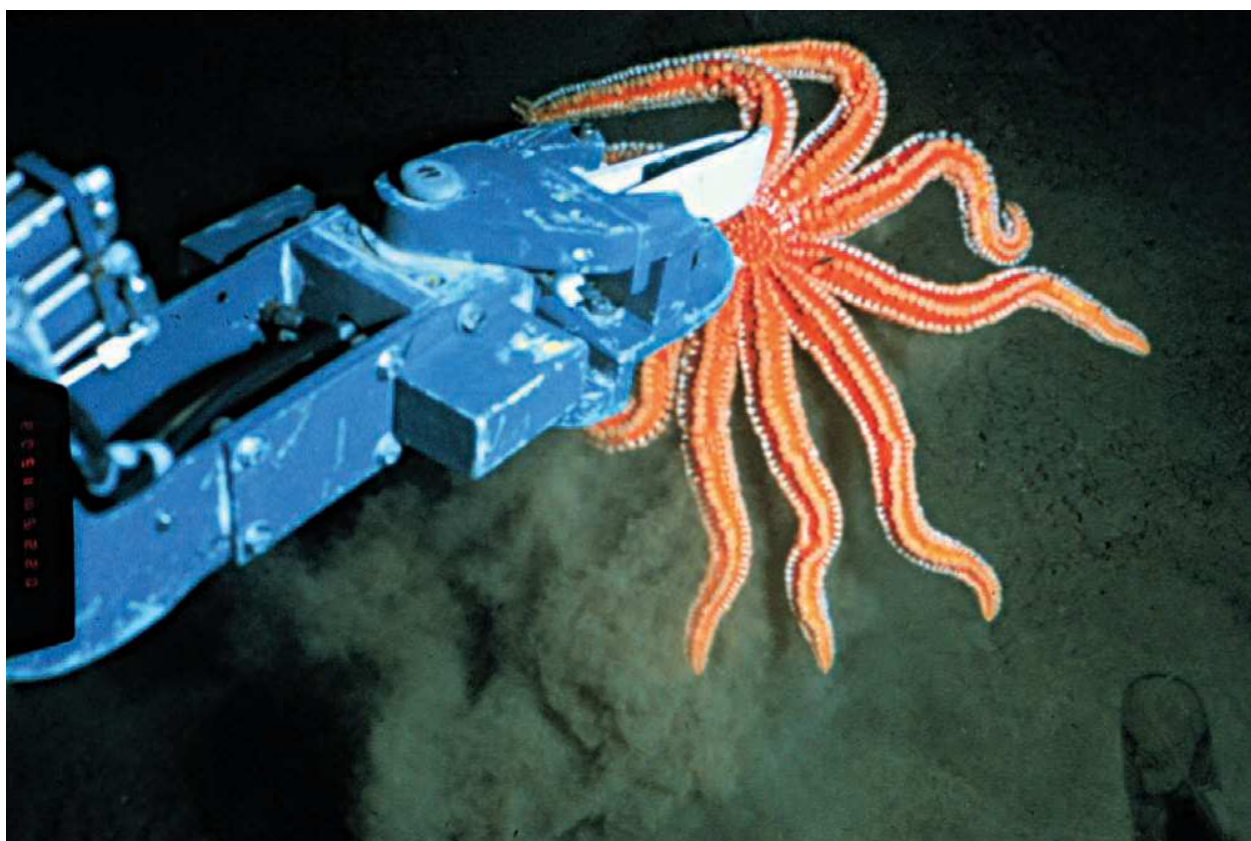
Regulations may classify activities as follows:

- Permitted – permitted activities may proceed provided they comply with the conditions specified in the regulations. The Exclusive Economic Zone and Continental Shelf (Environmental Effects – Permitted Activities) Regulations 2013⁴⁶ set out activities which are permitted and the conditions they must comply with.
- Discretionary (notified or non-notified) – discretionary activities may only proceed if a marine consent is granted. The Regulations⁴⁷ identify non-notified discretionary activities. All unspecified activities (identified in section 20) are also discretionary and will require marine consent.
- Prohibited – prohibited activities may not proceed under any circumstances. There are currently no regulations which specify prohibited activities, although the EEZ Act does not allow some dumping activities in any circumstances.

The current classifications are shown in Figure 3.3.

Permitted	Non-notified Discretionary	Discretionary	Prohibited
Marine scientific research Submarine cables Seismic surveys Prospecting for petroleum and minerals Exploration for petroleum and minerals, except petroleum drilling	Petroleum exploration drilling	All other activities identified in section 20 (see Figure 3.1)	Dumping of radioactive, toxic or hazardous waste

Figure 3.3 Status of activities under the EEZ Act



Seastar being picked up by the arm of a National Undersea Research Program submersible (*National Undersea Research Program*)

Environment	Description
Stony coral thickets or reefs	• Are most commonly found between 200 and 2,000 metres water depth and at temperatures between 4°C and 12°C • Are slow-growing and long-lived • Require a hard substrate for attachment as well as high nutrient and food supply for growth, currents or mixing to deliver the food and nutrients, and low sedimentation rates to allow efficient feeding and to avoid physical burial • Can provide habitat for some species of fish and invertebrates • Can produce three-dimensional matrix colonies that can coalesce to form 'reef', 'mound' or 'thicket' structures, extending over kilometres in length and up to 35 metres in height. Small patch reefs or thickets have been directly observed on many seamounts around New Zealand, including the Chatham Rise. • Are fragile and the biodiversity associated with these coral structures can be high • Are considered to be 'vulnerable marine ecosystems' that require protection from the impacts of deepwater fishing, drilling and mining
Xenophyophore beds	• Are made up of very large, single-celled protozoans (xenophyophore) • Live on the seabed and form complex clumps up to 25 centimetres in diameter made up of mineral grains, sponge spicule fragments and organic debris • Are associated with higher infaunal densities • Seven species have been recorded in New Zealand, three of which are endemic, and it is expected that more species will be recorded • Have been located on the eastern, northern and western continental slopes of New Zealand and on the Chatham Rise
Bryozoan thickets	• Are made up of suspension-feeding bryozoans (commonly called moss animals) that live in colonies, either on the seafloor or on algae, seagrass and animals • Individuals that comprise a colony are called zooids and are less than a millimetre in size • Colonies have varying growth forms and their size varies. They can get as large as 1,000 millimetres across. • Provide a larger surface area than the area of seafloor they occupy • Alter local physical processes such as current speed and trap sediments within their structures • Are particularly abundant and diverse in New Zealand with 27 species providing habitat over hundreds of square kilometres of seafloor
Calcareous tube-worm thickets	• Consist of clusters that result from the growth of intertwined calcareous tubes which form mounds or patch reefs • May occur on soft and hard seabed but generally need some hard substrate on which to establish • New Zealand has a number of tube-worm species that secrete tubes of calcium carbonate and they are most common in coastal waters, but occur at all depths • Are found from Taranaki south to Stewart Island but are likely to be rare in New Zealand's EEZ
Chaetopteridae worm fields	• A number of tube-worm species, which do not form calcareous tubes, reach sufficient sizes and/or densities to provide biogenic habitat for other species • In New Zealand, virtually nothing is known about the potential role of tube worms in forming biogenic habitat for other species • Fields of <i>Phyllochaetopterus socialis</i> are known off the east coast of the South Island though they may also occur around the North Island

Environment	Description
Macro-algae beds	• Occur on hard rocky substrates within the photic zone (layer that receives sunlight), to depths of 200 metres, depending on water clarity • Kelps beds are recognised as key contributors to reef ecosystems, providing highly structured three-dimensional habitats critical for other species • Within the EEZ, rocky reefs in the photic zone are very rare, and thus beds of macro-algae are likely to be equally rare • Have been recorded in the EEZ on the Mernoo Bank east of Banks Peninsula, on the Ranfurly Bank off East Cape, and on at least one seamount on the Kermadec Ridge
Brachiopod beds	• Some species form dense beds, two or three layers deep, with up to 1,000 individuals per square metre • In some areas dead brachiopod shells contribute to habitat complexity and provide abundant interstices for small invertebrates and fish • Occur throughout New Zealand at all depths • Known diverse or abundant brachiopod assemblages occur off the Three Kings Islands, off Ranfurly Bank (East Cape), on parts of the Chatham Rise where rare species are associated with coral thickets, and on areas of the Campbell and Bounty Plateaux
Deep-sea hydrothermal vents	• Occur where cold seawater percolates down through the seabed, is heated by geothermal energy, becomes buoyant and rises back up to the seabed. The hot water dissolves metals and sulphides from the surrounding rocks. • Venting can occur from point sources, from chimneys (made from precipitated minerals) or from cracks and fissures in the seabed. It can also occur as a percolation in a diffuse fashion through sands or muds. • Are associated with the subduction zone of the Pacific plate under the Australian plate to the north of New Zealand • Provide habitat for unique communities of organisms which are dependent on the sulphide-rich vent fluids that support chemosynthetic bacteria (primary producers which use chemicals, instead of light, to create energy) at the base of the food web • Vent specialists can only exist in close proximity to the active venting. They are surrounded by a distinct non-vent community at the periphery of the vent site. • Seabed communities found at hydrothermal vents are considered to be sensitive to physical disturbance. Their levels of endemism suggest that impacts to these habitats may have significant effects, including species extinction. • Some vent species in New Zealand waters are recognised by the Department of Conservation's Threat Classification System, including <i>Vulcanolepis osheai</i>
Methane or cold seeps	• Occur where methane-rich fluids escape into the water column from underlying sediments • Are usually associated with areas where gas hydrates (methane frozen within an ice matrix) occur within the sediments, generally in water depths of at least 500 metres • Most of these sites support live communities of obligate seep-associated fauna • Have been identified at many sites on the Hikurangi Margin along the east coast of the North Island. Clusters of active methane seeps have also been confirmed within five areas on the east coast of the lower North Island. • Much of the seep-associated fauna from the Hikurangi Margin appears to be new to science, or endemic to New Zealand seeps, suggesting the region may be a new biogeographic province for cold-seep fauna

Figure 3.4 Description of sensitive environments (*Adapted from MacDiarmid A et al., 2013*)

Level 2 surveys are smaller-scale seismic investigations such as those associated with scientific research. These are subject to the same general mitigation requirements as Level 1 surveys with some differences such as:

- Seismic survey vessel crew can be considered qualified observers
- Passive acoustic monitoring is optional
- Start-up will be delayed or the source shut down if an observer detects a cetacean with a calf within 1 kilometre of the source, a 'Species of Concern' (which are listed in Schedule 2 of the Code of Conduct) within 600 metres of the source, or any marine mammal within 200 metres of the source

Level 3 surveys are small-scale and have noise levels lower than commercial shipping. They are not subject to the provisions of the Code of Conduct.

Scientific research, prospecting and exploration

The EEZ regulations permit marine scientific research, prospecting for petroleum and minerals and exploration for petroleum and minerals (excluding drilling for petroleum)⁵⁵ subject to the requirement that the party undertaking the activity must undertake a number of actions specified in the EEZ Permitted Activities Regulations (see Figure 3.5).



Prospecting and exploration within the EEZ, such as the sampling of seafloor massive sulphides shown here, is a permitted activity so long as it complies with a set of requirements (*Secretariat of the Pacific Community, 2013c – Photo courtesy of Nautilus Minerals*)

- Notify the EPA of the activity at least 40 working days before the activity commences
- Notify iwi, hapū, customary marine title groups and protected customary rights groups at least 25 working days before the activity commences
- Provide the EPA with a report detailing the notification to those groups, any response received, and the outcome of any dialogue entered into at least five working days before the activity commences
- Provide the EPA with details of any change that is proposed to the activity as a result of the notification process at least five working days before the activity commences
- At least five working days before commencing the activity, carry out an initial environmental assessment and provide this to the EPA with a sensitive environments contingency plan. Notify the EPA within 24 hours of the date on which the activity is commenced.
- Maintain a log book that includes details of the person(s) carrying out the activity, the activity, the location, every sensitive environment encountered including likely effects and measures taken to avoid, mitigate or remedy adverse effects, or a description of area if no sensitive environments are found
- Ensure all reasonable measures are taken to avoid, mitigate or remedy adverse effects of the activity on any sensitive environment encountered
- Ensure no more material is removed from the seabed or subsoil than is reasonably necessary to undertake the activity
- Ensure no person sells any non-living natural material removed from the seabed or subsoil for the purpose of the activity
- Notify the EPA within 24 hours of completing the activity
- Provide the EPA with a post-activity report detailing the activity undertaken, the dates, the location, estimates of the activity/s environmental footprint (including an estimate of volume of material removed), a description of and the location of every sensitive environment encountered, details of action taken to avoid, remedy, mitigate adverse effects of the activity on every sensitive environment encountered and the reasons for choosing those actions

Figure 3.5 Requirements for scientific research, prospecting and exploration

Application of marine consent timeframes

The *Trans-Tasman Resources Ltd Marine Consent Decision* noted that the EEZ Act establishes 'demanding timeframes for the processing of and decision-making on applications for marine consents' which are 'a source of considerable challenge for all parties'.⁵⁹ If you are participating in a marine consent process it is important that you are fully aware of the relevant deadlines and make the most of the time available.

Submissions on the Trans-Tasman Resources marine consent application initially closed on 19 December 2013 but this was extended until 28 January 2014. The applicant's expert evidence was lodged on 17 February and submitter's expert evidence was lodged on 24 February. The hearing commenced on 10 March and adjourned on 8 May so that the Decision-Making Committee could more fully consider the information provided during the hearing. The Committee exercised its delegated powers to extend the time period for completing a hearing and formally closed the hearing on 19 May. The decision was publicly released on 18 June 2014.

The following sections describe the marine consent process in more detail. They apply to both the notified and non-notified process tracks unless otherwise specified.

Application

An application for a marine consent must be lodged with the EPA and must include an impact assessment.⁶⁰ The impact assessment must be in such detail that corresponds to the scale and significance of the environmental effects and must:⁶¹

- Describe the activity for which consent is sought
- Describe the current state of the area where it is proposed that the activity will be undertaken and the environment surrounding the area
- Identify the effects of the activity on the environment and existing interests
- Identify persons whose existing interests are likely to be adversely affected by the activity
- Describe any consultation undertaken
- Include copies of any written approvals to the activity
- Specify any possible alternative locations for, or methods for undertaking, the activity that may avoid, remedy, or mitigate any adverse effects
- Specify the measures that the applicant intends to take to avoid, remedy, or mitigate the adverse effects identified

The level of detail must correspond with the scale and significance of the effects that the activity may have on the environment and existing interests. It must be sufficient to enable the EPA and persons whose existing interests may be affected to understand the nature of the activity, and its effects on the environment and existing interests.⁶²

The EPA may return an application as incomplete, if it decides that it does not include any information that is required by the Act, or does not comply with the impact assessment requirements.⁶³

Case law on the completeness of marine consent applications

In *Greenpeace of New Zealand Inc v The Environmental Protection Authority* [2013] NZHC 3482 Greenpeace sought judicial review of a decision by the EPA to accept as complete an impact assessment submitted by Anadarko. Greenpeace alleged that the EPA erred in law as the impact assessment included as an appendix the discharge management plan (required by Maritime New Zealand under the Maritime Transport Act 1994), but did not include its annexes, which included oil spill modelling reports, the emergency response plan, and the oil spill contingency and well control contingency plan. The High Court found that the EPA's role under section 41 does not involve any assessment of the merits of the content of the impact assessment, and is limited to assessing whether the application contains information about the required matters. The application for judicial review was unsuccessful.

The EPA may require briefs of evidence to be provided prior to the hearing. In such a case, the applicant will need to provide briefs of evidence at least 15 working days prior, and submitters will need to provide briefs of evidence at least 10 working days prior. The EPA may direct that evidence and submissions be recorded, taken as read, limited to matters in dispute or may direct the applicant or submitters to present within a time limit.

Making a decision

The EPA may grant a consent (in whole or in part) or refuse an application. The EPA may refuse an application if it considers that it does not have adequate information to determine it. If the EPA grants the application, it may issue the consent subject to conditions. The decision on an application for a discretionary marine consent must be issued within 20 working days after the conclusion of the hearing. For non-notified discretionary activities, the decision must be made as soon as is reasonably practicable, and no later than 50 working days after the application is accepted as complete.

Decision-making criteria

When the EPA is considering an application for marine consent and submissions it must 'take into account' a number of criteria as identified in section 59(2) of the EEZ Act (see Figure 3.7).

- Any effects on the environment or existing interests of allowing the activity, including cumulative effects and effects that may occur in New Zealand or beyond our jurisdiction
- The effects on the environment or existing interests of other activities undertaken in the area covered by the application or in its vicinity, including the effects of activities that are not regulated under the EEZ Act and effects that may occur in New Zealand or beyond our jurisdiction
- The effects on human health that may arise from effects on the environment
- The importance of protecting the biological diversity and integrity of marine species, ecosystems, and processes
- The importance of protecting rare and vulnerable ecosystems and the habitats of threatened species
- The economic benefit to New Zealand of allowing the application
- The efficient use and development of natural resources
- The nature and effect of other marine management regimes
- Best practice in relation to an industry or activity
- The extent to which imposing conditions under section 63 might avoid, remedy, or mitigate the adverse effects of the activity
- Relevant regulations
- Any other applicable law
- Any other matter the EPA considers relevant and reasonably necessary to determine the application

Figure 3.7 Matters to be taken into account when considering a marine consent application

The requirement to 'take into account' matters in section 59(2) is a deliberate legislative contrast with the language 'recognise and provide for' which is used in section 6 of the RMA. 'Recognise and provide for' means that actual provision must be made for the listed matters. In contrast, the obligation to 'take into account' requires the decision-maker to consider each matter, to weigh it up with other relevant factors, and to give it the weight that is appropriate in the circumstances.⁷⁰

There is no explicit internal hierarchy between the matters listed in section 59(2). The importance of each matter will vary depending of the factual context of each application, the nature of the environment, and the extent and nature of existing interests. However, the words 'the importance of protecting' indicate an emphasis is to be placed on the protection of the intrinsic value of important biological resources.⁷¹

The definition of 'effect' includes cumulative effects. Therefore, the specific reference to 'cumulative effects' in this subsection, in addition to its inclusion in the effects definition, emphasises the importance of considering cumulative impacts on the marine environment. There are two reasons why this emphasis is required. First, there are practical difficulties in identifying and managing cumulative effects such that it would be tempting to put such assessments in the 'too hard box'. Second, there are no planning

Cross-boundary activities

The EEZ Act provides for a joint application process for cross-boundary activities which are managed under both the RMA and the EEZ Act. A person who intends to undertake a cross-boundary activity may:⁸⁰

- Prepare a joint consent application that complies with the requirements of the RMA and the EEZ; or
- Apply for a marine consent and a resource consent separately (either concurrently or at different times)

Where consents are applied for separately, the EPA may decide that the marine consent application ought to be processed and heard with an application for a resource consent. The EPA can only make such a decision before or during the processing of an application for marine consent. It is not clear whether this allows such a decision to be made after a marine consent hearing has commenced. The EPA also has the power to sever a joint application if the applications are sufficiently unrelated that a joint process is not necessary, the hearing processes are sufficiently different, or joint processing would not be administratively efficient.⁸¹

Any joint application process is to be administered by the EPA. However the EPA and the consent authority must make separate decisions based on the relevant criteria under the EEZ Act and RMA respectively. There is also a process set out for when an application for consent for a nationally significant cross-boundary activity has been referred to the Board of Inquiry.⁸²

Comment on relevance of RMA documents

The Trans-Tasman Resources Decision-Making Committee noted that there is no requirement for marine consent applications to give effect to the RMA and its subordinate documents. However it observed that the proposed activity would cause effects in the coastal marine area and the NZCPS, regional policy statement and regional coastal plan give guidance as to the important values within the coastal marine area and how sustainable management (in the RMA sense) is to be achieved.⁸³ The Committee determined that the proposed activity would not satisfy the relevant policies of these documents. It stated:

*While that finding does not preclude us granting marine consents under the EEZ Act ... they provide important guidance as to whether the potential effects of the proposal, as mitigated, are acceptable in the marine environment.*⁸⁴

Existing interests

The EEZ Act defines an existing interest to mean an interest in any lawfully established existing activity, any activity authorised by an existing marine consent or resource consent, the settlement of a historical or contemporary claim under the Treaty of Waitangi, or a protected customary right or customary marine title.⁸⁵ A consent application that is publicly notified must also be served on any iwi authorities, customary marine title groups or protected customary rights groups the EPA considers may be affected by the application and any other persons the EPA considers have existing interests that may be affected by the application.⁸⁶ When a marine consent application is considered, the EPA must take into account the effects on existing interests of allowing the activity and the effects on existing interests of other activities undertaken in the application area or in its vicinity.⁸⁷ When determining the extent of adverse effects on existing interests the EPA must have regard to: the area the activity would have in common with the existing interest; the degree to which the activity and the existing interest must be carried out to the exclusion of other activities; whether the existing interest can only be exercised in the area to which the application relates; and any other relevant matter.⁸⁸

Exercising marine consents

Commencement and duration

A marine consent that has been granted commences when the time for lodging appeals expires and no appeal has been lodged, when the High Court determines all appeals, or when all appeals are withdrawn. A marine consent has a duration of 35 years unless otherwise specified in the consent.⁸⁹

Review of marine consent

The EPA may review the duration or the conditions of a marine consent:⁹⁰

1. At any time specified in the consent to deal with any adverse effects on the environment or for any other purpose specified in the consent
2. To ensure the conditions are consistent with regulations prescribing standards, methods or requirements

Further reading and references

MacDiarmid A, D Bowden, V Cummings, M Morrison, E Jones, M Kelly, H Neil, W Nelson and A Rowden, 2013, *Sensitive marine benthic habitats defined*, Prepared for the Ministry for the Environment, NIWA, Wellington, available at <http://www.mfe.govt.nz/publications/oceans/managing-our-oceans/sensitive-marine-benthic-habitats-defined.pdf>

Ministry for the Environment, 2012, *Managing our oceans: A discussion document on the regulations proposed under the Exclusive Economic Zone and Continental Shelf (Environmental Effects) Bill*, Ministry for the Environment, Wellington

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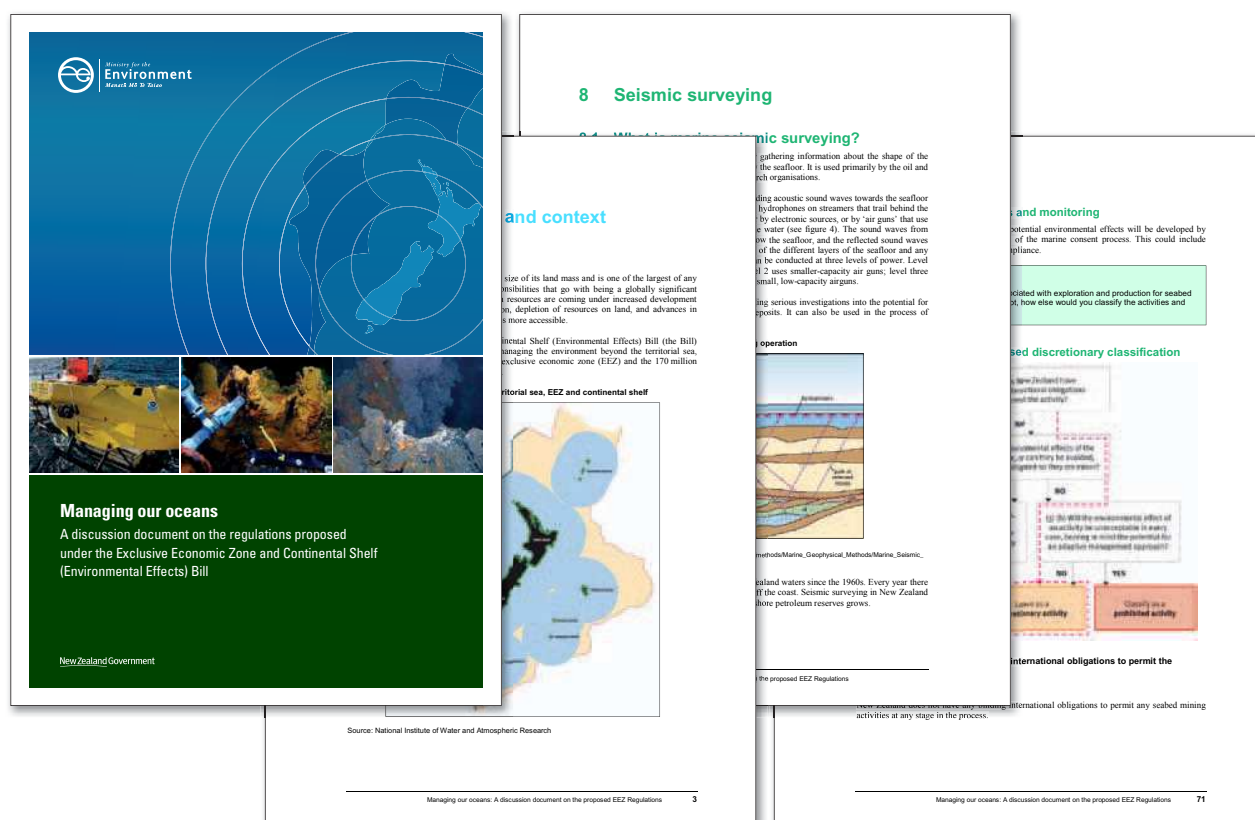
Secretariat of the Pacific Community, 2013a, *Sea-Floor Massive Sulphides: A physical, biological, environmental, and technical review*, Baker E and Beaudoin Y (eds), Deep Sea Minerals Volume 1A, Secretariat of the Pacific Community, Noumea

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Secretariat of the Pacific Community, 2013c, *Cobalt-rich Ferromanganese crusts: A physical, biological, environmental, and technical review*, Baker E and Beaudoin Y (eds), Deep Sea Minerals Volume 1B, Secretariat of the Pacific Community, Noumea

Secretariat of the Pacific Community, 2013d, *Deep sea minerals and the green economy*, Baker E and Beaudoin Y (eds), Deep Sea Minerals Volume 2, Secretariat of the Pacific Community, Noumea

Secretariat of the Pacific Community, 2013e, *Summary highlights*, Deep Sea Minerals Volume SH, Secretariat of the Pacific Community, Noumea



This discussion document, produced by the Ministry for the Environment, provides useful background to the classification of different activities within the EEZ under regulations

57 Exclusive Economic Zone and Continental Shelf (Environmental Effects – Permitted Activities) Regulations 2013, clause 8

58 EEZ Act, Section 68(2)

59 *Trans-Tasman Resources Ltd Marine Consent Decision* (June 2014), at [34]

60 EEZ Act, section 38

61 EEZ Act, section 39

62 EEZ Act, section 39(2)

63 EEZ Act, section 41

64 http://www.epa.govt.nz/EEZ/trans_tasman/application/Pages/Further-information-from-TTR.aspx

65 http://www.epa.govt.nz/EEZ/trans_tasman/reports-advice-evidence/Pages/Independent-review-of-technical-reports.aspx And http://www.epa.govt.nz/EEZ/trans_tasman/reports-advice-evidence/Pages/Reports-on-the-effects-of-the-TTR-application.aspx

66 EEZ Act, section 44A

67 EEZ Act, section 44B

68 EEZ Act, sections 50, 51 and 52

69 EEZ Act, sections 159 and 160

70 *Bleakley v Environmental Risk Management Authority* [2001] 3 NZLR 213 (HC). See also *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [113]

71 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [106]

72 Biosecurity Act 1993, section 2

73 http://www.epa.govt.nz/eez/EEZ000004/EEZ000004_01%20DOC%20-%20Eleanor%20Jamieson%20-%20Opening%20submission.pdf

74 EEZ Act, section 59(5)

75 EEZ Act, section 63

76 *Newbury DC v Secretary of State for the Environment* [1981] AC at [578] and *Housing NZ Ltd v Waitakere City Council* [2001] NZRMA at [202] (CA)

77 <https://www.mfe.govt.nz/publications/rma/enforceable-consents-jun01/html/principles.html>

78 <http://www.qualityplanning.org.nz/index.php/consents/conditions>. See for example *Mount Field Ltd v Queenstown Lakes DC* [2012] NZEnvC 262, at [77]

79 *Trans-Tasman Resource Marine Consent Decision* (June 2014), at [783]

80 EEZ Act, section 90

81 EEZ Act, section 94

82 EEZ Act, section 99

83 *Trans-Tasman Resources Marine Consent Decision* (June 2014), at [754]

84 *Trans-Tasman Resources Marine Consent Decision* (June 2014), at [760]

85 EEZ Act, section 4

86 EEZ Act, section 45(1)(c) and (d)

87 EEZ Act, section 59(2)(a) and (b)

88 EEZ Act, section 60

89 EEZ Act, section 73

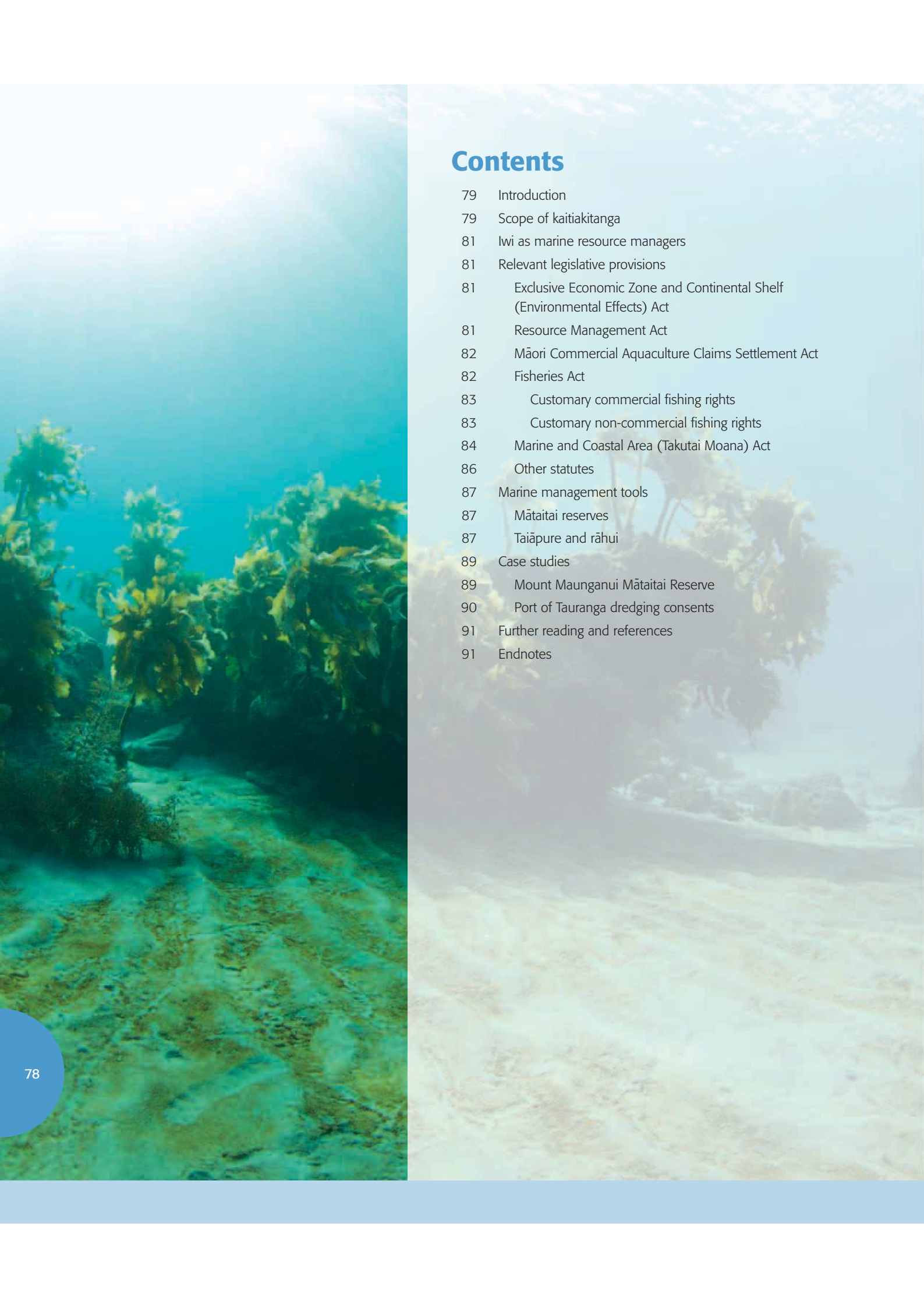
90 EEZ Act, section 76

91 EEZ Act, section 82

92 EEZ Act, section 87

93 EEZ Act, sections 85 and 86

94 EEZ Amendment Act, section 74



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Kaitiakitanga is already acknowledged in legislation and is defined as follows: '... the exercise of guardianship by the tangata whenua of an area in accordance with tikanga Māori in relation to natural and physical resources, and includes the ethic of stewardship' (section 2, RMA); and '... the exercise of guardianship; and, in relation to any fisheries resources, includes the ethic of stewardship based on the nature of the resources, as exercised by the appropriate tangata whenua in accordance with tikanga Māori' (section 2, Fisheries Act 1996).

Concern has often been expressed that present legal definitions do not fully express what kaitiakitanga is about, and that any attempt to define it in anything other than te reo Māori will always be insufficient.

The Aquaculture Steering Group offered the following ideas in order to generate discussion on the concept of kaitiakitanga. Kaitiakitanga contains many elements that can be described as:

- Mahi tapu – god-given and handed down through our tipuna
- Founded in whakapapa – the relationship between everything and everybody in the natural world. There is no distinction between people and their environment.
- Exercised on behalf of, and for the benefit of, all who are related through whakapapa
- A set of inalienable responsibilities, duties and obligations that are not able to be delegated or abrogated
- A web of obligations: to the taonga, to the atua, to ourselves and to our uri (descendants). Kaitiaki have a responsibility to provide for everyone and ensure everyone benefits.
- Independent of 'ownership' in a European sense. As on land, kaitiaki responsibilities are independent of others who hold 'ownership' or use rights under the law. For example, although as kaitiaki, iwi/hapū may 'own' only a percentage of the total marine farming space in a region under existing law, they still hold kaitiaki responsibilities over the whole area in accordance with tikanga
- Seamless and all-encompassing – making no distinction between moana (sea) and whenua (land)
- Given effect at whanau and hapū level
- Expressed in ways that are appropriate to the place and to the circumstances, according to tikanga
- Wider and more complex than existing legal definitions
- Given practical effect by exercising control over access to resources and sharing the benefits of the use of those resources
- Enabled through rangatiratanga, which includes the authority that is needed to control access to and use of resources, and to determine how the benefits will be shared. This means that it can be expressed in part through the concepts of 'ownership', 'property', 'title' or 'stewardship'. However, it is much wider than any of these.

Figure 4.1 Description of kaitiakitanga

There are a number of key messages in the description in Figure 4.1. Kaitiakitanga is integrated with the spiritual, cultural and social life of tangata whenua; is holistic across land and sea; includes people within the concept of environment; is locally defined and exercised; does not focus on ownership, but on authority and responsibility; and is concerned with both sustainability of the environment and the utilisation of its benefits. This concept of kaitiakitanga contains the purpose of the RMA (sustainable management of resources); and that of the Fisheries Act (sustainability and utilisation), but is broader than each of these concepts.

Traditionally, kaitiaki had a guardian and stewardship role for natural resources. Whilst individuals may have had specific roles, these were all exercised in terms of a collective responsibility determined through whakapapa and tikanga. The allocation and distribution of the benefits from those resources were further kaitiaki responsibilities. Tribal mana has, in most aspects of natural resources, been supplanted by legal ownership and statutory regulation. Adapting traditional kaitiakitanga to today's changed circumstances is a major challenge for tangata whenua.

In practice, natural resources are managed either directly by those having formal legal ownership, or indirectly through the exercise of statutory authority. The proportion of land and resources owned by Māori is small compared with that in non-Māori private ownership and included in the Crown Estate. However, Māori ownership of resources is more significant in the marine area than on land.

This provides greater opportunity to exercise kaitiakitanga in respect of marine resources.

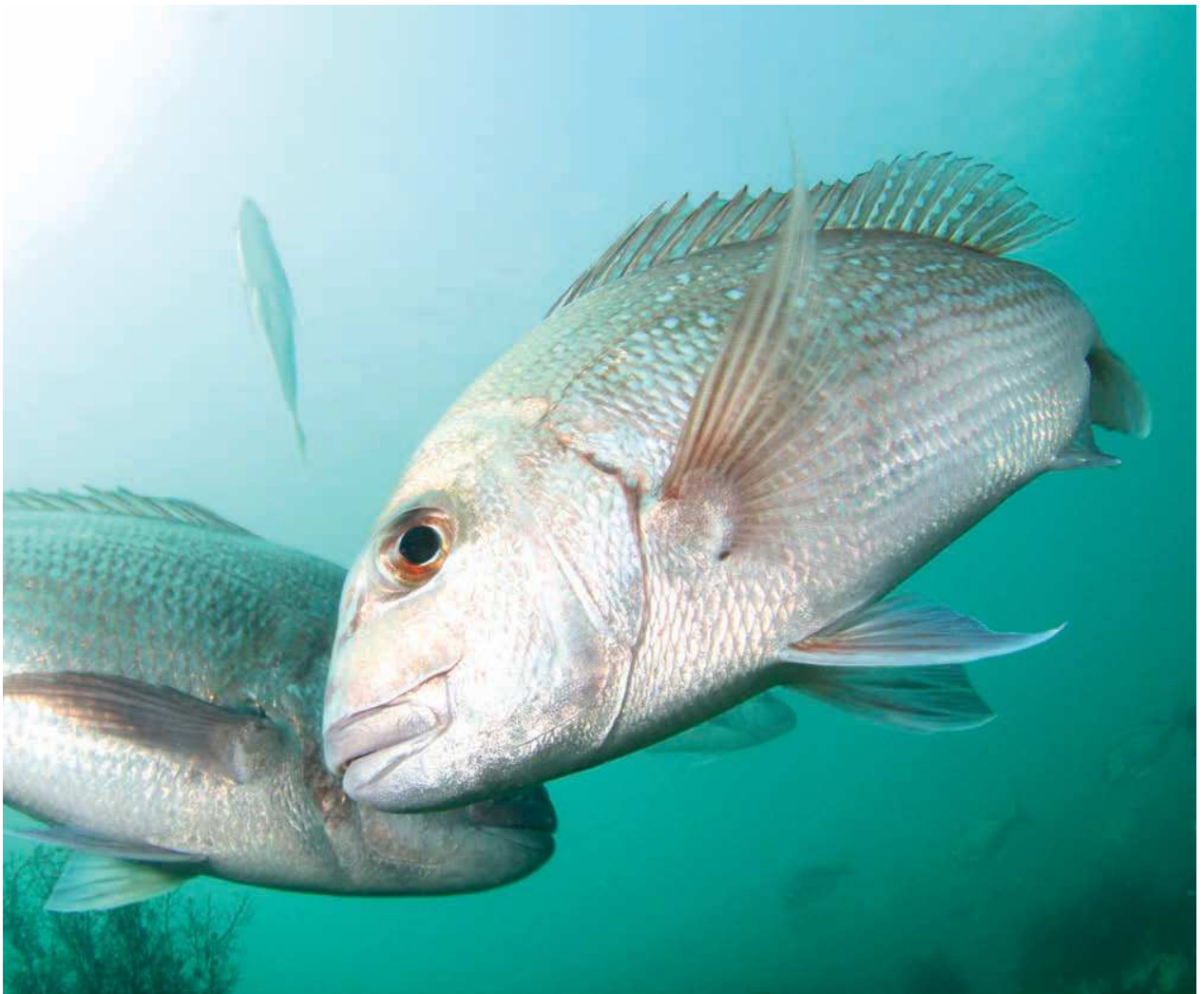
been established since the RMA came into force in 1991, and most public agencies now recognise that they have obligations to respond to them. Developers and their advisors also now have a greater appreciation of the nature of kaitiakitanga. In addition, the open public processes of the RMA have enabled public debate on the application of kaitiakitanga. However, because Māori organisations have frequently entered into RMA processes to defend their values in opposition to development or planning initiatives, kaitiakitanga has been characterised principally as a mechanism for conservation or protection. The utilisation aspect of kaitiakitanga is generally overlooked.

Māori Commercial Aquaculture Claims Settlement Act

Separate aquaculture settlement legislation, the Māori Commercial Aquaculture Claims Settlement Act 2004, requires 20 per cent of marine farm space created since 1992 to be allocated to iwi. The 2011 aquaculture law reforms provide for 'the 20% obligation' to be delivered through regional agreements which may include space, cash, or anything else that is agreed. The Minister responsible for aquaculture has a power to issue a *Gazette* notice to set aside space in the coastal marine area while regional agreements are being negotiated. These are called 'Aquaculture Settlement Areas'. Within such areas, authorisations will give iwi the exclusive right to apply for consent for aquaculture activities in that space.

Fisheries Act

The exercise of Māori fishing rights, and application of kaitiakitanga to them, is undertaken within the framework of the Fisheries Act and related regulations.



Māori have ownership of a substantial amount of fisheries quota, including for snapper, which is shown here (*Daniel Sharp*)

Application of kaitiakitanga to Te Whata Kai o Rakihouia i Te Tai o Marokura – Kaikōura Marine Area

The Kaikōura (Te Tai o Marokura) Marine Management Act 2014 puts in place management measures for the Kaikōura marine environment. These were developed through a process spearheaded by Ngāti Kurī called to Korowai. The measures include a marine reserve, whale and New Zealand fur seal sanctuaries, five customary fisheries areas, an advisory committee and fishing regulations. The three mātaimai and two taiāpure reserves are open to recreational fishing, though may be subject to special rules as determined by their governing committees. The Act establishes the Kaikōura Marine Guardians. The Minister of Conservation and Minister for Primary Industries may appoint and remove members and may disestablish the group. The Guardians may provide advice on any biosecurity, conservation or fisheries matter related to Te Whata Kai o Rakihouia i Te Tai o Marokura – Kaikōura Marine Area to relevant Ministers. The Ministers are required to take the Guardians' advice into account.



Marine and Coastal Area (Takutai Moana) Act

In 2003, the Court of Appeal found that Māori customary rights in the foreshore and seabed had not been extinguished and that the Māori Land Court had jurisdiction to conduct investigations into the title of the foreshore and seabed.¹⁵ The Crown's response was the enactment of the Foreshore and Seabed Act 2004. This reflected Government concerns that the great majority of New Zealanders understood that the foreshore and seabed was owned by the Crown on behalf of all New Zealanders. The Act vested ownership of the foreshore and seabed, except for those areas already in freehold title, in the Crown, while providing limited recognition of customary title through customary rights orders and territorial customary rights orders.

In its report on the Crown's Foreshore and Seabed Policy (known as Wai 1071), the Waitangi Tribunal found that the policy underpinning the legislation was in breach of the Treaty of Waitangi. It also failed in terms of wider norms of domestic and international law, including the rule of law, and the principles of fairness and non-discrimination against a particular group of people. Criticism was also voiced by the United Nations Commission on Human Rights which recommended that the Foreshore and Seabed Act be repealed.

Any claims to the common marine and coastal area made before the commencement of the Act are to be transferred to the High Court whose jurisdiction to consider such claims is codified in the legislation. The Act also provides for recognition of customary rights through:

- Protected customary rights – a right that has been exercised since 1840 and continues to be exercised in accordance with tikanga by the applicant group and is not extinguished as a matter of law¹⁸
- Customary marine title – where the applicant group holds the specific area in accordance with tikanga, and has exclusively used and occupied it from 1840 to the present day without substantial interruption, or received it after 1840 through a customary transfer

The implications of these rights for coastal management include:

- The holders of customary marine title have greater rights of participation in plan-making and the protection of protected customary rights is a matter of national importance that must be recognised and provided for in all plan-making
- There are additional rights associated with notification. Local authorities are required to assess whether protected customary rights groups or customary marine title groups will be affected by a resource consent application, and give limited notification to groups where they are affected
- There are financial implications as no coastal occupation charge may be imposed on a protected customary rights group or customary marine title group
- The local authority is required to monitor the exercise of protected customary rights in its region
- There are limitations on other activities occurring in protected customary rights or customary marine title areas

Other statutes

The exercise of kaitiakitanga has been enhanced through recognition in a variety of statutes. The Conservation Act does not refer to kaitiakitanga, but in section 4 requires that the 'Act shall so be interpreted and administered as to give effect to the principles of the Treaty of Waitangi'. While in principle this can empower kaitiakitanga, it has also led to conflict. For example, the Department of Conservation often works constructively with iwi on management of threatened species. But conflict has arisen over some marine reserve proposals where the protectionist approach of excluding all harvest is seen as being inconsistent with tangata whenua perspectives of customary fisheries rights. The permanence of marine reserves differs significantly from traditional marine management mechanisms such as rāhui. Under a rāhui, closure of harvest could be imposed to rebuild stocks, but rarely with permanence.

The Heritage New Zealand Pouhere Taonga Act 2014 does not specifically include reference to kaitiakitanga, but its purposes in section 3 and principles in section 4 include 'recognising the relationship of Māori and their culture and traditions with their ancestral lands, water, sites, wāhi tupuna, wāhi tapu, and other taonga'. In practice much of the work of iwi resource management units is with heritage management, under both the RMA and the Heritage New Zealand Pouhere Taonga Act.



Waharoa (customary gateway) on the Rangitoto Wharf, Hauraki Gulf (Raewyn Peart)

The purpose of taiāpure is to make better provision for the recognition of estuarine or littoral coastal waters that have customarily been of special significance to iwi or hapū either as a source of food or for spiritual or cultural reasons. A management committee representative of the local Māori community is appointed by the Minister for Primary Industries and it advises the Ministry on regulations to manage the area. These may include restrictions on the species and quantities that may be harvested, size limits, when fish may be taken, the fishing methods that can be used, and the areas from which species may be taken. Unlike in a mātaihai reserve, taiāpure allow commercial fishing.

Section 186A of the Fisheries Act provides for the temporary closure of fishing areas or restriction on fishing methods to recognise and make provision for the use and management practices of tangata whenua (similar to the traditional practice of rāhui). These provisions are used from time to time to allow depleted stocks to recover, often in the case of shellfish.



A rāhui was placed on the taking of green-lipped mussels in Ohiwa Harbour during 2008 (Raewyn Peart)

Port of Tauranga dredging consents

The Port of Tauranga is the largest export port by volume in New Zealand. The Port and its shipping channels are identified as regionally significant infrastructure in the Bay of Plenty Proposed Regional Policy Statement. The Port of Tauranga Limited holds consent to occupy areas of the coastal marine area to enable the management and operation of port-related commercial undertakings that it acquired under the Port



Consent was granted to the Port of Tauranga to dredge the seabed in order to widen and deepen shipping channels in 2013 (*Raewyn Peart*)

Companies Act 1988. The consented area includes a 10 metre radius around navigation aids and a strip from 30 to 60 metres wide along the extent of the wharf areas at both Sulphur Point and Mount Maunganui.²³ In early 2013, the Minister of Conservation granted resource consent for the Port of Tauranga to dredge sediment in order to widen and deepen its shipping channels, creating space to provide access for larger vessels.

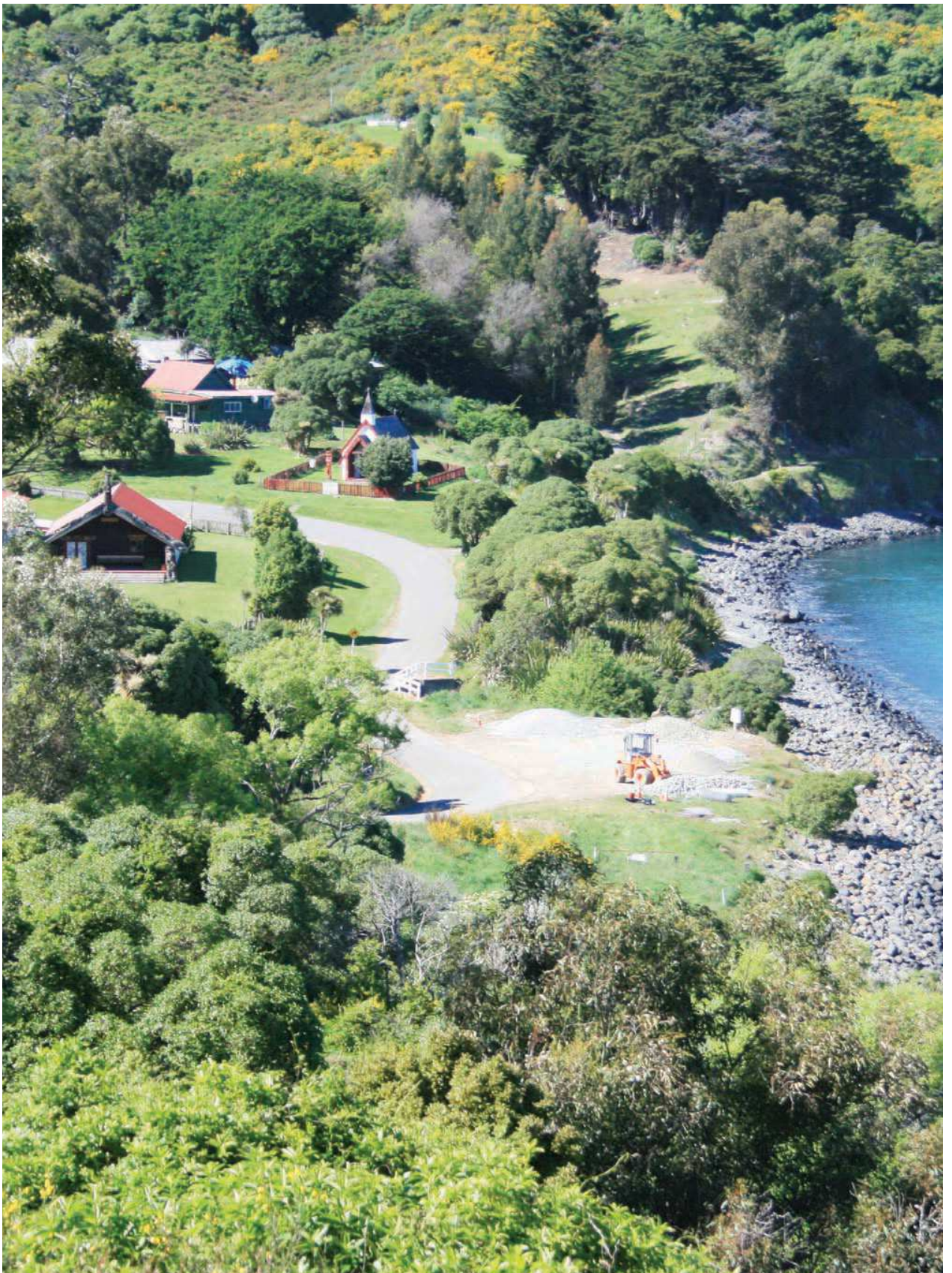
The decision came after the Port of Tauranga had spent \$2.5 million in legal fees on its proposal since lodging a resource consent application nearly four years previously. The bid had been opposed by Ngāi Te Rangi, Ngā Potiki and Ngāti Ruahine through the Environment Court in 2011. An eventual decision to grant resource consent for the deepening was later appealed to the High Court by Ngāti Ruahine hapū member Lance Waaka. The court did not grant Mr Waaka leave to take a further challenge to the Court of Appeal, allowing the Minister to finally grant consent.²⁴

The dredging project will widen and deepen the shipping channel from 12.9 metres to 16.0 metres depth at low water. Ships of up to 347 metres in length and 14.5 metres draught will be able to be accommodated in Tauranga Harbour.²⁵ The first stage of dredging will give access to ships with a capacity of 5,000 to 6,000 twenty-foot equivalent containers (TEUs). The biggest ships currently using the Port can carry around 4,500 TEUs. The second stage of dredging will accommodate 8,200 TEU ships, future-proofing the Port for the next 15 to 20 years. The expansion will also allow larger bulk cargo and cruise ships to visit Tauranga.

The Bay of Plenty Proposed Regional Coastal Environment Plan 2014 recognises that capital dredging is required for the Port of Tauranga to deepen or widen existing shipping channels or berths, or to establish new shipping channels or berths and that there is a potential for significant adverse effects when establishing new navigation channels or berths. The volume of dredgings from the Port of Tauranga makes full land-based disposal impractical. The dredged material will be predominantly clean sand, and the majority will be placed in existing offshore deposition sites which have been in use since 1968. There are also a number of near-shore sites that have been designed to replenish beaches at Mount Beach, Ocean Beach, and Pilot Bay.²⁶

There are also further works proposed to complete, extend and/or upgrade the wharves, berth areas and navigation channels. The redevelopment work includes 170 metres of additional berth at Sulphur Point, two new container cranes, six additional straddle carriers and some reconfiguration of container storage and handling facilities.²⁷

The resource consent conditions include the setting up of a trust with local iwi, the port and other stakeholders to set priorities and set aside funding for future harbour improvements. In addition, there is a minimum separation distance of the dredging from Te Kuia Rock (a sacred rock), the development of a Kaimoana Restoration Programme to mitigate the effects on local seafood especially the pipi beds, and the setting up of tertiary and post graduate research to promote better environmental health of the harbour. The Tauranga City Council is now in the process of preparing a resource consent application to the Bay of Plenty Regional Council to use some of the Port's dredged sand to renourish several harbour reserves including Kulim Park, Memorial Park, Fergusson Park and the Maxwells Road Esplanade Reserve to improve the coastal amenity in these high-use recreational areas.



Ōnuku Marae, Akaroa Harbour, Banks Peninsula (Raewyn Peart)



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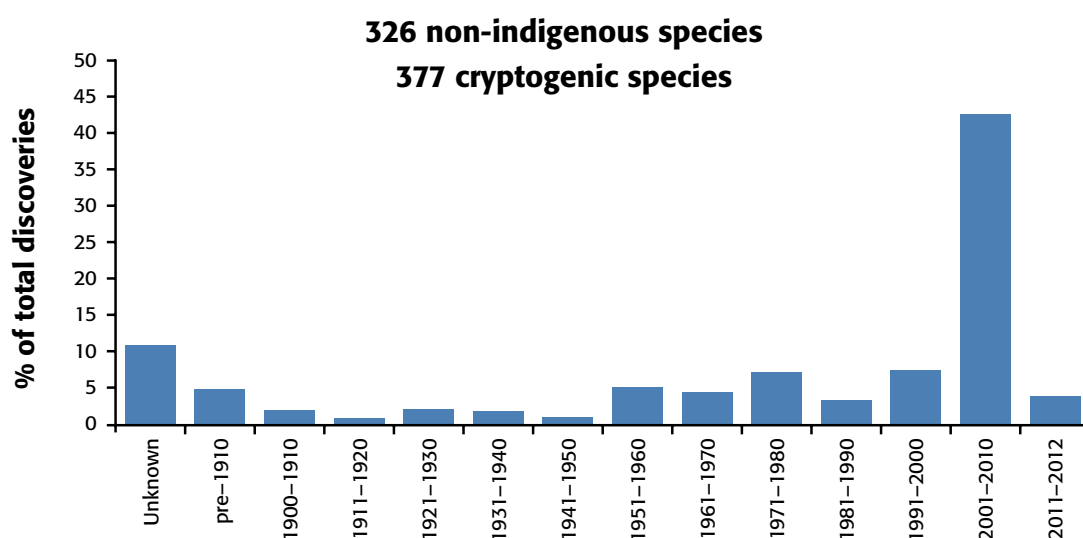


Figure 5.1 Marine non-indigenous species discovery in New Zealand (NIWA, 2013)
(Note: With constant taxonomic revisions these numbers can and will change)

Unwanted crab species

The Asian paddle crab (*Charybdis japonica*) was first reported in New Zealand in 2000. It is a very aggressive species and competes directly with New Zealand's native paddle crab. It was first noticed when some of the crabs were caught in flounder nets in the Rangitoto channel, on the outskirts of the Waitematā Harbour. It is now well established in the Auckland and Whangarei harbours, and appears to have displaced native crab species in these locations.³

The European shore crab (also known as the European green crab, *Carcinus maenas*) is an unwanted organism that has not yet reached New Zealand. It is a highly adaptable invasive



Asian paddle crab (Roger Grace)

species with a wide tolerance to different environments, which makes it a significant threat to New Zealand's inshore habitats. It is a voracious predator which eats mussels, crabs, oysters, limpets, barnacles and worms, as well as juvenile crabs and shellfish, including scallops. The European shore crab has the potential to significantly alter ecosystems, causing mortality in native crab and shellfish populations, and it has been implicated in the decline of native shellfish populations overseas. There is a risk that juvenile crabs or crab larvae could be transported via ballast water to this country.⁴

The Ministry for Primary Industries has identified six unwanted marine organisms which are highly invasive and of particular concern, being the Chinese mitten crab (*Eriocheir sinensis*), Mediterranean fan worm (*Sabella spallanzanii*), Northern Pacific seastar (*Asterias amurensis*), European shore crab, Asian clam (*Potamocorbula amurensis*) and *Caulerpa taxifolia* (a marine aquarium weed). Of these, the fan worm has become established in New Zealand since it was declared an unwanted organism. There are also numerous diseases of fish, molluscs and shellfish that New Zealand is mostly free from and risks importing.⁵ Some of these can have a devastating impact, such as the *Ostreid herpesvirus* (OsHV-1) which is discussed further in Chapter 10: Aquaculture.

The Ministry for Primary Industries, in collaboration with NIWA, has developed a new web portal called the Marine Biosecurity Porthole.⁶ This provides access to surveillance and other records of marine pests in New Zealand waters. These data come from a range of sources including a national series of Port Biological Baseline Surveys and the results of a Marine High Risk Site

Species	Status in New Zealand	Impacts
Asian paddle crab <i>Charybdis japonica</i>	Established	- Aggressive crab that can out-compete native crabs for space and food - Threat to aquaculture as it preys on shellfish - Carries white spot syndrome virus which can infect native and farmed prawns, crabs and lobsters
Pyura <i>Pyura doppelgaengera</i> previously <i>P. praeputialis</i>	Established	- Forms dense populations or mats, and can survive in a wide geographical range - Could displace important native New Zealand species, including green-lipped mussels
Australian droplet truncate <i>Eudistoma elongatum</i>	Established	- Unightly fouling of coastal environments and coastal marine infrastructure - Fouling nuisance for marine farming
Aquarium caulerpa <i>Caulerpa taxifolia</i>	Previously detected, not established in the environment	- Quickly spreads and smothers other algae, seagrasses and invertebrate communities - Out-competes native species for food or light and produces toxic compounds - Vast beds can destroy native species diversity and fish habitat - Tangles in nets and anchors - If aquarium caulerpa was to escape into the marine environment, there is a high risk of it establishing and severely damaging the marine environment
Asian clam <i>Potamocorbula amurensis</i>	Not detected	- Reduces planktonic food sources, causing decline in abundance and diversity of native species, and decline or collapse of commercial fisheries and farmed shellfish - Reaches extremely high population densities, altering the soft sediment community structure of an area by changing the sediment structure, and reducing the space available for other species
Chinese mitten crab <i>Eriocheir sinensis</i>	Not detected	- Burrowing weakens banks and causes accelerated erosion - Hosts liver fluke (<i>Paragonimus sp.</i>) that is harmful to human health - Consumes both plants and animals
European shore crab <i>Carcinus maenas</i>	Not detected	- Highly adaptable invasive species - Voracious predator – eats mussels, crabs, oysters, limpets, barnacles, worms, juvenile crabs and shellfish, including scallops - Potential to significantly alter ecosystems causing mortality in native crab and shellfish populations - Implicated in decline of native shellfish populations overseas, some of commercial importance
Northern Pacific seastar <i>Asterias amurensis</i>	Not detected	- Voracious predator of native species and economically important farmed shellfish - Potentially serious impacts on aquaculture, fisheries and wild shellfish populations

Figure 5.2: Summary of key impacts from particular invasive species

(Adapted from the Ministry for Primary Industries New Zealand's Marine Pest Identification Guide, 2012)⁹

Ballast water

Another major pathway for the introduction of marine pests is the movement and discharge of ballast water. This is seawater that has been pumped aboard a ship to provide stability and manoeuvrability. If this seawater is discharged in a different port from which it came, it can disperse the eggs and larvae of exotic marine organisms, which are able to establish in the new marine environment after discharge.

Other sources of risk

The rapid growth of the aquaculture industry, some of which involves the movement of juvenile animals (spat and smolt) and equipment around the coast, can play a role in the domestic spread of non-indigenous species once they have arrived in New Zealand. The aquaculture industry has codes of practice and industry standards in place to reduce this risk, as discussed in Chapter 10: Aquaculture.

Another source of risk is the aquarium trade. Although there are industry standards designed to avoid the release of exotic organisms, there is still considerable risk associated with the spread of marine species if captive fish or algae are released into the wild. Import health standards and risk analysis have been developed to address and mitigate these risks (discussed further below).

Legislative framework

Biosecurity activities are guided and influenced by domestic law, as well as by international agreements and standards to which New Zealand is a signatory nation. These set out the legal framework for responses to marine biosecurity risks.

Biosecurity Act

The purpose of the Biosecurity Act is to enable 'exclusion, eradication, and effective management of pests and unwanted organisms'. The Act provides the framework for border controls aimed at preventing unwanted organisms from entering the country, for establishing surveillance to detect organisms once they have arrived, and for the control and eradication of pests once they have become established. The Biosecurity Act was amended to apply within the EEZ by the Biosecurity Reform Act 2012.

Biosecurity functions are split between the Ministry for Primary Industries, other government departments and regional councils. The Ministry oversees the implementation of the legislation, undertakes border control, manages national surveillance programmes, carries out responses to incursions and manages several national control programmes. Section 12A of the Act requires the Director-General to provide overall leadership in activities that 'prevent, reduce, or eliminate adverse effects from harmful organisms that are present in New Zealand' through:

- Promoting alignment of pest management within the whole biosecurity system
- Overseeing pest management and measuring overall system performance
- Facilitating the development and alignment of national pest management plans and national pathway management plans
- Promoting public support for pest management
- Facilitating communication, co-operation and co-ordination among those involved in pest management to enhance efficacy, efficiency and equity of programmes

The role of regional councils is to undertake monitoring and surveillance of established pests and to prepare and implement regional pest management strategies. At this point in time, only a few regional councils are actively monitoring and undertaking surveillance for marine pests. Regional councils are also required by the Biosecurity Act and the National Pest Management Plan of Action to provide leadership by promoting co-ordination of pest management between regions.

Part 3 of the Biosecurity Act deals with risks associated with the importation of goods and the entry of craft into New Zealand. A set of import health standards specify requirements to be met to manage risks associated with the importation of goods (discussed further below). These relate to the importing and exporting of plants, animals and other materials which may represent risk goods, including all products that are derived from plant or animal material.

Type	Circumstances		Lead intervention decision-maker responsible for bringing parties with the necessary powers, functions and resources together	Reason for role
Population management (Species-led management)	Pest not previously detected in New Zealand		Ministry for Primary Industries	Manages border, national high-risk site surveillance and national incursion responses
	Pest already in New Zealand and an objective has been set to eradicate or contain nationally		Ministry for Primary Industries	Leads national pest programmes and national surveillance
	Pest already present in New Zealand and there has been a decision not to eradicate or contain nationally	Pests affecting public goods, and either not previously in the region or established, but tools to manage are available	Regional council to coordinate joint decision-making with Crown agencies and interested parties (depending on nature of the pest)	Accountable for regional public interest and has regional capacity to act, but multiple interests and beneficiaries will likely be involved
		Pests affecting a specific sector, industry or private interest, and either not previously in the region or established, but tools to manage are available	Industry and/or interested parties to co-ordinate joint decision making with those best placed to provide support	Industry is the primary beneficiary but may need capabilities of other parties to be effective
	Pests widespread in the region and there has been a decision not to eradicate or contain regionally		Becomes site management issue (see below)	Widespread pests that are not the subject of pest-led programmes can only be managed in specific places to meet site managers' priorities
Pathway/ vector management	Prevention of pest establishment in New Zealand (at border activity – ballast water, biofouling, hitchhiker organisms, goods and containers)		Ministry for Primary Industries	Manages border, national high-risk site surveillance and national incursion response
	Risk to any national or regional value associated with inter-regional vector movement			Requires national focus as automatically multiregional
	Risk to coastal marine areas of the Subantarctic Islands and Kermadec Islands (risks associated with vectors, in particular vessels and their equipment)		Minister of Conservation	Minister of Conservation has the responsibilities, functions and powers of a regional council under section 30(1)(d) of the RMA for these specific areas. The Department of Conservation may act on behalf of the Minister.

Type	Circumstances	Lead intervention decision-maker responsible for bringing parties with the necessary powers, functions and resources together	Reason for role
	Places recognised by formal regional policy as being of special value to regional communities (not being sites as above)	Regional councils	Accountable to regional community and have regional capacity and powers to act in the public interest
	Privately-owned structures occupying marine and other environments	Structure owners	Directly responsible as occupiers to meet rules under the Biosecurity Act and have capacity to act effectively on site in a way compatible with site use
	Other sites	Party or parties with the incentives to act and necessary powers to achieve desired objective for the site	Beneficiaries acting in their own interest

Figure 5.3 Default lead intervention decision-maker role for pests in the marine environment
(Ministry of Agriculture and Forestry and Biosecurity New Zealand, 2011)

Resource Management Act

Regional councils are responsible for controlling a number of activities in the coastal marine area which can affect biosecurity,¹⁷ including the establishment of structures, in-water cleaning of vessel hulls, and the discharge and deposit of substances into the marine environment. The NZCPS requires regional councils to provide, in regional policy and planning documents, for the control of activities that could have adverse effects on the coastal environment by causing harmful aquatic organisms to be released or otherwise spread.¹⁸ Resource consent conditions can also help to manage the risks of damage to the marine environment from such activities.



The invasive *Styela clava* growing in Northland (Northland Regional Council)

The activities identified in Policy 12 as likely to impact on the marine environment include:

- The introduction of structures likely to be contaminated with harmful aquatic organisms
- The discharge or disposal of organic material from dredging, or from vessels and structures, whether during maintenance, cleaning or otherwise; and whether in the coastal marine area or on land
- The provision and ongoing maintenance of moorings, marina berths, jetties and wharves
- The establishment and relocation of equipment and stock required for, or associated with, aquaculture

- The ballast water must be freshwater (not more than 2.5 parts per thousand sodium chloride); or
- The ballast water must have been treated using a shipboard treatment system; or
- The ballast water must be discharged into an onshore treatment facility

The International Maritime Organization is the United Nations' specialised agency with responsibility for the prevention of marine pollution by ships. The Organization is the global standard-setting authority for the environmental performance of international shipping, with a role to develop a regulatory framework that is universally adopted and implemented. In 2008 New Zealand became party to the International Maritime Organization's International Convention for the Control and Management of Ships Ballast Water and Sediments 2004.

A key feature of the Convention is a stringent performance standard requiring new ships, and eventually all ships, to operate ship-board treatment systems to disinfect ballast water taken up in foreign ports. Also included are procedures for member countries to take enforcement action where the regulations are violated. A new Ballast Water Levy will be created to cover the cost of an inspection regime to confirm compliance with the Convention's requirements. This annual levy will apply to ships, both international and New Zealand-flagged, bringing ballast water from outside the country for discharge in New Zealand waters.²² The Convention will come into force 12 months after ratification by 30 States, representing 35 per cent of world merchant shipping tonnage. As of 19 March 2013, 36 States have ratified the Convention, representing just over 29 per cent of the world merchant fleet tonnage. At the time of writing the Convention had not yet come into force.²³

Biofouling on vessels

The 'Craft Risk Management Standard for Biofouling on Vessels Arriving to New Zealand' was released by the Ministry for Primary Industries in 2014 and will come into force on 15 May 2018. It specifies the requirements to be met in respect of biofouling on the submerged parts of vessels that have come from, or recently visited, coastal waters of another country. It includes a requirement that vessels arrive in the country with 'clean hulls'.²⁴ The Standard is aligned with the 2011 International Maritime Organization Guidelines for Biofouling Management which means that much commercial shipping is already compliant. The Ministry will be using the four-year lead-in period to work with vessel operators and communicate the different measures available to shipping lines and other parties to ensure compliance.



Ships arriving into New Zealand are prohibited from discharging untreated ballast seawater taken on board in another country within New Zealand's marine area (*Raewyn Peart*)

Responding to incursions

Response is generally about minimising the impacts of a new marine pest. Ideally this means that eradication is considered but it may also include options to contain or slow the spread, or suppress the population. The case study below on the marine sea squirt *Pyura doppelgangera* provides an example of responding to a new marine pest.

If a new marine pest is found, the Ministry for Primary Industries will determine the most appropriate response according to a decisions framework that has been developed.²⁷ The Biosecurity Response Knowledge Base framework contains processes, standards and other resources for leading and managing biosecurity responses in New Zealand.

A Response Prioritisation Tool has been developed which is used to help determine whether or not to initiate a response. The tool provides an opportunity to evaluate the importance of the risk organism in terms of economic, environmental, health and socio-cultural impacts. The complexity of the response is also evaluated, including:

- What is the current distribution of the organism in New Zealand?
- What is the ability of the organism to spread and establish?
- To what degree do the methods exist for detection?
- To what degree will movement controls slow the spread or contain the organism?
- Do methods exist to control or eliminate the organism?
- What level and skill of human resource is required and are they accessible?
- Are the tools and equipment required for taking action available or accessible?

Developing new eradication methods

Northland Regional Council is currently collaborating with a science provider to test the use of ultra-sonic sound as a way of destroying biofouling on structures and hulls. Developing such innovative technologies is an important part of arming the Council, and the region, with a suite of marine biosecurity protection tools.

Public involvement and partnerships

Members of the public play an important part in keeping the marine environment free of unwanted organisms, and preventing or reducing any damage these may cause, should they occur. There are three key ways in which the public can help to keep a watch out for invasive marine species:²⁸

- Learn more about marine pests through accessing information and identification guides²⁹
- Keep up to date with the biosecurity campaigns or activities underway within a local area
- Call the free Exotic Pests and Diseases Hotline on 0800 80 99 66

Boat owners have a particular responsibility for marine biosecurity, as fouled boat hulls are a major source of marine pest spread. The boat owner's guide to preventing the spread of marine pests provides some useful guidance.³⁰ Divers can also help, by reporting anything unusual they see in an area, or a marine pest from their own area in a new place they are visiting.

Effective action requires co-operation and partnerships. National and regional partnerships are proving to be successful in improving pest management performance in the marine environment.³¹

Fiordland marine biosecurity partnership

The Fiordland Marine Guardians, Ministry for Primary Industries, Department of Conservation, Ministry for the Environment and Environment Southland are working in partnership to achieve marine biosecurity protection for Fiordland. The Fiordland Marine Biosecurity Programme is focused on addressing ways marine risk organisms can be transported into Fiordland, with supporting measures around surveillance, agency co-ordination, and communications for education and behaviour change. The partnership has developed a Fiordland-specific boat cleaning education resource.³²

Response to spread of *Sabella*³⁴

Mediterranean fan worm (*Sabella*) is an unwanted organism under the Biosecurity Act. It is a filter-feeding tube worm which can form dense colonies and has the potential to alter nutrient cycles and compete with native species for food and space. This fan worm grows rapidly in New Zealand waters, particularly in Auckland and Whangarei, where it infests vessel hulls, artificial structures and natural substrata. *Sabella* can potentially impact on marine farms and clog fishing nets and dredges.

Sabella was first detected in Lyttelton in 2008 during a targeted surveillance programme. During eradication attempts in Lyttelton, populations of fan worm were detected in Auckland's Waitematā Harbour, at densities suggesting it was well established there. The Ministry for Primary Industries decided that it was no longer feasible to attempt eradication of the organism from New Zealand. Since eradication efforts were abandoned in the Waitematā Harbour, *Sabella* has dramatically increased its range there. Recent incursions to new regions, such as Whangarei, suggest that *Sabella* has been moved by vessels coming from Auckland. However, individuals from the small population in Lyttelton Harbour are removed when found during a targeted surveillance programme, and this population has remained small (typically, no more than one or two individuals are found during each six-monthly survey).

Under the National Pest Management Plan of Action, regional councils assume the default lead role when responding to incursions of marine pests that are already established in New Zealand. The Ministry has decided to support a number of regional councils where *Sabella* has recently been detected, as outlined in Figure 5.7. The current approach is aimed at slowing the spread of *Sabella* while a national marine pathways management plan is developed.



The Mediterranean fan worm was first detected in Lyttelton Harbour (Raewyn Peart)

Region	Approach
Northland	<i>Sabella</i> was discovered in Whangarei Harbour in 2012 - In early 2014, further populations of the fan worm were discovered at additional sites, after two years of attempted eradication - Approaches used to slow the spread of the fan worm include developing charters with local vessel haul-out operators and marinas, profiling vessels suspected of harbouring <i>Sabella</i>, and ensuring infested vessels are defouled as soon as possible - The Northland Regional Council has recently sought science advice on the feasibility of managing <i>Sabella</i> at infected sites in Whangarei to ensure that it is not spread from the harbour
Waikato	- The Waikato Regional Council and the Ministry for Primary Industries jointly responded to the detection of heavy infestations of <i>Sabella</i> on two barges moored in the Coromandel Harbour following a notification from a member of the Marine Farming Association in April 2013 - Urgent action was taken to remove the fan worms as the organism spawns over a prolonged period during winter - A detection survey over the wider area was undertaken to ascertain whether there was an established population - All detectable mature fan worms were removed from the barges and no fan worms were found on the substrate after thorough searches of the wider area - Barges were then moved to a region where fan worm was established - Surveillance later in 2013 detected a small colony of fan worms on a mooring block in the vicinity of where the infested barges had been, which was removed - Surveillance will continue in the harbour - Council will explore ways of collaborating with adjacent regional councils to minimise the risk of further incursions into Coromandel and other harbours within its jurisdiction

Northland's portable biosecurity treatment facility³⁵

A large floating dock to treat hull biofouling has been built for the Northland Regional Council. A vessel can be brought into the dock and the gate is raised behind it enclosing all the seawater and preventing any marine pests escaping. The dock is 16 metres long and 5.5 metres wide with a 3 metre draft. It can accommodate a variety of vessel shapes, including many launch and small barge designs as well as yachts with a deep keel.

If a vessel hull is carrying a marine pest then the dock can be used to contain and treat the hull of the vessel before it spreads any further. Testing has been completed on the floating dock over the last year, which indicates that the dock can be a very effective and quick way of managing vessels which have hull fouling.

A marine pest treatment was recently undertaken in Auckland using the floating dock. A team of biosecurity staff from Northland Regional Council and NIWA scientists treated the hull of a vessel that was infested with the Mediterranean fan worm *Sabella* using the portable treatment facility. Aquaria trials have indicated that *Sabella* are very susceptible to chlorine over a short exposure time. Chlorine is known to disperse well in salt water and this treatment is quick and easier to administer once the vessel is enclosed within the dock. A resource consent was obtained to add chlorine to the water in the dock which was then left overnight to kill the *Sabella* (and other fouling). The treatment appeared to work very well and during the course of the day scientists were able to collect data regarding the survival of *Sabella* on the hull.

Most vessel owners are aware that clean hulls and fresh antifoul prevent marine pests from establishing and that slipping, cleaning and antifouling vessels on a regular basis will prevent marine pest transfer. However, in some remote harbours where slipping and cleaning facilities are not available (such as Whangaroa) the treatment dock can provide a safeguard which can be used in emergencies to treat an infected hull without moving the boat to another harbour where facilities are available.

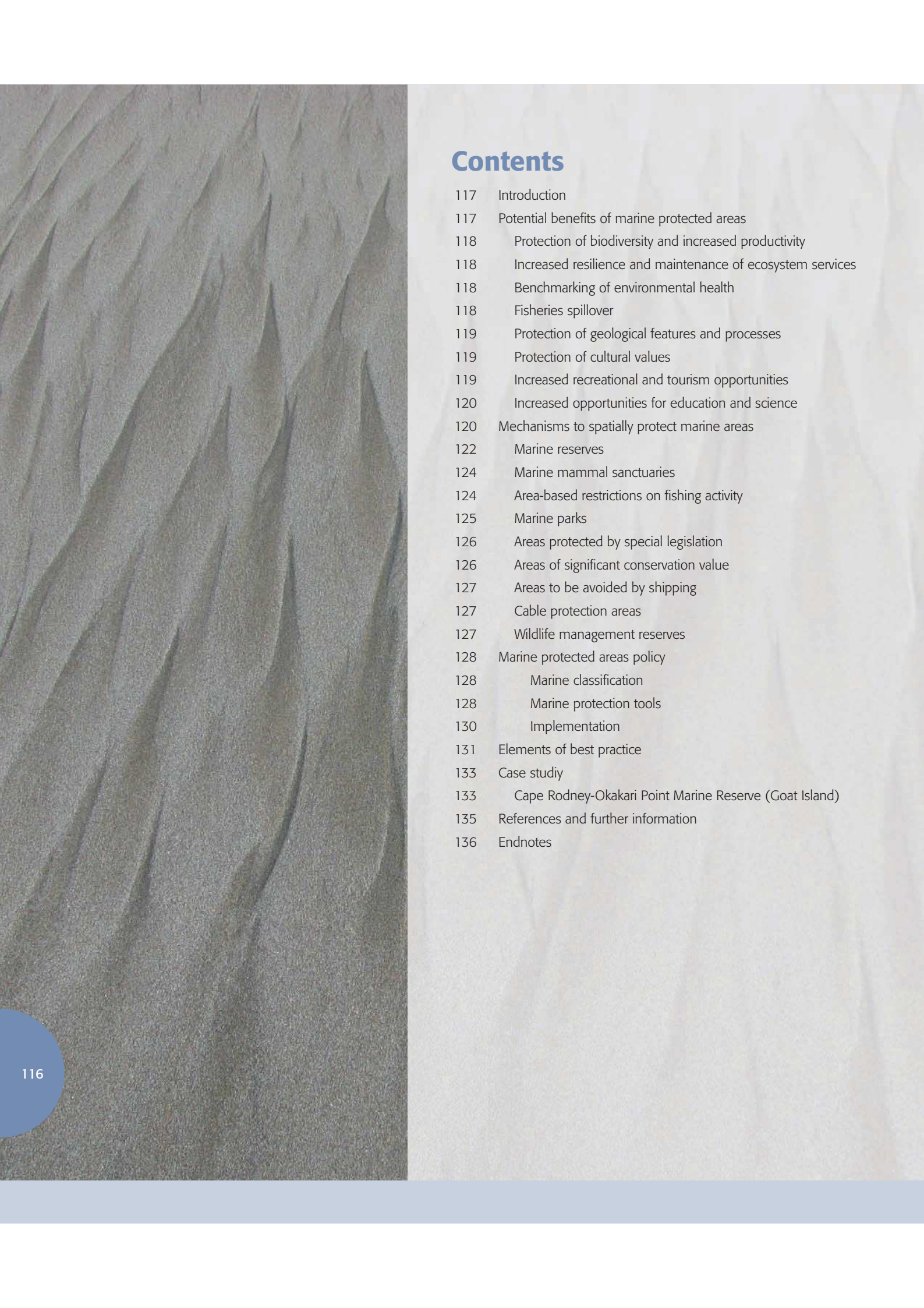
Northland now has two portable treatment facilities which have proven to be particularly useful in instances where there are no other facilities to slip a vessel, such as remote locations, bays or harbours that have no supporting infrastructure. This facility is expected to become a regular tool for dealing with biofouling on (smaller) vessels arriving in Northland ports and marinas from elsewhere in New Zealand and overseas.



Inflatable vessel quarantine facilities, such as the one shown in use at Westhaven, Auckland, can be used to treat the infected hulls of vessels in a variety of locations (Northland Regional Council)

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- 32 <http://www.biosecurity.govt.nz/fiordland>
- 33 Information for this case study has been provided by Ministry for Primary Industries staff
- 34 Information for this case study has been provided by Ministry for Primary Industries staff
- 35 Information provided by Northland Regional Council's Biosecurity team

The background of the page is a photograph of sand dunes, showing the rhythmic, undulating patterns of the sand under a soft light. The dunes are in shades of light brown and tan, with shadows that emphasize their texture and form.

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Protection of biodiversity and increased productivity

Somewhat surprisingly, there have as yet been no studies in New Zealand to establish the direct impact of marine reserves on biodiversity. However, research has conclusively shown that, where extractive activities are prohibited, abundance of previously exploited marine species in marine reserves can increase dramatically, and that this can occur within relatively short periods of time.²

An increase in the abundance of previously fished species can have a positive flow-on effect, through rebalancing predator–prey relationships, and the overall food web. For example, the increase in snapper and rock lobster within the Cape Rodney–Okakari Point Marine Reserve was associated with a marked decrease in the abundance of sea urchins (on which snapper and rock lobster prey), and a regeneration of kelp beds (on which the sea urchins prey). This, in turn, is thought to have increased productivity and diversity within the marine reserve. This is because kelp forests are major primary producers, support a richer and more diverse faunal community, and are ‘central to ecosystem function and diversity.’ Scientists have estimated that macro-algae primary productivity in the reserve has increased by 58 per cent over a 20 year period.³



Marine reserves can help to rebalance predator–prey relationships, and as a consequence reduce kina barrens, which are shown here in the Bay of Islands (Raewyn Peart)

Reserve was associated with a marked decrease in the abundance of sea urchins (on which snapper and rock lobster prey), and a regeneration of kelp beds (on which the sea urchins prey). This, in turn, is thought to have increased productivity and diversity within the marine reserve. This is because kelp forests are major primary producers, support a richer and more diverse faunal community, and are ‘central to ecosystem function and diversity.’ Scientists have estimated that macro-algae primary productivity in the reserve has increased by 58 per cent over a 20 year period.³

Increased resilience and maintenance of ecosystem services

Well-designed marine protected areas can help to promote healthy ecosystems which are resilient and able to adapt to stresses because they are complex systems with inbuilt buffers. Resilience enables marine communities to better withstand pressures whilst still retaining their basic function, including the delivery of the ecosystem services upon which we all rely.⁴ Critical services include climate regulation, oxygen production, nutrient recycling and production of protein for human consumption.

Benchmarking of environmental health

The complexity of marine ecological dynamics makes it hard for scientists to assess the impact of human-derived pressures on the marine environment.⁵ Modelling can assist, but cannot provide all the answers. The establishment of areas in which specific activities are excluded can play an important role in providing ‘benchmarks’ or ‘controls’ against which human impacts and management of the rest of the marine area can be evaluated.⁶

In particular, because marine reserves exclude fishing, they can provide useful information on the characteristics of an unfished population, which can then be compared with those of the fished stock. This can provide an indication of the extent of change caused by fishing, and therefore help inform fisheries sustainability decisions.⁷

Fisheries spillover

Marine protected areas can potentially benefit commercial and recreational fishers by contributing to fish stocks outside the protected area. This can occur if harvestable-size stock moves from the closed area to open areas where they can be caught, or if eggs and larvae cross the boundary and establish outside the closed area.

Increased opportunities for education and science

Marine reserves provide the opportunity for scientists to study marine communities which are largely intact, and which are not directly disturbed by fishing activity. For example, science undertaken in the Cape Rodney-Okakari Point Marine Reserve has made a major contribution to our understanding of rocky reef systems in New Zealand.¹⁸

Scientists are able to observe unharvested populations of fish in marine reserves. This is important to enable the development of understanding about the natural behaviours and population dynamics of fish species. In addition, reserves enable scientists to carry out longitudinal studies on individuals which are not possible where fishing takes place, as the individual studied will likely be lost to harvesting during the experiment.¹⁹

Educational facilities often co-locate with marine reserves, as the reserves provide an excellent place for students and members of the public to observe marine life. For example, the Goat Island Marine Discovery Centre is located near the marine reserve at Leigh and the Sir Peter Blake Marine Education and Recreation Centre is located adjacent to the Long Bay-Ōkura Marine Reserve.

Mechanisms to spatially protect marine areas

There are many types of spatially-defined marine areas which can be created under statute in New Zealand, and which may constitute marine protected areas. These include marine reserves, marine mammal sanctuaries, areas closed to selected fishing methods, marine parks, areas established under special legislation, areas of significant conservation value, areas to be avoided by shipping and cable exclusion areas. These mechanisms are summarised in Figure 6.2 and described in more detail in the following sections.

Traditionally, Māori used the temporary closures of rāhui, mātaihai and taiāpure to manage fisheries resources. While there is some overlap between the effect of rāhui and marine protected areas, their approach and purpose differs. Fully protected marine reserves impose permanent closures to fishing in order to protect biodiversity in the long term. Rāhui imposes a temporary closure to assist with the replenishment of kaimoana stocks so that they can be harvested in the future. As a result of these differences, proposals to establish marine protected areas can run into conflicts with manawhenua who wish to continue exercising traditional customary fishing rights.

Type of protection	Key features
Marine reserves <i>Marine Reserves Act 1971</i>	Purpose is for 'preserving them [areas of the sea] in their natural state as the habitat of marine life for scientific study' Cannot be created outside the territorial sea Only a small number of parties specifically identified in the Act can apply to create a marine reserve Can only be created if the Ministers of Conservation, Primary Industries and Transport agree Created by the Governor-General by Order-in-Council (meaning it must be approved by Cabinet) Provides a high level of protection – all fishing usually excluded Managed by the Department of Conservation
Marine mammal sanctuaries <i>Marine Mammals Protection Act 1978</i>	No specific purpose stated in the Act, but they are intended to protect marine mammals from harmful human impacts Can be created within the territorial sea and EEZ Created by the Minister of Conservation by <i>Gazette</i> notice. Requires the consent of other Ministers if they control any Crown-owned land, foreshore, seabed, or waters of the sea. May specify activities that may or may not be engaged in within the sanctuary, and may impose restrictions in respect of the sanctuary Managed by the Department of Conservation

Marine reserves

Highly protected areas, called marine reserves, can be established in New Zealand under the Marine Reserves Act. These can only be created within the territorial sea.

The purpose of establishing marine reserves under the Act focuses on scientific study rather than preserving biodiversity. The long title of the Act states that it is 'an Act to provide for the setting up and management of areas of the sea and foreshore as marine reserves for the purpose of preserving them in their natural state as the habitat of marine life for scientific study.'

Marine reserves are managed by the Department of Conservation with the aim of maintaining them as far as possible in their natural state. Section 3(3) of the Act provides that no person shall fish in a marine reserve unless specifically authorised. This means that they are usually 'no-take' areas. Other activities which could damage or remove features are also excluded. Boating, swimming and diving are usually permitted. The Marine Reserves Regulations 1993 apply to activities in marine reserves and address diving, anchoring, the use of vessels and scientific study within reserves.

Although non-commercial fishing can legally be permitted in a marine reserve, it is only currently permitted within a small part of the Kapiti Marine Reserve and is restricted to whitebaiting. Schedule 4 of the Crown Minerals Act specifically excludes mining activity from marine reserves (those declared a marine reserve under section 4(1) of the Marine Reserves Act).

An application to establish a marine reserve can only be made by a range of specified applicants, those being a university, a body administering a reserve adjacent to the coast, an organisation involved in the study of marine life and natural history, iwi or hapū with tangata whenua status over the area and the Director-General of Conservation. A significant proportion of marine reserves have been proposed by external applicants, including New Zealand's first at Cape Rodney-Okakari Point which was proposed by the University of Auckland. The process to establish a marine reserve is outlined in Figure 6.3.

A marine reserve may not be established if it would:

- Unduly interfere with any interest in land in or adjoining the reserve
- Interfere unduly with any existing right of navigation
- Interfere unduly with commercial fishing (within the broader marine area – commercial fishing will almost always be excluded from the marine reserve area itself)
- Interfere unduly with, or adversely affect, any existing usage of the area for recreational purposes
- Otherwise be contrary to the public interest



The Cape Rodney-Okakari Point Marine Reserve at Leigh was one of the first no-take marine reserves to be established in the world (Craig Potton)

Marine mammal sanctuaries

Marine mammal sanctuaries can be established under the Marine Mammals Protection Act. The purpose of the Act as set out in the long title is the 'protection, conservation, and management of marine mammals within New Zealand and within New Zealand fisheries waters'. The sanctuaries are managed by the Department of Conservation.

Marine mammal sanctuaries are created by the Minister of Conservation declaring the sanctuary by notice in the *Gazette*. The consent of other Ministers must also be obtained if they control any Crown-owned land, foreshore, seabed, or seawater. Within 28 days after publication of the notice, members of the public may lodge written submissions with the Minister, who may vary, redefine or abolish the sanctuary after considering the submissions. The Minister may specify restrictions on activities that may take place in the sanctuary at the time the declaration is made. Activities which have been controlled in marine mammal sanctuaries include fishing, mining and seismic surveying.



The Akaroa Marine Mammal Sanctuary protects the endangered Hector's dolphin
(Raewyn Peart)

Area-based restrictions on fishing activity

Areas can be closed to specified types of fishing activity by the Minister for Primary Industries, under the sustainability provisions of the Fisheries Act, after taking into account the effects of fishing on any stock and the aquatic environment. Before making a decision to restrict or prohibit fishing in an area, the Minister must consult with stakeholders, and take into account any conservation or fisheries services and any relevant fisheries plan.

The process to impose a restriction on fishing activity will normally include the following steps:

- The Ministry for Primary Industries will prepare an Initial Position Paper
- The paper is publicly released and written submissions called for. Any person is able to make a written submission.
- Once the submission period has closed the Ministry summarises and considers the written submissions and prepares a Final Advice Paper
- The Final Advice Paper is forwarded to the Minister who makes a final decision. The decision is recorded in a short letter which sets out the decision and the reasons for it.
- The decision is then given effect to, either through a notice in the *Gazette* or by regulations promulgated under section 298 of the Fisheries Act

There are no appeal rights, but the decision can be judicially reviewed in the High Court.

Figure 6.4 Process to impose an area-based restriction on fishing activity

Numerous area-based restrictions have historically been placed on fishing activities, particularly in harbours and coastal areas. Many of these were not originally intended to provide marine biodiversity benefits, but were put in place to deal with issues such as conflicts between recreational and commercial fishers. They may, however, in practice result in improvements in biodiversity. More recently, restrictions under the Fisheries Act have closed 17 areas around and over seamounts (and other bathymetric features) and 17 large benthic protection areas (primarily) within the EEZ to dredging and trawling activity as a result of biodiversity concerns. These areas cover around 30 per cent of the EEZ which equates to more than four times the area of New Zealand's land mass.

Areas protected by special legislation

Some areas are protected by other special legislation, which is tailored to each case, and does not have more general application. The Sugar Loaf Islands Marine Protected Area Act 1991 establishes the Ngā Motu/Sugar Loaf Islands Marine Protected Area covering an area of 8 square kilometres. The protected area is managed by the Department of Conservation. Mining is prohibited in the protected area and there are also restrictions on fishing activities under the Fisheries Act.

The Fiordland (Te Moana o Atawhenua) Marine Management Act establishes a Fiordland (Te Moana o Atawhenua) Marine Area and the Fiordland Marine Guardians which has an advisory and facilitative role, amongst other things. Members of the Guardians are appointed by the Minister for the Environment with the requirements that one member be nominated by Te Rūnanga o Ngāi Tahu, that at least five members are ordinarily resident in the Otago or Southland regions, and that the membership reflects a balanced mix of knowledge and experience in relation to the Marine Area. Unlike the situation in the Hauraki Gulf, the Fiordland legislation does not establish any common management objectives for the marine area.

Similar legislation now applies to Kaikōura. The Kaikōura (Te Tai o Marokura) Marine Management Act 2014 provides for the establishment of the Kaikōura Guardians who can provide advice to Ministers on matters affecting the Te Whata Kai o Rākohia i Te Tai o Marokura/Kaikōura Marine Area which is identified in the legislation. Members of the advisory body are appointed by the Minister of Conservation and the Minister responsible for the administration of the Fisheries Act (currently the Minister for Primary Industries). The members of the advisory committee must represent certain interests and areas of expertise including Te Rūnanga o Ngāi Tahu, the Kaikōura community, biosecurity, conservation, education, environment, fishing, marine science and tourism. The Act also creates a number of marine protected areas and other management areas including a marine reserve, a fur seal sanctuary, a whale sanctuary, mātaītai and taiāpure.

Areas of significant conservation value

Areas of significant conservation value are sometimes spatially identified in regional coastal plans under the RMA, and measures such as protective rules to ensure that these areas are not adversely affected are specified. Usually activities such as aquaculture, marinas and moorings, reclamations and building structures are to be avoided in these areas.

Such provisions are included in plans after a public process, which involves the public notification of the proposed plan provisions, opportunity for written public submissions, some opportunity for cross-submissions and then a public hearing. Decisions of the council on submissions can be appealed to the Environment Court.



The Mahurangi Harbour has been identified as a significant marine ecological area in the Proposed Auckland Unitary Plan
(Raewyn Peart)

Marine protected areas policy

In 2005 the Department of Conservation and former Ministry of Fisheries formally released a policy statement and implementation plan for marine protected areas. This was designed to provide a more strategic approach to identifying and creating the network of representative protected marine areas identified in the Biodiversity Strategy. The purpose of the policy is to 'protect marine biodiversity by establishing a network of MPAs [marine protected areas] that is comprehensive and representative of New Zealand's marine habitats and ecosystems'. The Ministry for Primary Industries and the Department of Conservation are jointly responsible for implementing the policy.

Marine classification

The policy takes a regional approach to the planning and establishment of a network of protected areas around New Zealand, although it is not aligned with regional council boundaries. A coastal and marine classification system, which identifies different types of marine habitat, has been developed, to help ensure that the network of marine protected areas is comprehensive and representative. Fourteen biogeographic regions have been identified (see Figure 6.6 for an example), which collectively cover the entire New Zealand marine area within the territorial sea (12 nautical miles seawards). They are intended to reflect major coastal biological patterns. These regions have been further classified, through the application of additional layering criteria, into 44 different types of habitat. The layering used to identify these areas included:

- Biogeographic region (14 regions)
- Environment (estuarine or marine)
- Depth (intertidal, shallow subtidal, deep subtidal)
- Exposure (low, medium, high)
- Substrate type (mud, sand, gravel, cobble, boulder, rocky, biogenic reef)

Marine protection tools

The policy provides that, in order to meet the requirements for a marine protected area, a management tool must enable the maintenance or recovery of the site's biological diversity at the habitat and ecosystem level to a healthy functioning state. In particular, the management regime must provide for the maintenance and recovery at the site of:

- Physical features and biogenic structures that support biodiversity
- Ecological systems, natural species composition (including all life-history stages), and trophic linkages
- Potential for the biodiversity to adapt and recover in response to perturbation

There are three types of protected area recognised in the policy based on the capacity of the tool to restrict human fishing and non-fishing impacts, with only Type 1 and 2 recognised as having sufficient protection to qualify as 'marine protected areas':

- Type 1: Marine protected areas that provide a high level of protection, including broad scope to restrict damaging activities (only marine reserves under the Marine Reserves Act)
- Type 2: Marine protected areas that provide only limited protection, but must as a minimum provide for prohibition of trawling, Danish seine netting and dredging
- Type 3: Other protection tools that are similar to marine protected areas, but which do not sufficiently protect biodiversity to meet the standard of Type 1 or 2 above. Type 3 areas are relevant when measuring progress toward the New Zealand Biodiversity Strategy target.²³

Type 2 marine protected areas can include a wide range of protective mechanisms including cable protection zones, marine mammal sanctuaries, rules in regional coastal plans and provisions under the Crown Minerals Act, Maritime Transport Act and Biosecurity Act.

In 2011, the Department of Conservation undertook a gap analysis of marine protected areas to determine how representative the network was. The report concluded that significant gaps existed. Of the ecosystem types that were represented in the existing network, only a very small proportion of their total area was subject to protection.²⁴ The report found that 6.9 per cent of all coastal marine bioregions were protected within the coastal marine area, but that the figure was heavily influenced by two large marine reserves (Kermadec Islands and Subantarctic Islands). Aside from those two bioregions, less than one per cent was protected to a Type 1 standard. Approximately 1.14 per cent of the territorial sea was estimated to be protected to a Type 2 standard. Most marine bioregions nationally were poorly represented in marine protection.²⁵ These areas are shown in Figure 6.7 below.

Biogeographic region	Total area (km ²) of marine reserves		Total area (km ²) of Type 2 MPAs		Total area (km ²)
Chatham Islands					12,318
East Coast South Island	112	1%			11,288
East Coast North Island	29	0.2%			11,637
Fiordland	104	1%	378	3.7%	10,241
Kermadec Islands	7675	100%			7149
North Cook Strait	30	0.2%	241	1.8%	13,671
North Eastern	85	0.2%	898	2.4%	38,073
Snares Islands					2154
South Cook Strait	39	0.3%	139	1.1%	12,241
Southern South Island	11	0.01%	88	0.4%	20,986
Subantarctic Islands	9408	79%			11,936
Three Kings					2226
West Coast South Island	174	1.3%			13,158
West Coast North Island	33	0.2%	329	2.3%	14,589
Total	17,698	10%	2,073	1.1%	181,699

Figure 6.7 Areas of marine protected areas in coastal marine biogeographic regions (Department of Conservation)²⁶

Note: Blank indicates no marine protected areas, data rounded to the nearest km²

Implementation

Recent implementation efforts for the policy have been concentrated on the territorial sea. As part of the agreement with the fishing industry to establish the 17 benthic protection areas in the EEZ, the government agreed not implement the marine protected areas policy within the EEZ until 2013. It is anticipated that planning in this area, when it does occur, will be undertaken by an expert offshore panel.

Each of the biogeographic regions is to be the focus of a planning process to be undertaken by community-based marine protection planning fora supported by staff from the Department of Conservation and the Ministry for Primary Industries. The fora will have up to 14 members including representatives from tangata whenua, commercial fishers, recreational users including fishers, charter fishers and divers, conservation groups, tourism, the aquaculture industry, marine science and the minerals industry.

The fora are tasked with identifying potential sites and tools for area-based protection of biodiversity. They are expected to seek consensus on proposed areas and to make recommendations to the Ministers of Conservation and Primary Industries. They are expected to constructively involve and engage stakeholders in each region and are required to seek written submissions on any recommendations being made. It was initially expected that each forum will produce a set of recommendations within 18 months of being appointed. In practice it has taken much longer than this.

The 'Marine Protected Areas: Classification, Protection Standard and Implementation Guidelines', prepared by the former Ministry of Fisheries and the Department of Conservation, provide design guidelines to assist the fora in identifying and selecting potential protected areas. These include guidance on identifying potential protected areas, on selecting candidate areas from among potential sites and for selecting appropriate tools for protecting the areas. Figure 6.8 summarises these details.

- Does each MPA include a variety of depths and transition zones and areas which cover the life-stage movements of species?
- Are the MPAs large enough to be effective, self-sustaining ecological units? Do they protect and maintain ecological processes such as nutrient flows and food-web interactions? Are they large enough for populations to be self-seeding?
- Does the MPA network include replicates of each representative habitat? It is recommended that at a minimum three replicates of each habitat type should be included.
- Does the MPA network include ecologically significant areas such as rocky reefs, oyster reefs, mussel beds, bryozoan reefs, salt marshes, seagrass beds and mangroves?
- Does the MPA network include unique habitats?
- Does the MPA network protect spawning aggregations of fish, egg sources and nursery areas?
- Does the MPA network protect roosting sites, foraging grounds, socialising areas and migratory routes for marine mammals and seabirds?
- Does the MPA protect source populations including larger, older individuals who have a greater reproductive capacity?
- Are the MPAs spaced so that larvae from as many species as possible can reach other MPAs with appropriate habitat?
- Are the MPAs in the network located so that they are connected by the exchange of eggs, larvae recruits, other propagules, juveniles and adults?
- Are individual MPAs large enough to cover a range of movement patterns of adults, juvenile fish and invertebrates and to include larval dispersal distances?
- Are MPAs located in a wide variety of places in relation to prevailing currents to facilitate larval dispersal?



Te Whanganui-A-Hei (Cathedral Cove) Marine Reserve protects rich and varied habitats associated with rocky reefs and sandy seafloor areas *(Raewyn Peart)*

More recent monitoring in 2011 has shown that butterfish, red moki, snapper, John Dory, banded wrasse, blue cod and silver drummer are all present within the reserve in higher numbers than outside it. Snapper abundance was four times greater within the reserve and overall species diversity was higher.³¹

The marine reserve attracts hundreds of thousands of visitors a year. It supports a number of tourism enterprises including the glass-bottom boat *Aquador*, which provides tours within the reserve and Goat Island Dive and Snorkel which provides snorkel and dive equipment, guided snorkelling tours and dive courses. In addition the University of Auckland has recently opened the Goat Island Marine Discovery Centre, next to their marine laboratory, which provides a touch tank, interactive displays and a wealth of information on the marine environment.



The marine reserve attracts hundreds of thousands of visitors a year and supports marine tourism enterprises (Raewyn Peart)



The Goat Island Marine Discovery Centre, on the shores of the marine reserve, provides a wealth of information on the marine environment for the public (Raewyn Peart)

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Differences between spatial planning of land and sea

Although the concept of spatial planning has been applied to land (such as in the form of zoning in district plans) as well as to the sea, the approaches taken in these two domains necessarily differ. The marine environment is more fluid and has greater ecological interconnections than land, so what happens in one area has greater impact on other areas. In addition, these interconnections are spatially more extensive. The lifecycles of marine species generally extend over much larger areas than terrestrial species, with larvae travelling long distances in ocean currents, before settling out and developing.

We know much less about the marine environment, with environmental impacts in the sea being less visible than those occurring on land and harder to detect. Marine management is much more reliant on scientific information, rather than visual observation, but this is more expensive to obtain as it often requires the use of research vessels and sophisticated underwater equipment.

Many marine activities are reliant on ecosystem health including commercial fishing, recreational fishing and diving, aquaculture, cultural harvesting, some types of research and ecotourism. At the same time, a much smaller proportion of the sea is fully protected, with around a third of New Zealand's land being in conservation areas, but less than one per cent of the coastal marine area surrounding the mainland being protected by marine reserves.⁴

All these factors make it particularly important that the marine area is managed well, but at the same time create significant challenges in achieving this. Marine spatial planning is an approach which uses the information which is available on the marine environment, and stakeholder values, to guide future decision-making. The resultant plans rarely include detailed zoning like those on land, but usually do identify significant areas which contribute to ecosystem health and need to be protected, and locations for key activities such as aquaculture, renewable energy and petroleum or minerals extraction.

Application of marine spatial planning

One of the earliest applications of marine spatial planning was to assist with the management of marine protected areas. The first spatial plan for Australia's Great Barrier Reef Marine Park was developed in the early 1980s. It identified a number of marine zones where a different range of activities could take place. A similar approach was applied to the Florida Keys National Marine Sanctuary during the 1990s where a comprehensive management plan, including zoning, was developed.

During the early 2000s, Australia and Canada laid the foundations for applying marine spatial planning to large bioregions, which extended out to 200 nautical miles from the shore. While the original intention in both cases was to develop comprehensive zoning plans, this proved difficult in practice, and the main tangible output has been the identification of candidate areas for marine protection.

Marine spatial planning has more recently focused on the management of conflicts between competing marine activities, particularly in the heavily congested marine areas of Europe. Belgium has been progressively implementing a Master Plan for its portion of the North Sea since 2003, the Netherlands developed an overarching spatial planning framework for its North Sea area in 2005 (and revised it in 2009), and in 2008 Germany finalised a spatial plan for its EEZ.

These European planning processes were largely prompted by the need to better manage growing demands from new ocean uses, such as wind energy, oil and gas extraction and aquaculture, as well as the need to implement European Union directives on nature conservation. The European Commission is promoting the adoption of marine spatial planning as 'a stable and transparent way to improve the competitiveness of the EU maritime economy and to deal with complex trans-national issues such as the effects of climate change.' The European Parliament endorsed a Directive on marine spatial planning in April 2014.

There has been a recent high-level initiative in the USA, supported by President Obama, to implement marine spatial planning nationwide. Similar to the approach taken in Australia and Canada, marine spatial planning is to be applied to large marine ecosystems. The spatial plans will identify:

Areas most suitable for various types or classes of activities in order to reduce conflicts among uses, reduce environmental impacts, facilitate compatible uses, and preserve critical ecosystem services to meet economic, environmental, security and social objectives.⁵

How marine spatial plans are developed

Elements of the plan development process, common to many marine spatial planning initiatives, include:

- Identification of goals, principles and strategies to guide the process
- Collation of spatial information on the marine environment
- Bioregionalisation of the planning region (i.e. spatially delineating the marine area into different categories based on broad habitat type)
- Identification of ecologically important and sensitive areas
- Collation of spatial information on human uses of, and values attached to, the marine environment
- Spatial analysis of compatibilities and conflicts
- Consideration of management approaches and options
- Identification of spatial management areas
- Finalisation of the plan
- Implementation of the plan
- Evaluation and monitoring

What a marine spatial plan might contain

There is a wide variety of material that can be included in marine spatial plans. But as a minimum, such a plan would be expected to include:

- A clear vision, including measurable goals of what the plan seeks to achieve
- Spatial identification of the 'ecological backbone' of the marine area including ecologically important marine areas and connections between them such as areas of high biodiversity, fish nursery areas, shellfish beds, important benthic habitats, important migratory routes for fish and marine mammals, important habitats for seabirds, seamounts, hydrothermal vents, salt marshes, seagrass beds, sponge gardens, horse mussel beds and mangrove forests. It could also include areas providing valuable ecosystem services such as estuaries and coastal wetlands.
- Spatial identification of different uses and values attached to the area including economic, cultural and social. This could include activities such as kaimoana gathering, fishing, shipping, mining, oil and gas, aquaculture and recreational boating.
- Spatial identification of areas of conflict: such as those between catchment activities and ecologically important marine areas, between marine users and those areas, and between different marine users
- Spatial identification of areas of opportunity such as places where activities support each other (e.g. marine conservation and tourism or aquaculture and good catchment management)
- Strategies and tools to address conflicts and maximise opportunities including the spatial identification of marine (and associated catchment) areas to be



Marine spatial planning can identify preferred locations for marine activities such as salmon farming, shown here at Ruakākā Bay, Marlborough
(Raewyn Peart)

Sea Change – Tai Timu Tai Pari: The Hauraki Gulf marine spatial plan

The Hauraki Gulf covers 1.2 million hectares of ocean, stretching from Mangawhai in the north to Waihi on the Coromandel Peninsula and hosting New Zealand's largest city. It is one of this country's most prized marine resources, for many reasons. The Hauraki Gulf generates more than \$2.7 billion every year in economic activity and supports the greatest number of recreational fishers and boaties in the country. It has a particularly rich diversity of seabirds, marine mammals, fisheries and marine habitats. In addition, it provides a wide variety of sanctuaries, marine reserves and islands. The Hauraki Gulf was designated a marine park in 2000.⁷

The Sea Change – Tai Timu Tai Pari project had its inception in the 2011 'State of Our Gulf' report⁸ released by the Hauraki Gulf Forum. This reported that 'the Gulf is experiencing ongoing environmental degradation, and resources are continuing to be lost or suppressed at environmentally low levels.' In addition, through taking a historical perspective on the changes to the state of the Gulf over time, the report was able to highlight 'the incredible transformation the Gulf has undergone over two human lifespans'. The report made it clear that current management approaches were ineffective in addressing the scale of the challenge and that a step change was required to turn the situation around.

The Forum released an updated report in 2014 that confirmed that 'pressures on the Gulf remain high and increasing' and that although there had been some environmental improvements, 'many values are continuing to be degraded or suppressed'.⁹

The preparation of a marine spatial plan was seen as a mechanism through which all the significant issues impacting on the Gulf could be considered together as an integrated whole. It also provided the opportunity for manawhenua and stakeholders to be placed at the centre of the process through the adoption of a collaborative process. A collaborative process is a powerful mechanism through which stakeholders gain an understanding of each other's values and perspectives, jointly scrutinise available scientific information, and seek to develop joint solutions.



Port Fitzroy, Great Barrier Island, is part of the area to be covered by the Hauraki Gulf marine spatial plan (Raewyn Peart)

Rhode Island Ocean Special Area Management Plan

This plan, which was completed in 2010, covers a marine area of around 3,800 square kilometres located off the coast of the state of Rhode Island, USA. It includes both state and federally managed marine areas. The area is heavily used for recreational boating, commercial and recreational fishing and shipping. It is located on the convergence of two biogeographic areas and is important for migratory fish, marine mammals, birds and sea turtles.¹¹

The key impetus for preparing the plan was the interest in developing offshore wind farms in the area. Spatial planning was seen as a tool which could be used to optimise the locations of these structures.

The plan was prepared by the Rhode Island Coastal Resources Management Council which is the state government's regulatory agency for the coastal area. The University of Rhode Island was closely involved in the plan management team providing several senior advisors and researchers.

A stakeholder group was established for the planning process, and had an independent chair. The group met monthly and initially refined and approved the goals and principles established for the plan. The group then received a comprehensive overview of the ecology and human use of the area. Following on from this, members considered and commented on each draft chapter of the plan which was prepared by agency staff. The group reviewed material as it was produced but was not asked to formally accept or reject it.

A science advisory task force was established to provide expertise and input into the science and research-based elements of the plan. Separate technical advisory committees were established for each chapter of the plan and they focused on improving and refining the content of each chapter.

The spatial elements of the plan, which subsequently emerged, include three broad 'zones' within the overall oceans area. The 'Renewable Energy Zone' is the preferred site for large-scale renewable energy projects in state waters. Where a project wishes to locate outside these areas it must be shown to have no significant adverse impact on natural resources or human uses.

A number of 'Areas of Particular Concern' were identified and mapped. These include:

- Historic shipwrecks, archaeological or historic sites and their buffers
- Offshore dive sites
- Glacial moraines which are important habitat areas for fish (see Figure 7.1)
- Navigation, military and infrastructure areas including designated shipping lanes and anchorages
- Areas of high fishing activity
- Heavily used recreational boating and sailboat racing areas
- Naval fleet submarine transit lanes

Applications for offshore developments are required to avoid Areas of Particular Concern. Where they cannot be avoided, the applicant must minimise to the greatest extent possible any impact and mitigate any significant impact. Applicants are also required to demonstrate why these areas cannot be avoided or why no alternatives are available.

'Areas Designated for Preservation' have also been identified as needing protection for their ecological value. These include sea duck foraging habitat. Large-scale offshore development, mining or other development which is in conflict with the 'intent and purpose' of an Area Designated for Preservation is prohibited.

The plan is to be accompanied by a 'Science Research Agenda' and a 'Progress Assessment and Monitoring Process'. This monitoring system will be designed to 'record decisions, capture lessons learned, note achievements, and document policy and management adaptations. This process will be ongoing, available on the project web sites, and formally reported to the public on a biannual basis'. The oceans plan itself will undergo a major review every five years.

In order to maintain stakeholder engagement with the implementation of the plan, a public forum is to be held every two years. This is intended to highlight projects underway, report on new research findings, review progress towards goals and objectives and recognise contributions to the plan's implementation. The public forum will also address emerging issues and any needed changes to the plan.¹²

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Endnotes

- 1 This chapter draws heavily from the Hauraki Gulf Forum report 'Spatial Planning for the Gulf' which includes an international review of marine spatial planning initiatives. We would like to thank the Hauraki Gulf Forum for making the material available for inclusion in this publication.
- 2 <http://www.aucklandcouncil.govt.nz/EN/AboutCouncil/representativesbodies/haurakigulfforum/Documents/Spatialplanforthegulf.pdf>
- 3 Interagency Ocean Policy Task Force, 2009, 1
- 4 Department of Conservation and Ministry of Fisheries, 2011
- 5 http://www.noaa.gov/factsheets/new%20version/marine_spatial_planning.pdf
- 6 UK-MSP Working Group, 2005, *Added value of marine spatial planning*, County Agencies, Interagency MSP Working Group, United Kingdom, cited in Douvere F and C N Ehler, 2009, 78
- 7 <http://www.aucklandcouncil.govt.nz/en/planspoliciesprojects/plansstrategies/seachange/Pages/home.aspx>
- 8 Hauraki Gulf Forum, 2011b
- 9 Hauraki Gulf Forum, 2014, 11–12
- 10 <http://www.seachange.org.nz/News-Events/Media-release-library/Sea-Change-underway-for-Hauraki-Gulf/>
- 11 Rhode Island Coastal Resources Management Council, 2010, 11–3
- 12 Rhode Island Coastal Resource Management Council, 2010, 11–6



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Preparing policies and plans

Policy/plans	Application	Opportunity for public involvement
Resource Management Act 1991		
New Zealand Coastal Policy Statement Other national policy statements	Activities on land and the territorial sea of the country	- Written submissions on proposed policy statement - Presentation of submissions and evidence to Board of Inquiry (where the Board of Inquiry process is adopted)
Regional policy statements ¹	Activities within catchments and the territorial sea of the region	- In some cases, written feedback on draft policy or plan - Written submissions on proposed policy or plan
Regional coastal plans	Activities within the territorial sea of the region	- Written further submissions by some parties - Presentation of submissions and evidence to council hearing²
Regional plans	Activities within the catchments of the region	Appeal to Environment Court – unless nationally significant proposal decided by Board of Inquiry
District plans ³	Activities within the district	- Presentation of submissions and evidence to Environment Court hearing - Appeal to High Court on points of law
Fisheries Act 1996		
Fisheries plans	Fisheries or groups of fisheries	- No statutory rights to participate - Minister must consult with persons or organisations representative of those classes of persons having an interest in the stock or effects of fishing on the aquatic environment in the area concerned including Māori, environmental, commercial and recreational interests - In practice, a draft plan is usually circulated for public written submissions
Conservation Act 1987		
General policy	Conservation areas and natural and historic resources applied nationally	- Written submissions on draft policy - Presentation of submissions to the Director-General of Conservation
Conservation management strategies Conservation management plans	Regional conservation areas and natural and historic resources	- Written submissions on draft strategy - Presentation of submissions at a meeting of representatives of the Director-General of Conservation and affected conservation boards
Marine Mammals Protection Act 1978		
Population management plans	Marine mammals	- Written submissions on draft plan - Presentation of submissions to the Director-General of Conservation or his or her representatives
Wildlife Act 1953		
Population management plans	Wildlife including seabirds	- Written submissions on draft plan - Presentation of submissions to the Director-General of Conservation or his or her representatives
Reserves Act 1977		
Reserve management plans	Land-based reserves	- Written submissions where intention to prepare plan is publicly notified - Written submissions on draft plan - Presentation of submissions to reserve administration body or nominee

Once the proposed policy statement or plan is formally notified, members of the public can make written submissions. The following persons are then able to lodge further submissions in support or opposition to the submissions already lodged:

- Any person representing a relevant aspect of the public interest. This is likely to include public interest environmental groups.
- Any person that has an interest in the proposed policy statement or plan greater than the interest that the general public has. This is likely to include owners of land and users of resources directly affected by plan provisions. It is also likely to include iwi and hapū where their interests are directly affected.
- The local authority itself

All submitters then have the opportunity to present their case before a council-initiated public hearing which is required under the RMA.

When preparing a plan or a plan change, councils must take into account relevant iwi planning documents which have been lodged with them. This provides an opportunity for tangata whenua to have proactive input into planning initiatives.

Once the council has made decisions on submissions, any submitter has the right to appeal the decisions to the Environment Court. The Court will consider the matter afresh at a public hearing.

The process is slightly different for the preparation of the Auckland Unitary Plan in the Auckland region. An Independent Hearing Panel is considering submissions. The Panel must make recommendations to Auckland Council within a defined timeframe. Recommendations accepted by Auckland Council can only be appealed to the High Court on points of law. Where Auckland Council rejects a recommendation of the Hearings Panel and decides on an alternative solution the right to appeal to the Environment Court is available.⁴

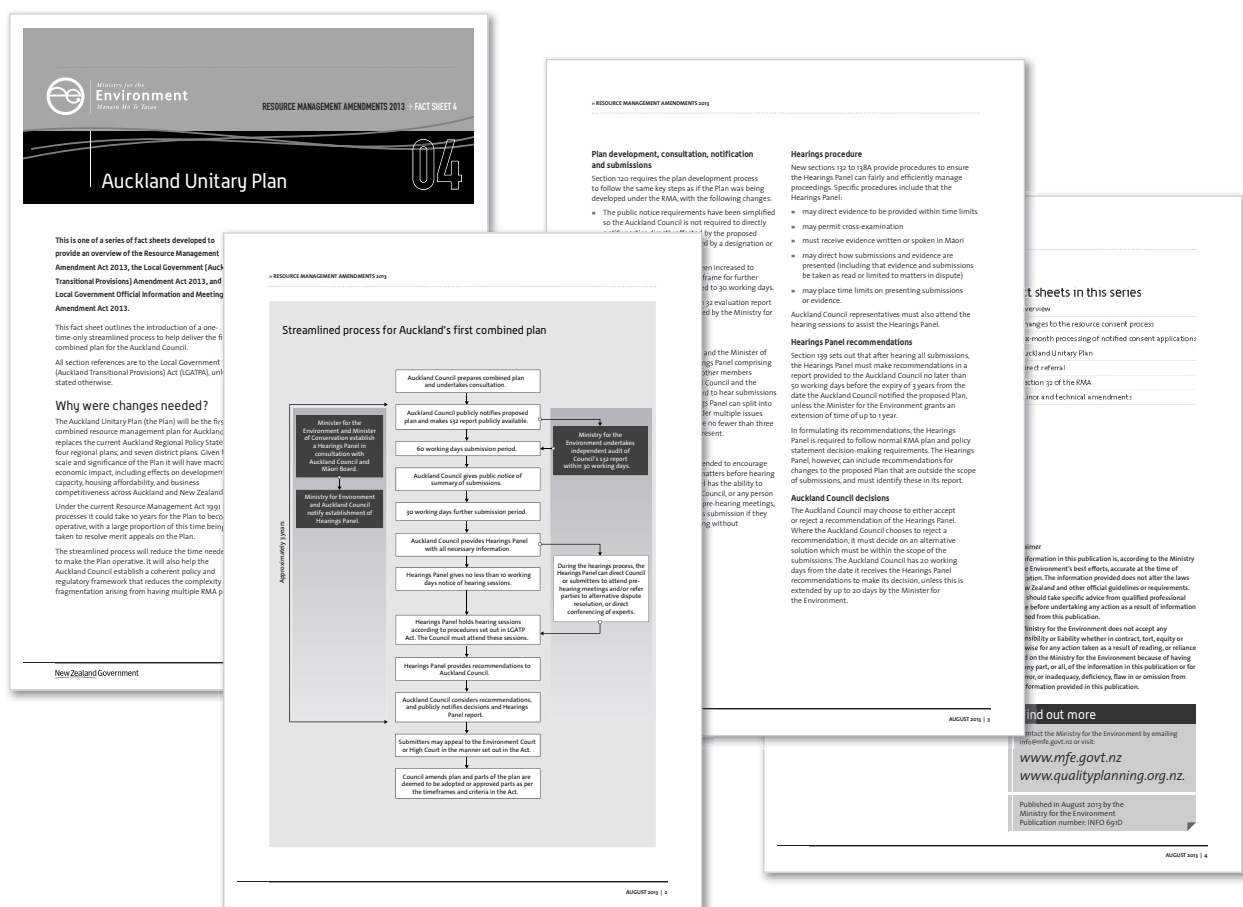


Figure 8.2 Public process applying to the preparation of the Auckland Unitary Plan (Ministry for the Environment)⁵

Making rules and regulations

There are a large number of processes through which rules and regulations are made to manage activities affecting the marine environment.

Rules and regulation	Application	Opportunity for public involvement
Resource Management Act 1991		
National environmental standards	Activities within catchments and the territorial sea	• Must be adequate time and opportunity for public comment • Usually written submissions on the proposed standard will be sought
Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act 2012		
Regulations	Activities within the EEZ and continental shelf	• Must be adequate time and opportunity for public comment • Usually written submissions on the proposed regulations will be sought
Fisheries Act 1996		
Sustainability measures Total allowable catch Total allowable commercial catch	Fish stock Fishing activity within the territorial sea and EEZ	• No statutory rights to participate • Minister must consult with persons or organisations representative of those classes of persons having an interest in the stock or effects of fishing on the aquatic environment in the area concerned including Māori, environmental, commercial and recreational interests • In practice, written submissions are sought on an initial advice paper
Mātaihai reserves	Spatial marine area	• Written submissions from local community and people having a fishing interest in the area on proposed reserve • Local community consultative meeting
Taiāpure	Spatial marine area	• Written submissions on proposed taiāpure • Presentation of submissions to judge of Māori Land Court • Appeal to the High Court on points of law
Maritime Transport Act 1994		
Marine protection rules	Vessels and/or offshore installations	• Written submissions on the proposed rule
Marine Mammals Protection Act 1978		
Marine mammal sanctuaries	Spatial marine area	• Written submissions on sanctuary after it has been notified by the Minister
Marine Reserves Act 1971		
Marine reserves	Spatial marine area	• Written objections to the proposal to establish a marine reserve
Crown Minerals Act 1991		
Minerals programme	Individual minerals	• Written submissions on draft minerals programme
Biosecurity Act 1993		
Import health standards	International vessels	• Consultation with persons who are representative of classes of persons having an interest in the standard • In practice, written submissions are sought from the public on draft standards

Figure 8.3 Summary of opportunities to participate in the development of rules and regulations affecting the marine environment

Mātiatai and taiāpure

When mātiatai reserves are proposed, consultation is undertaken with members of the local community and people with a fishing interest in the area. Public participation provisions for taiāpure proposals are broader. Any person can make written submissions on a proposal and present their submissions to a Māori Land Court judge. There are also rights to appeal the decision of the judge to the High Court on points of law. These processes are described in more detail in Chapter 4: Kaitiakitanga.

Marine reserves

When an application is made to establish a marine reserve, the application is publicly notified and any person can make a written objection to the proposal. The objections are lodged with the Director-General of Conservation. The final decision is made by the Minister of Conservation with the concurrence of the Ministers of Transport and Primary Industries. There is no right of appeal although judicial review is an available remedy.

Biosecurity

There are no statutory rights for the public to participate in the preparation of import health standards, although there is an obligation in the Biosecurity Act for the chief technical officer to consult with representatives of persons having an interest in the standard. In practice, written submissions are sought from the public on draft standards before they are finalised.

Other processes

When marine protection rules are developed under the Maritime Transport Act, when minerals programmes are developed under the Crown Minerals Act and when marine mammal sanctuaries are established by the Minister of Conservation, there are statutory provisions requiring members of the public to be given the opportunity to make written submissions, but no requirement for public hearings and no appeal rights.



Any person may make a written objection to a proposal for a marine reserve; the opposition shown here is to the proposed marine reserve at Nugget Point, Catlins Coast, Southland (*Raewyn Peart*)



Members of the public must be given the right to make written submissions on proposals for marine mammal sanctuaries, which may be proposed to protect endangered populations such as the bottlenose dolphins shown here in the Bay of Islands (*Raewyn Peart*)

Marine consents

The EEZ Act establishes that activities can be classified as permitted, non-notified discretionary, discretionary or prohibited. All marine consent applications for discretionary activities will be publicly notified and any person will have the opportunity to make written submissions and present to a hearing held by a Decision-Making Committee appointed by the EPA. Decisions can be appealed to the High Court on points of law only.

The ways that the public can become involved in a marine consent process, and suggestions for how this might be done, are provided in Figure 8.5.

Stage	Detail
Application lodged	
Assessing a marine consent application	In order to decide whether or not to lodge a submission, you will need to assess a marine consent application to determine the effects of the proposed activity on the environment and any existing interests.
Ensure that you have the full application	Before assessing a marine consent application, ensure that you have access to the full documentation including the marine consent application, the impact assessment, and any supporting technical reports.
Check that effects have been identified and addressed	Check that the effects of concern to you have been adequately identified and assessed. This may include impacts on the benthic environment, marine mammals or fishing interests.
Identify any modifications to mitigate effects	Identify how the application could be modified or measures taken to reduce adverse effects to acceptable levels. Identify other mechanisms for addressing adverse effects, such as environmental offsetting.
Consult with others	Arrange to meet with other interested parties to discuss the application. They may have similar concerns to yours and be able to provide different perspectives on the application. Explore the possibility of lodging a joint submission and pooling resources to obtain professional assistance. If possible, discuss the application with relevant professionals (such as a marine ecologist or oceanographer) who can provide initial guidance on the adequacy of the application and potential impacts of the proposal.
Submission period	
Identify aspects to support or oppose	Before preparing a submission on a marine consent application you will need to assess the application to identify which aspects of the application you wish to support or oppose. Once you have identified issues you would like to raise, you can prepare a written submission and lodge it with the EPA.
Form of submission	The EPA provides a submission form particular to each marine consent application. This will be available on the EPA website when the submission period opens.
Content of submission	• Contact details • Whether you have an existing interest that may be affected by the application • Your position on the proposed application (do you support in full, support in part, are neutral, oppose in part, oppose in full, or have a range of views on the application?) • Your reasons for your position. (When writing your reasons you should consider the Decision-Making Criteria⁹ and Information Principles¹⁰ that the Decision-Making Committee will need to take into account. These are discussed in Chapter 3: Economic zone legislation.) • The decision you would like the EPA to make • In what form you would prefer correspondence. (There will be a large amount of correspondence so we recommend you use electronic methods if possible.) • Whether you wish to speak to your submission by presenting at the hearing. You should attend the hearing to present your submissions if possible. • Whether you will be available to participate in meetings or mediation • Whether you will be providing expert witnesses
Lodging of submission	Submissions may be lodged by mail or email. Directions will be set out on the EPA website when the submission period opens. There is no charge for lodging a submission.

Stage	Detail
Appeals	
	• The applicant or any submitter on an application for a marine consent may appeal to the High Court against the decision of the EPA on a question of law only¹¹ • The purpose of an appeal on a question of law is to determine whether the EPA has acted within its powers. The weight given to various relevant considerations is for the EPA to determine and cannot be reconsidered by the High Court. • The decision of the High Court may be appealed to either the Court of Appeal or the Supreme Court • If you are considering an appeal to the High Court you will need to consult with a lawyer

Figure 8.5 Participating in a marine consent process under the EEZ Act

Publicly-initiated processes

In most cases, the initiation of a process to develop new policy or rules, or to amend existing instruments applying to the marine environment, is left to the statutory authority concerned. Members of the public will need to persuade the relevant Minister or agency that a change is needed. There are, however, two statutory processes that can be initiated by any person.

Private plan change

Any person can request a change to a regional or district plan, including a regional coastal plan, and this is called a 'private plan change'. Such a request can only be made, however, when the plan is operative. The council can only reject the request if:

- It is frivolous or vexatious
- The local authority or the Environment Court has considered the substance of the request within the last two years
- The request is not in accordance with sound resource management practice
- The request would make the plan inconsistent with the provisions of the RMA
- The policy statement or plan has been operative for less than two years

If none of these criteria apply the council must process the private plan change, and has the option of adopting the plan change as though it was initiated by the council. This is likely to occur when the council supports the change sought.

Requesting a private plan change can be an expensive exercise as the application will need to include an assessment of environmental effects of the proposal and an assessment of its benefits and costs. Preparing these assessments is likely to require the assistance of environmental professionals. In addition, the council can charge the applicant the full costs of processing the proposed plan change including the costs of the submission process and hearing.

Taiāpure

It is legally possible for any member of the public to propose the establishment of a taiāpure under the Fisheries Act. The application is lodged with the Minister for Primary Industries in the first instance. Support, however, is required from the Minister after consulting with the Minister of Māori Affairs, before the proposal can proceed.

The purpose of a taiāpure is to make 'better provision for the recognition of rangatiratanga and of the right secured in relation to fisheries by Article II of the Treaty of Waitangi'.¹² This means that, in practice, a person would need to be acting on behalf of an iwi or hapū that had an Article II right in order to succeed.



Although anyone can technically apply for a taiāpure, such as the Akaroa Taiāpure shown here, in practice only those acting on behalf of an iwi or hapū exercising Treaty rights are likely to succeed (Raewyn Peart)

Akaroa Marine Reserve

In June 2014, on World Oceans Day, the Akaroa Marine Reserve was established. It was a milestone in a process that had taken nearly two decades and which had struggled to secure consensus within the community affected.

The marine reserve's inception was in 1990, when Forest and Bird undertook a series of exploratory surveys in the Akaroa Harbour as part of a baseline study of its biodiversity. Near the Akaroa Heads the area was described as being 'profuse with marine life'. The area was already within a marine mammal sanctuary where set netting was banned for most of the year.¹⁶ Community interest groups, environmentalists and recreational bodies identified that the establishment of some form of marine protection within the harbour was the best course of action.¹⁷ The Akaroa Harbour Marine Conservation Working Group, initiated by the North Canterbury Conservation Board, was formed in 1993. Representatives from local community groups, dive clubs, recreational and commercial fishing interests, the five local rūnanga of the Banks Peninsula area, marine conservationists, aquaculturalists, recreational surface users and commercial tour operators were involved in the Group and began to process the proposal further.¹⁸

In 1995 the Akaroa Harbour Marine Protection Society was formed to facilitate the proposal for the marine reserve in the Dan Rogers area, a site selected for its important coastal value, including the diverse array of fish species supported, and the opportunity to protect across the land–sea boundary because of existing scenic reserves. The consultation period took 10 months and involved schools, dive clubs, resident groups, adjacent land owners and local iwi, as well as commercial and recreational fishermen.

In 1996 a formal application for a 560 hectare reserve, covering 10 per cent of the Akaroa Harbour, was submitted. The area of the marine reserve applied for was in the south-eastern area of Akaroa Harbour extending around the head of the harbour to Gateway Point. Over 2,300 supporting submissions were received along with more than 700 objections. A second round of consultation held later that year received a further 48 submissions of support along with 25 objections. Primary opposition came from recreational fishers, concerned about the loss of access to fishing, and Ngāi Tahu, who wished to see the area set aside as a taiāpure. A taiāpure was set in place prior to the marine reserve being created.

In 2010 the marine reserve application was declined by the former Minister of Conservation on the basis that the establishment of a reserve would have an adverse effect on the area's recreational use including fishing. The decision was challenged by the Akaroa Harbour Marine Protection Society (supported by the Environmental Defence Society) by way of judicial review and was subsequently reversed by the High Court in the same year. After agreeing to reduce the size of the reserve to 475 hectares, to take into account customary and recreational fishing areas, the Akaroa Marine Reserve was finally approved by the Minister of Conservation in 2013.



An active community group in Akaroa was successful in promoting the establishment of a marine reserve in the harbour
(Raewyn Peart)

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Peart R, 2008, *The New Zealanders' guide to the Resource Management Act 1991*, 3rd edition, Craig Potton Publishing, Nelson

Endnotes

- 1 In some areas plans are combined, such as the Proposed Auckland Unitary Plan which includes a regional policy statement, coastal plan, regional plans and district plans
- 2 For the Proposed Auckland Unitary Plan, a number of changes to standard processes were made by the Local Government (Auckland Transitional Provisions) Act 2010
- 3 Changes can be made to regional plans, coastal plans and district plans as part of nationally significant proposal decided by the Board of Inquiry, but not to regional policy statements
- 4 Local Government (Auckland Transitional Provisions) Act 2010, sections 156 and 158
- 5 <https://www.mfe.govt.nz/publications/rma/rma-amendment-act-factsheets-2013/factsheet-4.pdf>
- 6 Fisheries Act 1996, section 12
- 7 Fisheries Act 1996, section 21(2)
- 8 <http://www.mfe.govt.nz/publications/rma/annual-survey/2012-2013/rma-survey-2012-2013.pdf>
- 9 EEZ Act 2012, section 59
- 10 EEZ Act 2012, section 34
- 11 EEZ Act 2012, section 105
- 12 Fisheries Act 1996, section 174
- 13 <http://www.aucklandcouncil.govt.nz/EN/ENVIRONMENTWASTE/POLLUTION/Pages/reportpollutionhome.aspx>
- 14 Kirman C and C Whata, 2005, 1024
- 15 Known as the *Wednesbury* test
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- 17 http://www.saveakaroaharbour.co.nz/downloads/Akaroa_Harbour_MR_application.pdf
- 18 https://researcharchive.lincoln.ac.nz/bitstream/10182/2140/4/barr_msc.pdf
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- 25 <http://www.stuff.co.nz/waikato-times/news/2564378/Coromandel-town-still-divided-over-new-marina>



Black sea urchins, Great Mercury Islands (Daniel Sharp)



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Legend

— EEZ boundary

Trawling effort 1990-2008

Area trawled (km² per cell)

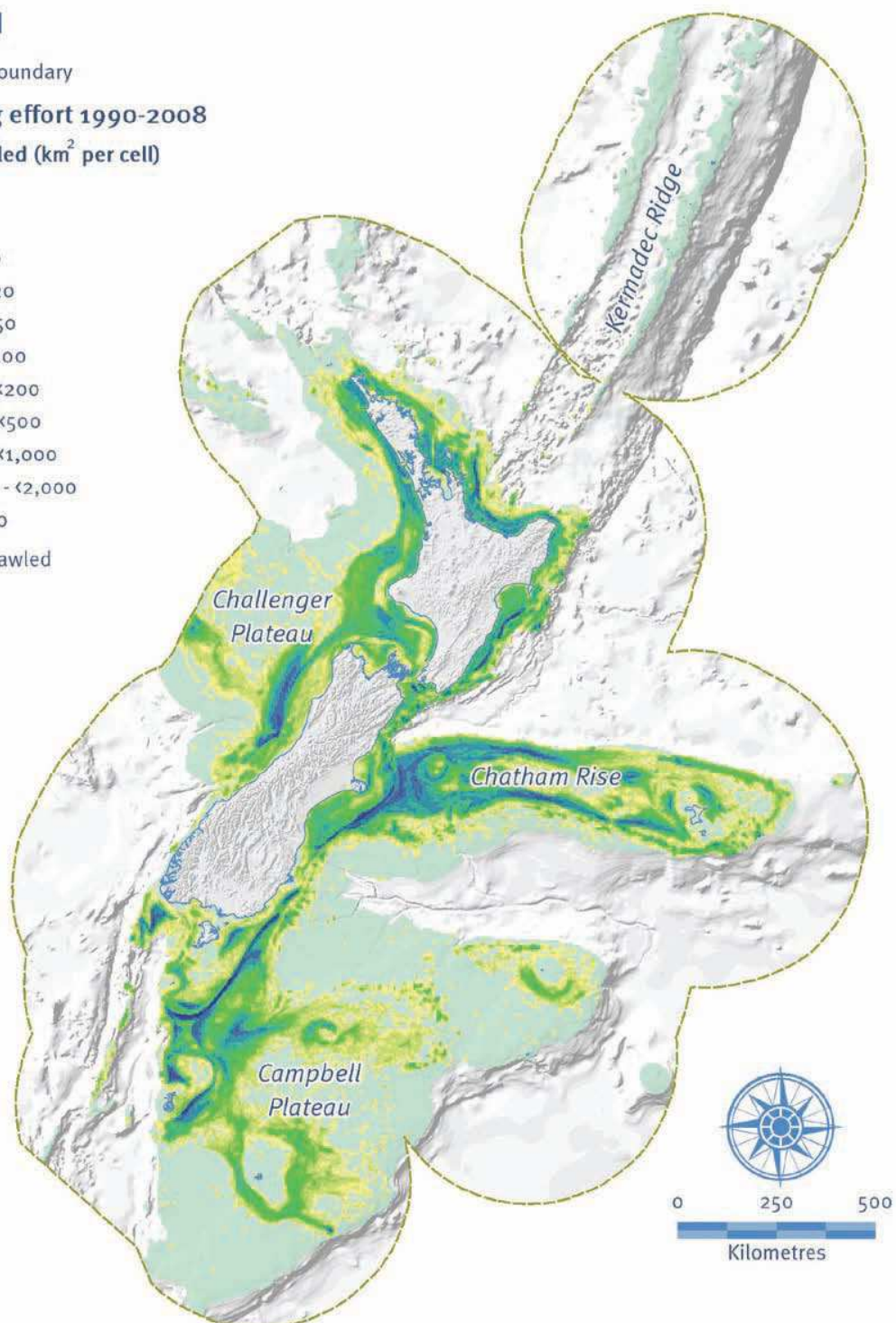
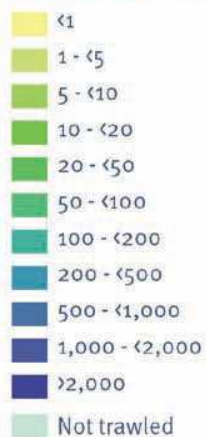


Figure 9.1 The general spatial pattern of commercial inshore trawl fishing activity⁶ (Ministry of Fisheries, NIWA and MfE)

General potentially adverse environmental impacts

Fishing activity can potentially impact on the marine environment in many ways. Some of these effects can be mitigated through careful fisheries management which is discussed further below in the section on elements of best practice.

The potential impacts can be classified into four broad categories as the effects on:

- Target fish populations
- Non-target species
- Seabed habitats
- Marine ecosystems

Impacts on target fish populations

Fishing activity can significantly reduce the population size of a targeted fish species. In fact, fisheries management has long been aimed at achieving such a reduction. This is because it is thought that 'fishing down' a stock will increase its productivity, through removing older, slower-growing individuals and freeing up ecological 'space' for younger faster-growing cohorts. But if fishing levels are not well managed, then sometimes biomass can reduce to a level where the stock has difficulty recovering. This can be a particular problem with long-lived species which reproduce slowly, such as orange roughy and sharks. It can also occur when species are hard to measure and therefore monitor.

In the main, fishing preferentially removes larger and older fish, which might affect the reproductive potential of the stock. This is because larger fish produce more eggs in relation to their body weight than smaller fish. Fishing also changes the size and age structure of exploited populations and potentially reduces their genetic diversity.

Some fish species, such as orange roughy, have a set of biological characteristics which mean that they are only able to sustain very low exploitation rates. In addition, if such stocks do become depleted, recovery takes a long time and may not occur at all. The characteristics include:⁷

- Maturation at relatively old ages
- Slow growth
- Long life expectancies
- Low natural mortality rates
- Spawning that may not occur every year (not all fish may spawn in any one year)

Impacts on non-target species

Most fishing methods result in the catch of species that are not targeted by the fisher (bycatch). Non-targeted fish species that are in the QMS (discussed below) cannot generally be discarded at sea and must be landed. In general, this bycatch is a more serious problem for fishing nets (including gillnets and trawling) than for hook and line methods and fish traps, which are more selective.



Non-targeted species such as the male pigfish shown here off the Alderman Island in the Hauraki Gulf, can be inadvertently caught as bycatch in fishing nets (Daniel Sharp)

Potential effects of specific fishing methods

A wide range of methods are used to harvest seafood, which are described below and shown in Figure 9.2. Each has different impacts on the marine environment, and these are of varying significance depending on the location and intensity of the activity. There are a number of methods which can be used to mitigate many of these impacts which are outlined in the section on elements of best practice.

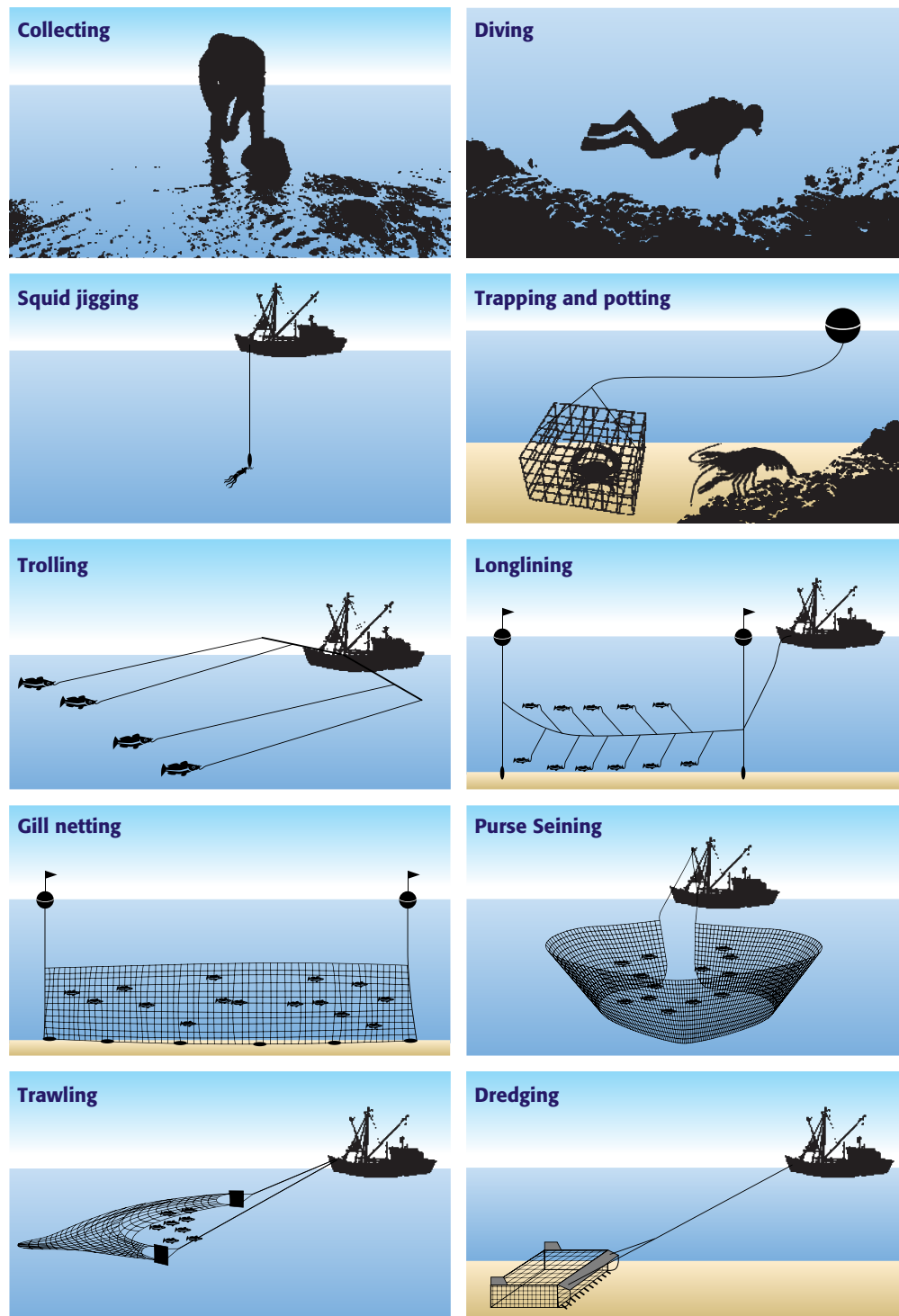


Figure 9.2 Sketches of different fishing methods

attached to the seafloor with an anchor at one end and a weight at the other. The lines are left on the bottom for between six and 12 hours and are then hauled in. The number of hooks may range from 1,000 to 30,000 per day. This method results in little damage to the fish caught, which sometimes can then be sold for higher prices, but can incur higher costs to catch. Longlines generally have little impact on seafloor habitats, although if lines and weights are dragged across the seafloor they can damage sensitive benthic organisms. The size and shape of the hook and type of bait can be tailored to attract specific species, thereby reducing bycatch. Because this method of fishing uses a large amount of bait, it can place pressure on stocks of bait fish if these are not managed sustainably. Longlining can result in the death of seabirds, turtles and marine mammals which drown as a result of becoming caught on the hooks or entangled in the lines.



Tuna fishing vessel berthed in Nelson (Raewyn Peart)

Gill netting

Gill nets are used to catch a variety of inshore fish including flounder, rig shark, elephant fish, butterfish and grey mullet. The nets are normally narrow with the bottom weighted down and the top held up with floats. Driftnets are left to drift in water currents and are usually set at or near the water surface. Set nets are anchored to the seafloor and are usually set near the bottom. The net effectively forms a wall in the water which fish swim into and get caught in, often by their gills. Using a net with a large mesh size can help avoid small fish being caught. This has little impact on benthic habitats. However, compared to other methods, there is a higher risk of catching non-target species with gill nets including dolphins, penguins, shags, shearwaters, seals, sharks and fish. Such nets are a particular problem for the endangered Hector's and critically endangered Maui's dolphins which live in the shallow coastal waters where nets are commonly set.

Purse seining

Seining is a method where a net is used to encircle fish and capture them. Purse seining is used to catch fish that feed on the surface such as tuna, trevally, kahawai or mackerel. When a school of fish has been sighted from the main fishing vessel, a small boat is launched which tows one end of the net around the school to encircle the fish. Ropes at the bottom of the net are drawn in (pursed) to completely enclose the fish. The net is then pulled close to the larger boat and the fish are scooped or pumped into the hold. Because purse seining targets schooling fish, it usually results in little bycatch, although dolphins can be caught. It has little impact on benthic habitats as the net does not normally come in contact with the seafloor.

Beach and Danish seining

Other variants of seining are beach seining and Danish seining. Beach seining, also called drag netting, is used to target mullet, flatfish, snapper, trevally and crabs. The beach seining net has long ropes attached to each end. One of the ropes is left on the beach and the net is then extended out into the sea and brought around in a semi-circle until the other rope reaches further down the beach. The net is then hauled in towards the shore. Danish seining is similar, but is undertaken in deep water with the net and ropes slightly weighted and drawn along the seafloor. It is used to target species such as snapper, flat fish, gurnard and John Dory. Beach and Danish seining can be less selective than purse seining, depending on the extent to which schools of fish are targeted. Danish seining can damage sensitive benthic habitats as a result of the weighted net and ropes being dragged across the seafloor.



Danish seine fishing vessel berthed at Whitianga wharf (Raewyn Peart)

Fishing method	Main features	Potential environmental effects		
		Bycatch	Protected species	Benthic habitats
Collecting and diving	Harvesting by hand or with a mechanical digger or rake	Low	Low	Generally low
Squid jigging	Lighting up the sea to attract squid and jigging hooked lures up and down	Low	Low	Low
Trapping and potting	Baited pots and traps are placed on the seafloor	Low	Low – seals and shags can be drowned and marine mammals can become entangled in pot lines	Low – damage can occur to sensitive species located under pots
Trolling	A set of hooked lures are slowly dragged through the water	Low	Low – occasionally fur seals and seabirds are caught	Low
Longlining	Numerous baited hooks are placed in the sea on small branch lines attached to long main lines	Medium – sharks and billfish can be caught	High – seabirds, marine mammals and turtles can become hooked or entangled; can also impact on benthic species such as corals and sponges	Low – although lines and weights dragged across the seafloor can damage sensitive benthic habitats
Gill netting	Nets are placed in the sea and fish swim into them	High – catches a wide range of species which swim into the net	High – Hector's and Maui's dolphins, seabirds and seals can become entangled	Low
Seining	Nets are used to encircle fish	Purse seining – low as targets fish schools Beach and Danish seining – low to high depending on the extent to which fish schools are targeted	Low	Purse seining – low Beach and Danish seining – low but may impact on sensitive benthic habitats
Trawling	A large net is dragged through the water or along the seabed	Low to high and is species dependent. May be very highly targeted but may catch wide range of non-target species.	High – seals, sea lions and dolphins can be caught	Mid-water trawling – low if net does not touch seabed Bottom trawling – high
Dredging	A steel-framed, toothed structure is dragged along the seafloor	High	Low	High

Figure 9.3 Potential environmental impacts of commercial fishing methods *(Adapted from Forest and Bird, 2012)*

The Ministry for Primary Industries' 'Harvest Strategy Standard 2008'¹⁵ is a Minister-approved policy statement of best practice for the setting of fishery and stock targets and limits. This serves as a statement of intent as to how stock management will be carried out within the discretion provided by the Act. The standard provides reference points for the size of stock biomass that trigger requirements for management action, such as the development and implementation of a stock rebuilding plan. Operational Guidelines developed by the Ministry in 2011 provide more clarity on how the standard will be applied. These indicate that the fished stock should be around 35 per cent of the unfished stock size for medium productive species such as cod and hake.¹⁶

Total allowable commercial catch

The Minister is required to set a total allowable commercial catch, which specifies how much of the total allowable catch can be harvested by commercial fishers, and may adjust it from time to time. This must not exceed the total allowable catch and the Minister must 'allow for' Māori customary non-commercial fishing interests and recreational interests before setting or adjusting the total allowable commercial catch. Where fish stocks are shared between commercial, recreational and/or customary fishers, the Act provides no guidance as to what proportion should be allocated to each sector. This is a matter which is left up to the discretion of the Minister for Primary Industries.¹⁷

Sustainability measures

The total allowable catch is the primary sustainability measure for a stock, but the Minister may also establish other sustainability measures to control the effects of commercial and other fishing on a fish stock or on the marine environment more generally. These measures may include restrictions on fishing methods, the size of fish taken, and where and when fishing may be undertaken.

Where there are fishing conflicts with marine mammals, the Minister may set a limit on fisheries-related mortality for a species. For example, in the squid fishery surrounding the Auckland Islands (SQU6T) the Minister has imposed a fisheries-related mortality level for the New Zealand sea lion for more than 20 years, the most recent number being 68 sea lions during the fishing year. If the limit is exceeded, the fishery is closed for the year.¹⁸



The Minister sets limits on fisheries-related mortality for sea lions under the sustainability measures of the Fisheries Act (Raewyn Peart)

Individual transferable quota

The main means of allocating commercial rights to fishing is through the creation of individual transferable quotas in individual fish stocks. Individual transferable quota are expressed as 'quota shares' and they provide a right in perpetuity to harvest a proportional share of the total allowable commercial catch for a fish stock.

Annual catch entitlement

Each quota share generates an annual catch entitlement which is the right to harvest that share of the total allowable commercial catch during one fishing year. For instance, if a party owns 10 per cent of the quota shares for snapper in the quota management area known as SNA1, the annual catch entitlements generated by such quota share, gives the right to harvest 10 per cent of whatever the total allowable commercial catch for SNA1 is in that year (expressed in kilograms). The total allowable commercial catch may change from year to year, but unless the quota owner buys or sells the shares, it will continue to have rights to 10 per cent of it. Both individual transferable quota and annual catch entitlements are freely traded. The latter are often leased to fishers who do not own quota, to enable them to harvest particular species.

Deemed values

A system of 'deemed values' encourages commercial fishers to ensure that they have sufficient annual catch entitlements to cover the species and amount of fish caught. It also provides an incentive to minimise bycatch of other species for which catch

Fisheries decision-making processes for deepwater, highly migratory species, inshore finfish, inshore shellfish and freshwater fisheries are guided by National Fisheries Plans. These plans are designed to build on the Fisheries 2030 Government strategy for the seafood sector which has a long-term goal of 'New Zealanders maximising benefits from the use of fisheries within environmental limits'.²⁰ They are important, because the Minister for Primary Industries is required to take them into account before setting or varying relevant sustainability measures, or making other decisions under the Fisheries Act.²¹ As well as setting out the broader management approach, the plans may contain fishery-specific chapters which provide more detail on the way that individual fisheries are to be managed in accordance with the overall fisheries plan. These chapters also provide operational objectives for managing bycatch species. Figure 9.6 outlines the wider context of the fisheries plans. Information about National Fisheries Plans is available on the Ministry's website.²²

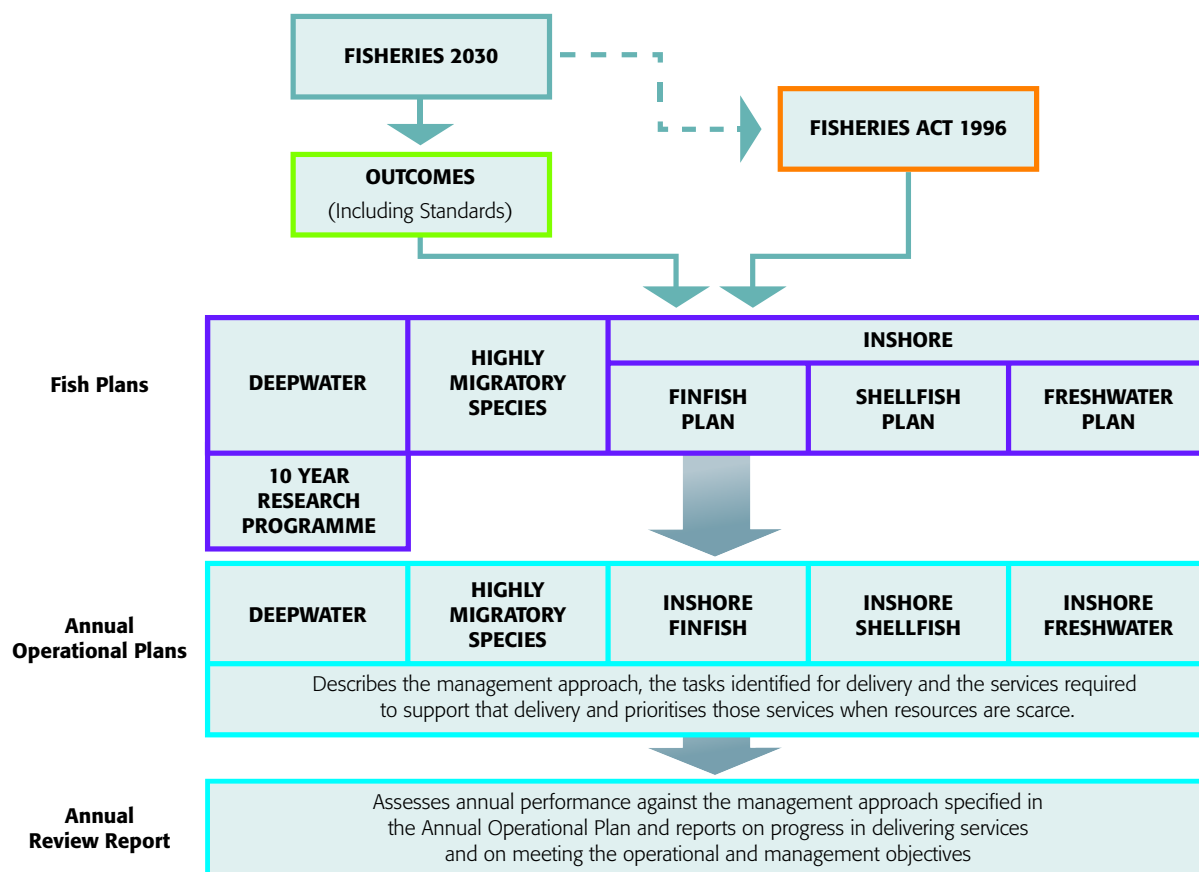


Figure 9.6 Fisheries planning structure (Ministry for Fisheries, 2010b)

Research planning and contracting

The Ministry for Primary Industries contracts around \$20 million of research each year (as at 2011).²³ Much of this cost is recovered from the fishing industry. Research is determined by fisheries managers, who take advice from science staff, and is carried out by different fisheries. For example, deepwater fisheries have had a 10 year research programme in place for four years. The projects identified in the research plan are contracted out to science providers. The draft results of each research contract are considered by the relevant science working group (see below) before finalisation. Research is important, as it is often the basis on which fisheries management decisions are made.

Stock assessment

Science working groups meet regularly to evaluate research and to contribute to the assessment of fish stocks. The Ministry for Primary Industries Science Working Groups cover a wide range of sector interests including deepwater, shellfish, middle-depth, Antarctic fisheries, Northern and Southern inshore, highly migratory species, rock lobster, eel, stock assessment and fisheries data.

Each round takes five to six months and consists of the following steps:

- An Initial Position Paper is prepared for each stock that is being reviewed. This is normally prepared by the Ministry, but in some cases is prepared by a fisheries management group, for example the National Rock Lobster Management Group. This will set out the issues to be addressed, identify and assess options to address them, and make recommendations as to what management measures should be adopted. The paper will also often include a summary of the relevant science related to the fish stock in question, usually drawing on the information contained in the latest plenary report.
- The Initial Position Paper is publicly released and written submissions called for. Usually six weeks is allowed for submissions, but this time may be extended for more contentious issues. Any person is able to make a written submission.
- Once the submission period has closed, Ministry staff summarise and consider the written submissions. The Ministry then prepares a Final Advice Paper, which is forwarded to the Minister. This provides the information on which the Minister will base his or her decision. This paper will normally have attached to it the Initial Position Paper and a summary of written submissions received.
- A notice is then placed in the *Gazette* to give legal effect to the final decision

There is no legal right to appeal the Minister's decision on its merits. It can only be legally challenged through initiating judicial review proceedings in the High Court.

Regulatory rounds

In addition to the sustainability rounds, the Ministry operates regulatory rounds, which function in a similar manner but apply to changes that require regulations to be adopted by the Governor-General on the recommendation of Cabinet (rather than becoming operative through a notice in the *Gazette* after the decision of the Minister). The regulatory round is more appropriate for decisions that will have longer-term implications.

Addressing environmental impacts

In order to help address the environmental impacts of fishing activity, the former Ministry of Fisheries prepared a report in 2005 titled 'Strategy for Managing the Environmental Effects of Fishing'.²⁷ The core approach adopted in the Strategy was the development of environmental standards to specify the limits of acceptable effects of fishing on the aquatic environment. Such standards were to have no statutory force, but were to be used as a guide to decision-making under the Fisheries Act. No environmental standards have as yet been finalised, but the following standards have been prepared:²⁸

- Harvest Strategy Standard 2008 – discussed further below
- Stakeholder Consultation Process Standard (draft) – outlines how stakeholders will be consulted during decision-making
- QMS Introduction Process Standard (draft) – defines the process for introducing new fisheries into the QMS
- Research and Science Information Standard 2011 – focuses on the delivery of high-quality research and science to inform fisheries management

The Harvest Strategy Standard, finalised in October 2008, provides guidance for the setting of the target sizes for fish stocks. It contains three core elements: a specified target about which a fishery or stock should fluctuate; a soft limit that triggers a requirement for a formal, time-constrained rebuilding plan; and a hard limit below which fisheries should be considered for closure. The Operational Guidelines for New Zealand's Harvest Strategy Standard (Revision 1) was prepared in June 2011 to support the implementation of the Standard.²⁹ This document includes guidance on calculating biological reference points which assist with setting fishing targets that form the basis for the default limits specified in the Standard.

Much of the work on Standards has now been superseded by National Plans of Action. For example, the National Plan of Action Seabirds 2013 sets out a number of long-term objectives for the next five years with the goal of reducing the seabird interactions with the fishing industry. These are broken down into practical, biological risk and research and development objectives, both on a high-level and five-year basis, and include the following examples:³⁰

- 'The level of mortality of New Zealand seabirds in New Zealand commercial fisheries are reduced so that species currently categorised as at very high or high risk from fishing move to a lower category of risk.'

Elements of best practice for fisheries

Ecosystem approach

There has been a growing awareness that fishing activity needs to be managed by an approach which considers the ecosystems within which the fisheries are located. An ecosystem-based management approach takes into account the interactions of fishing activity with the marine ecosystem and incorporates such considerations into fisheries management decisions. The aim is to allow the utilisation of fisheries while maintaining the natural structure, balance and functioning of marine ecosystems and important species. It also focuses on protecting habitats of importance to fisheries.

The fisheries management challenge can be conceptualised as consisting of a number of levels, which extend from simpler single species management up to more complex and comprehensive multi-sector approaches, as shown in Figure 9.7.³³

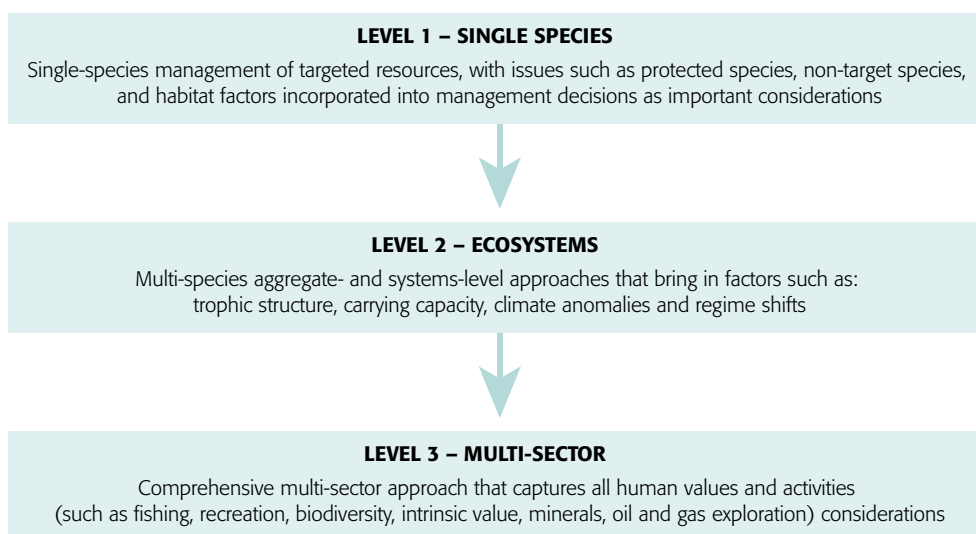


Figure 9.7 Levels of complexity within fisheries management

The fisheries management system currently applied by the Ministry for Primary Industries is based on Level 1, being primarily single-species focused. Applying a more comprehensive ecosystems approach requires an understanding of the ecosystems within which fishing is taking place, and the interactions between fishing and the affected ecosystems – knowledge which is gradually being developed. An ecosystems-based approach to fisheries management could be expected to take into account the issues identified in Figure 9.8.



Current fisheries management is focused on single species, but increasing knowledge about ecosystem interactions should enable a more comprehensive approach to be applied in the future. Fishing boats are shown here moored at the Viaduct Basin, Auckland (Raewyn Peart)

Issue	Best practice considerations
Monitoring and enforcement	• Is adequate monitoring being undertaken? • Is there adequate enforcement of fisheries regulations?
Management	• Have clear management goals been established? • Have management plans been developed to ensure progress towards those goals? • Do these management plans include clear targets and timelines for progress?

Figure 9.8 Best practice considerations for fishing activities (*Sourced and adapted from the Australian Department of the Environment and Water Resources, 2007; Marine Stewardship Council, 2002; and Ward et al., 2002*)

Strengthening links between science and the fishing industry

As an industry science provider Trident Systems are forging important links between science and the fishing industry. Trident is supporting fishing vessel crews with scientific protocol training. This is a partnership programme, with funding from a number of partners and co-funding from Seafood Innovations Limited. Working with a number of science providers, the company's research and development focus covers three key areas:³⁴

- Efficient sampling of inshore stocks, especially finfish fisheries
- Realising greater value from fisheries data
- Evaluating management procedures for lower-information stocks

A valuable component of Trident's work is the application of any scientific information gathered directly to processing plants and fishing vessels in order to improve fisheries practices.

Reducing seabird bycatch

Seabird mortalities are one of the major challenges associated with longlining. A number of measures can be implemented to reduce seabird attraction and subsequent entanglements or capture during fishing activity. These include:

- Batching offal – offal is held onboard and periodically discarded when there is less danger from the fishing activity. It can also be minced before it is discarded, making it less attractive to some birds. Offal management in the bottom longline sector is mandatory under fisheries regulations.³⁵



Southern Buller's Albatross can become caught on longline hooks (*Tamar Wells Photography*)

In response to sea lion bycatch in the squid trawl fishery, the fishing industry has developed a Sea Lion Exclusion Device, and this has been deployed in squid trawl nets over recent years. It consists of a grid that is placed across the net in front of the cod-end where the fish accumulate. The gaps in the grid are large enough for squid and other fish to swim through, but are too small for the larger sea lions. The device is designed so that marine mammals which reach the grid are forced to swim up, forward and out of the exit hole on top of the net. Research work on efficacy suggests that the device can increase the likelihood of survival for sea lions that enter a trawl net. More information can be found in the 'Aquatic Environment and Biodiversity Annual Review 2013'⁴² and in the Ministry for Primary Industries 'Fisheries Assessment Plenary May 2014'.⁴³

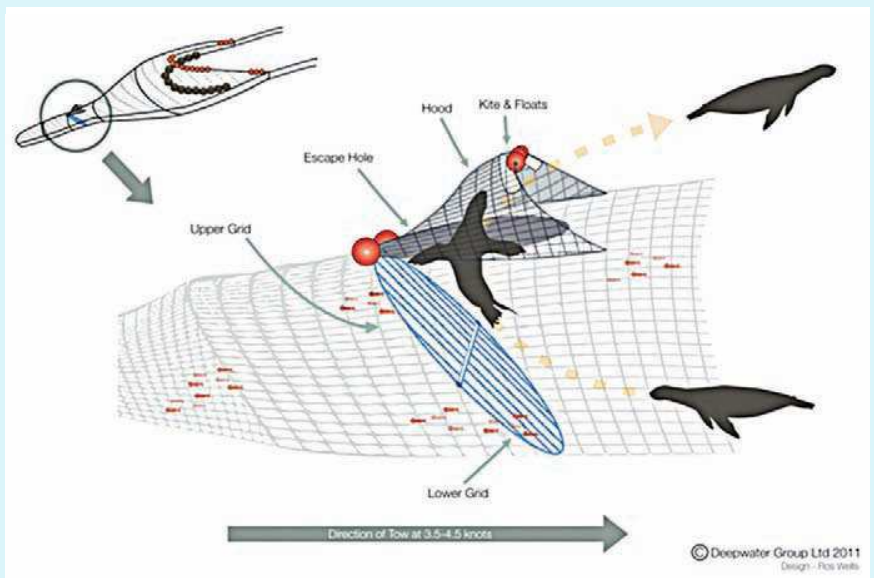


Figure 9.9 Diagram of a Sea Lion Exclusive Device inside a trawl net
(Ministry for Primary Industries, 2013b – image courtesy of the Deepwater Group)

Reducing fish bycatch

Bycatch of other fish can also be a problem for the trawling industry but does tend to be lesser where one species is specifically targeted. Best practice measures to reduce bycatch of juvenile and unwanted fish species include:

- Using longline or purse seine methods to harvest fish rather than trawling
- Using shorter more carefully targeted trawls
- Using larger mesh sizes for nets to enable juveniles to escape
- Not fishing in areas where juveniles predominate the catch



Juvenile snapper, such as the one shown here, can inadvertently become caught in trawl nets (Daniel Sharp)

Reducing benthic impacts

One approach to managing fishing impacts on the seabed is through the use of spatial controls. A wide variety of controls apply in New Zealand waters. Some of these were introduced specifically to manage the impacts of fishing on the seabed (see Figure 9.10). Other closures exist for other reasons such as for protecting cables, but have the effect of protecting certain areas of seabed from disturbance. These are discussed further in Chapter 6: Marine protected Areas.

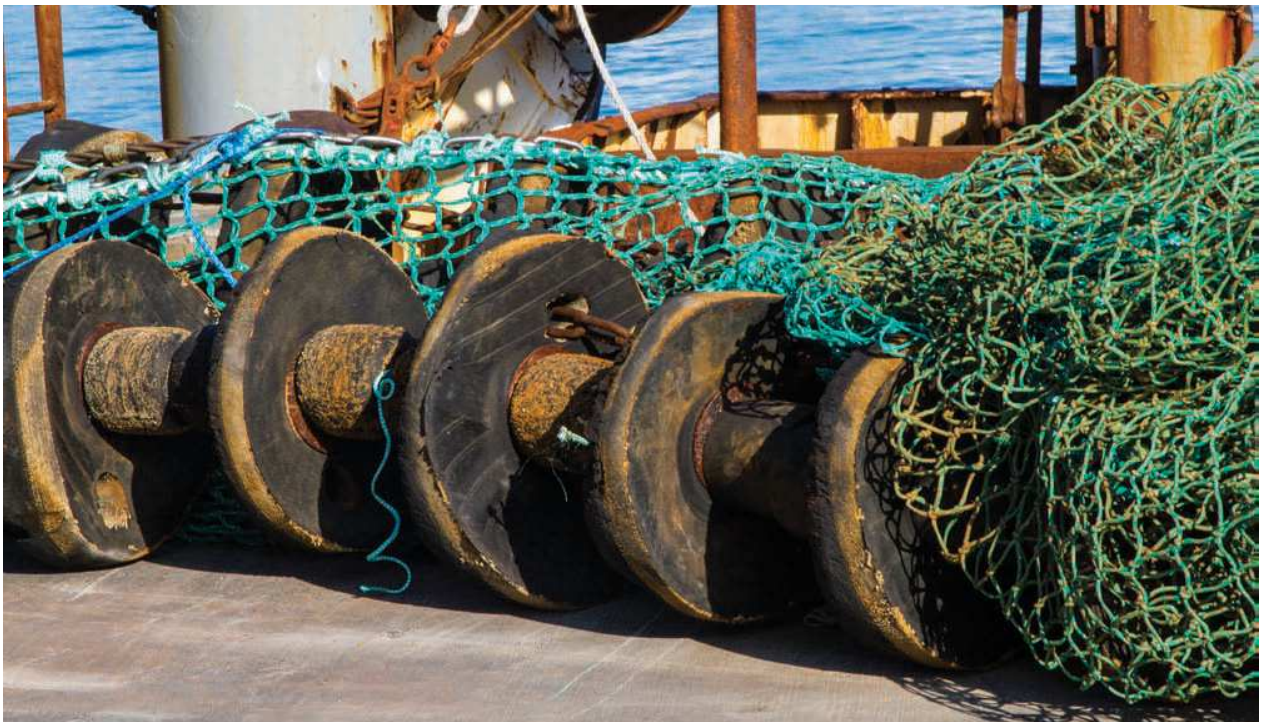
Other approaches to reducing the impacts of trawling on benthic habitats include:

- Ensuring that trawl equipment does not come into contact with the seafloor

Vulnerable marine ecosystems

The Food and Agriculture Organization's 2009 international guidelines on deep-sea fishing provide criteria to assist with identifying vulnerable marine ecosystems which are recommended to be specifically protected from the impacts of fishing. These guidelines are not binding and are focused on implementing the United Nations General Assembly Resolution on Sustainable Fisheries (UNGA61/105). The criteria provided by the guidelines include:⁴⁴

- a) Uniqueness or rarity – an area or ecosystem that is unique or that contains rare species whose loss could not be compensated for by similar areas or ecosystems. These include:
 - habitats that contain endemic species;
 - habitats of rare, threatened or endangered species that occur only in discrete areas; or
 - nurseries or discrete feeding, breeding, or spawning areas.
- b) Functional significance of the habitat – discrete areas or habitats that are necessary for the survival, function, spawning/ reproduction or recovery of fish stocks, particular life history stages (e.g. nursery grounds or rearing areas), or of rare, threatened or endangered marine species.
- c) Fragility – an ecosystem that is highly susceptible to degradation by anthropogenic activities.
- d) Life-history traits of component species that make recovery difficult – ecosystems that are characterised by populations or assemblages of species with one or more of the following characteristics:
 - slow growth rates;
 - late age of maturity;
 - low or unpredictable recruitment; or
 - long-lived.
- e) Structural complexity – an ecosystem that is characterized by complex physical structures created by significant concentrations of biotic and abiotic features. In these ecosystems, ecological processes are usually highly dependent on these structured systems. Further, such ecosystems often have high diversity, which is dependent on the structuring organisms.

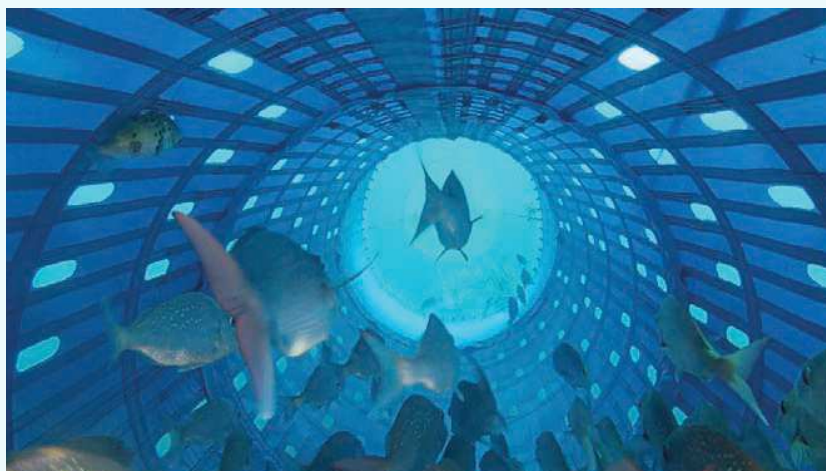


Gear dragged along the seafloor, such as the trawl net bobbins shown here on the Viaduct Basin wharf, Auckland, can damage vulnerable marine habitats (Raewyn Peart)

Precision seafood harvesting

Innovative research is underway to identify new technologies that can reduce the environmental impacts of fishing. Precision seafood harvesting, also known as modular harvesting, is the latest advancement in trawling and is a significant step change in harvesting technology.

This replacement system for the old trawl technology, which has been in use for over a century, has resulted from 10 years of development effort as part of a joint-funded research programme between the Government, Sanford Limited, Sealord Limited and Aotearoa Fisheries Limited. The new technology



New technology for reducing the environmental impacts of fishing being trialled
(Precision Seafood Harvesting)

has been based on understanding fish physiology and behaviour. This has enabled the development of a trawling system that is intended to reduce the number of non-target species taken, better target selected species, and land them in an improved condition. This could potentially provide multiple benefits including obtaining a higher price for the fish caught, reducing the numbers of juvenile fish killed and increasing the probability of bycatch species being released alive.⁴⁷

The new system means that traditional trawling nets are replaced with a flexible PVC landing liner which is dotted with escape portals. During a trawl, undersized and non-target species are more likely to be able to escape the net through the portals. This means that they can freely return to the sea without being boarded on a vessel, and this greatly reduces any physiological stress. Targeted fish continue to swim at a natural pace within the liner, until such time as they are landed. This method is far gentler than traditional methods, where fish are forced down to the cod-end of the net and hauled along at speed, and fish remain alive until they are landed. The target fish are often very close to the bottom and the new equipment will contact the seabed, but the performance of the new technology is significantly improved in other areas.

Trials using the new method have shown that if juvenile snapper are able to swim out of the net before it is hauled aboard, they have a 90 per cent survival rate in the first 24 hours post-harvest.⁴⁸ For targeted species, the much greater probability of being alive when the net is hauled improves catch quality and may result in more opportunities to export to live fish markets. In addition, the technological development means that any bycatch, which can include species too large to escape through the escape portals such as shark or rays, can be rereleased with a good chance of survival.⁴⁹ If the technology meets the shorter-term expectations for performance, it will allow smaller fishing vessels to significantly improve the quality and value of their catch. In the longer term it is expected there will be vessel design changes to optimise the handling systems for the new system and changes to fishery laws that will bring improved management of the fishery, including bycatch and wastage of juvenile fish.

The participating fishing companies have been trialing the new technology on board their fishing vessels. Vessel crews were initially skeptical about the new technology, but are beginning to see its value in the long run:⁵⁰

- *'Replacing the old trawl technology is really important for the industry. We're going to see better stock recruitment and better stock in the water'* – Nathan Reid, Vessel Manager for Aotearoa Fisheries Limited
- *'When we talk to them (the crews) now, when we see their reactions to the fish coming up, we know we're on to something. I know we're doing something unique and great when I look at the crews'* – Bill Healey, Vessel Manager for Sealord Limited

While the test results to date have been very encouraging, the validation of these results as a data set will take some time. This validation step is a necessary process that will help promote the successful deployment across a wider range of commercial fishing vessels at later stages of the programme.

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